

1(a) Functions $f(x)$ and $g(x)$ are defined as follows.

$$f(x) = \sqrt{x} \text{ for } x > 0 \text{ and } g(x) = x^3 - x - 6 \text{ for } x > 2.$$

The function $h(x)$ is defined as

$$h(x) = fg(x).$$

Find $h(x)$ in terms of x and state its domain.

[2]

(b) Find $h(3)$.

[1]

(c) Fig. 15 shows $h(x)$ and $h^{-1}(x)$, together with the straight line $y = x$.

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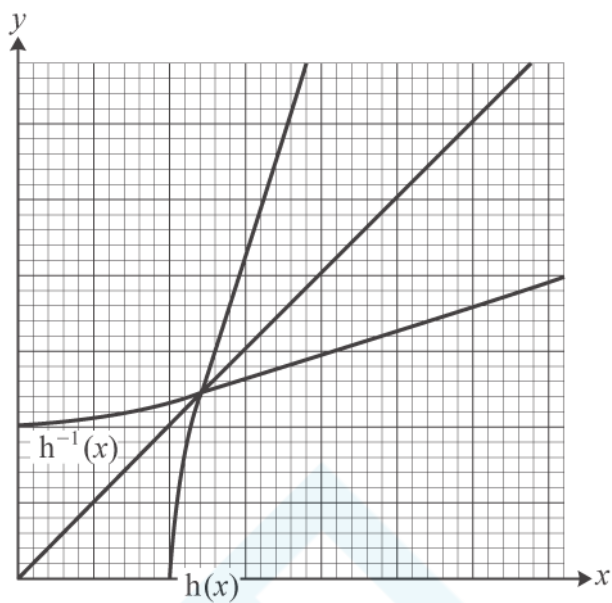


Fig. 15

Determine the gradient of $y = h^{-1}(x)$ at the point where $y = 3$.

[4]

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2 In this question you must show detailed reasoning.

Fig. 14 shows the graphs of $y = \sin x \cos 2x$ and $y = \frac{1}{2} - \sin 2x \cos x$.

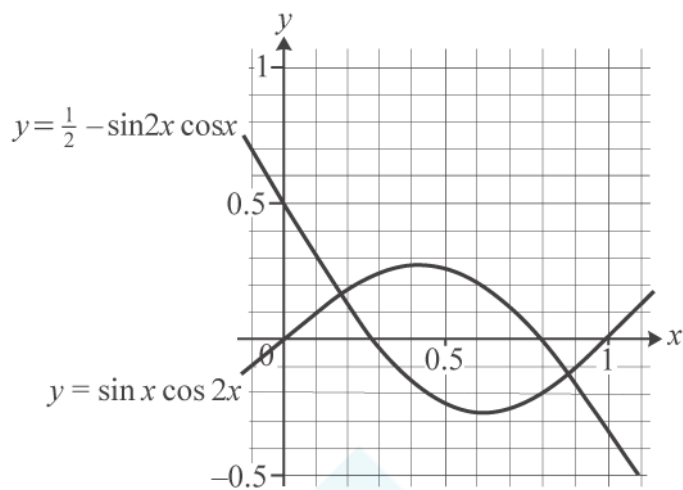


Fig. 14

Use integration to find the area between the two curves, giving your answer in an exact form.

[8]

3(a) In this question you must show detailed reasoning.

The equation of a curve is

$$y = \frac{\sin 2x - x}{x \sin x}.$$

Use the small angle approximation given below and in the Formula Booklet:

$\sin \theta \approx \theta$, $\cos \theta \approx 1 - \frac{1}{2}\theta^2$, $\tan \theta \approx \theta$ where θ is measured in radians to show that

$$\int_{0.01}^{0.05} y \, dx \approx \ln 5.$$

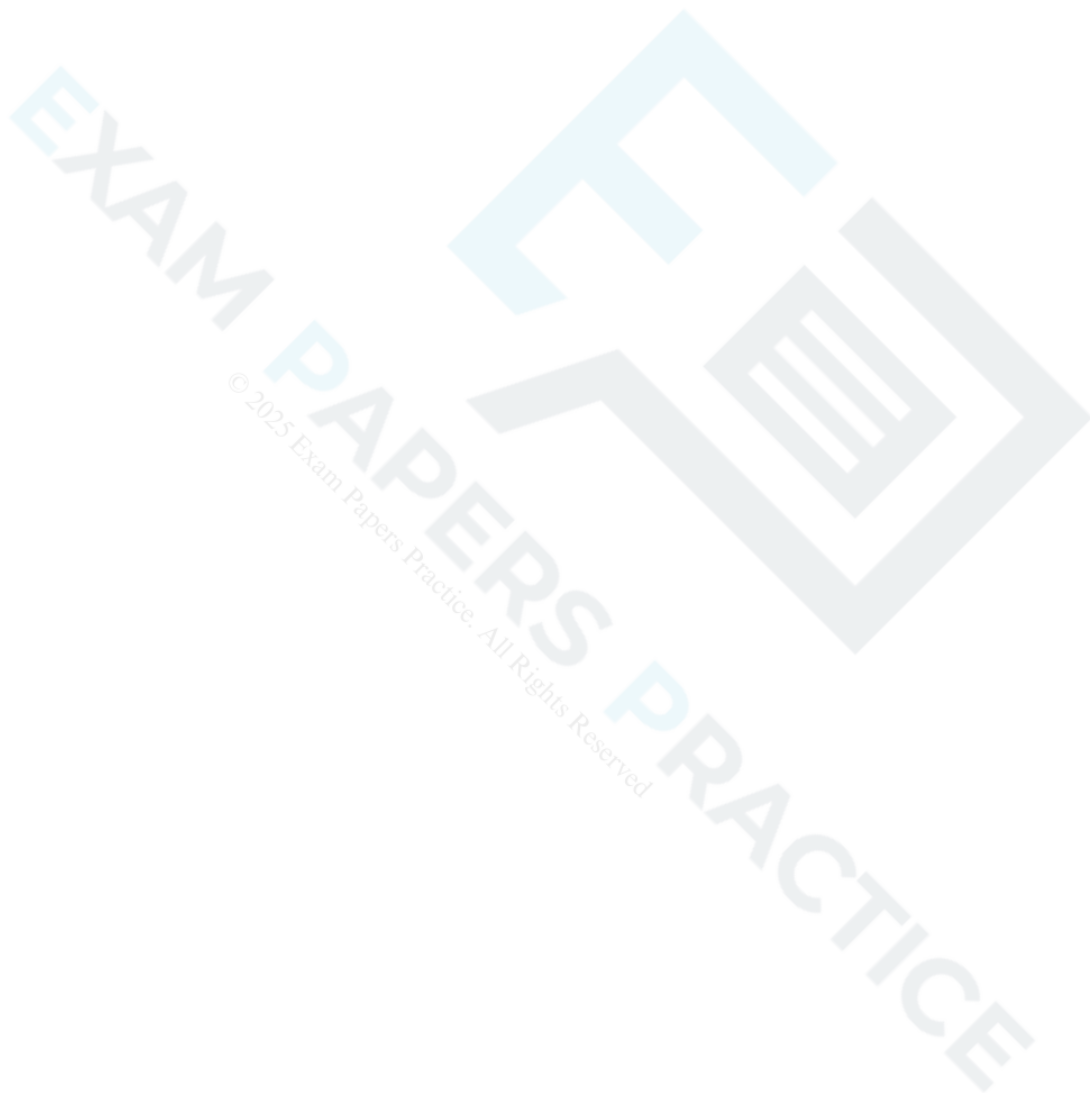
[4]



(b) Use the same small angle approximation to show that

$$\frac{dy}{dx} \approx -10000 \text{ at the point where } x = 0.01.$$

[2]



- (c) The equation $y = 0$ has a root near $x = 1$. Joan uses the Newton-Raphson method to find this root. The output from the spreadsheet she uses is shown in Fig. 10.1.

n	0	1	2	3	4	5	6	7
x_n	1	0.958509	0.950084	0.948261	0.94786	0.947772	0.947753	0.947748

Fig. 10.1

Joan carries out some analysis of this output. The results are shown in Fig. 10.2.

x	y
0.9477475	-7.79967E-07
0.9477485	-2.90821E-06
x	y
0.947745	4.54066E-06
0.947755	-1.67417E-05

Fig. 10.2

Consider the information in Fig. 10.1 and Fig. 10.2.

- Write 4.54066E-06 in standard mathematical notation.
- State the value of the root as accurately as you can, justifying your answer.

[3]

4(a) You are given that $y = 4x + \sin 8x$.

Find $\frac{dy}{dx}$.

[2]

(b)

Find the smallest positive value of x for which $\frac{dy}{dx} = 0$, giving your answer in an exact form.

[2]

- 5 Use integration to show that the area bounded by the x -axis and the curve with equation $y = (x - 1)^2 (x - 3)$ is $\frac{4}{3}$ square units.

[6]



6(a) Please refer to Insert H640-03 Maths B November 2020 to answer this question.

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Show that the only stationary point on the curve $y = \frac{\ln x}{x}$ occurs where $x = e$, as given in line 45.

[3]

(b) Show that the stationary point is a maximum.

[3]

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(c)

It follows from part (b) that, for any positive number a with $a \neq e$, $\frac{\ln e}{e} > \frac{\ln a}{a}$.

Use this fact to show that $e^a > a^e$.

[2]

7 Please refer to Insert H640-03 Maths B November 2020 to answer this question.

Show that e^x is an increasing function for all values of x , as stated in line 39.

[2]

Show that $\int_e^{\pi} \frac{1}{x} dx = \ln \pi - 1$ as given in line 37.

[2]



9(a)

Use the substitution $x = \tan \theta$ to find the exact value of $\int_{-1}^1 \frac{1}{(1+x^2)^2} dx$.

[8]

(b)

The curve $y = \frac{1}{(1+x^2)^2}$ is shown in Fig. 8.

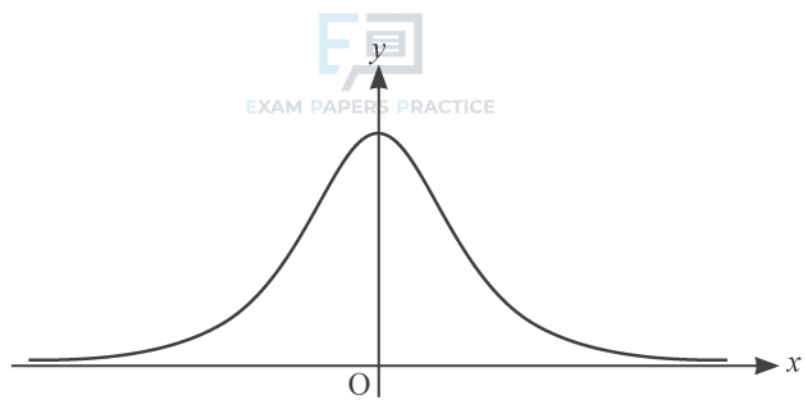


Fig. 8

(i)

Show that $\frac{d^2y}{dx^2} = \frac{20x^2 - 4}{(1 + x^2)^4}.$

[5]

(ii) In this question you must show detailed reasoning.

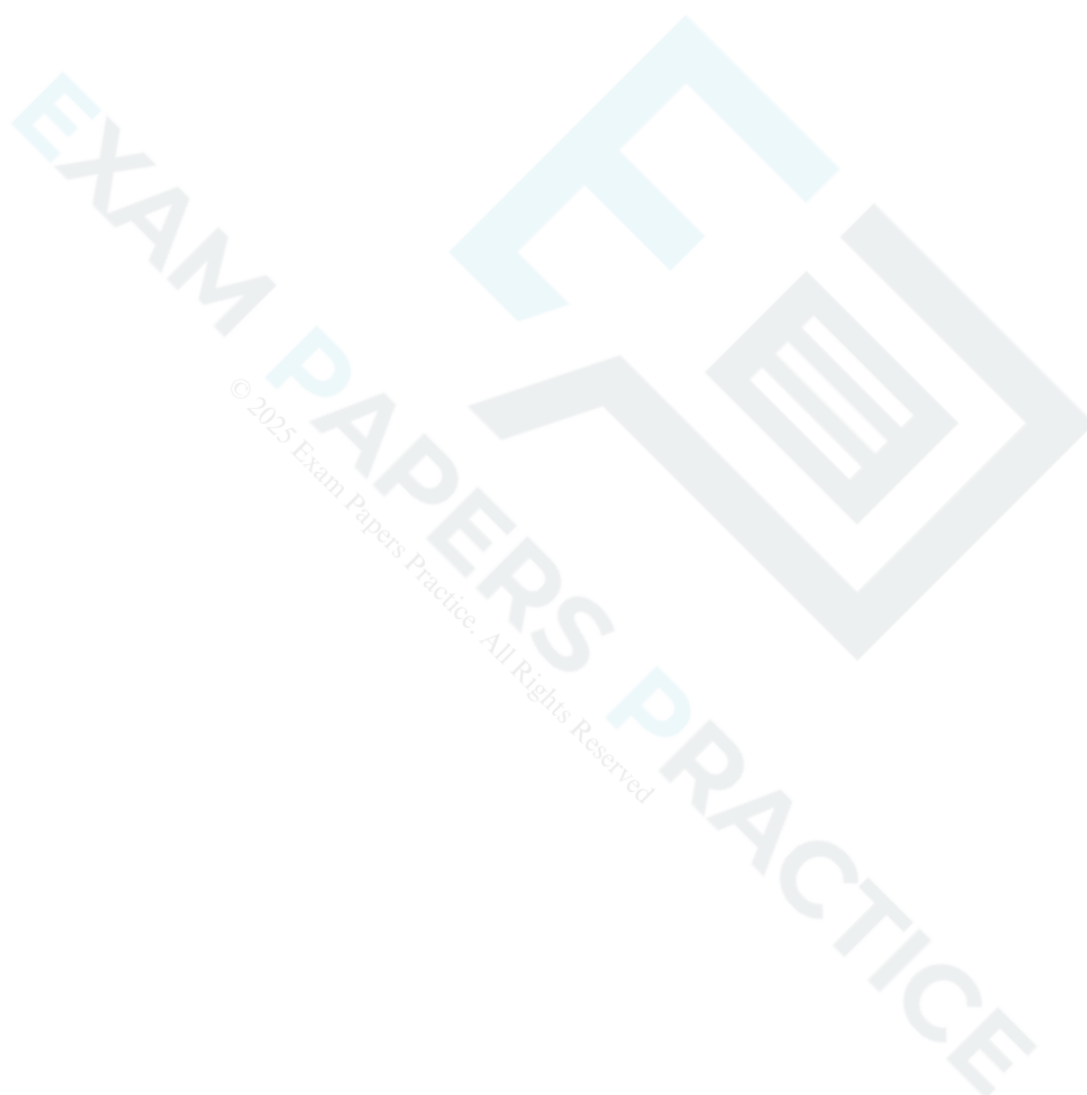
Find the set of values of x for which the curve is concave downwards.

[3]

10(a)

Express $\frac{1}{x} + \frac{1}{A-x}$ as a single fraction.

[1]



- (b) The population of fish in a lake is modelled by the differential equation

$$\frac{dx}{dt} = \frac{x(400 - x)}{400}$$

where x is the number of fish and t is the time in years.

When $t = 0$, $x = 100$.

In this question you must show detailed reasoning.

Find the number of fish in the lake when $t = 10$, as predicted by the model.

[8]

11(a) In this question you must show detailed reasoning.

Fig. 12 shows part of the graph of $y = x^2 + \frac{1}{x^2}$.

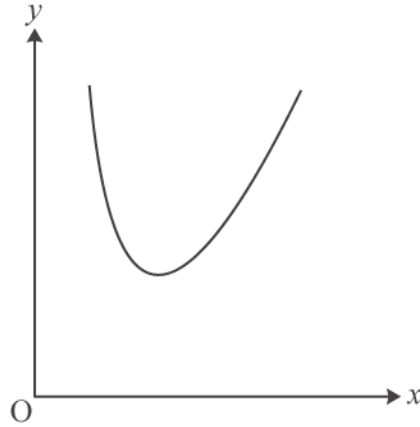


Fig. 12

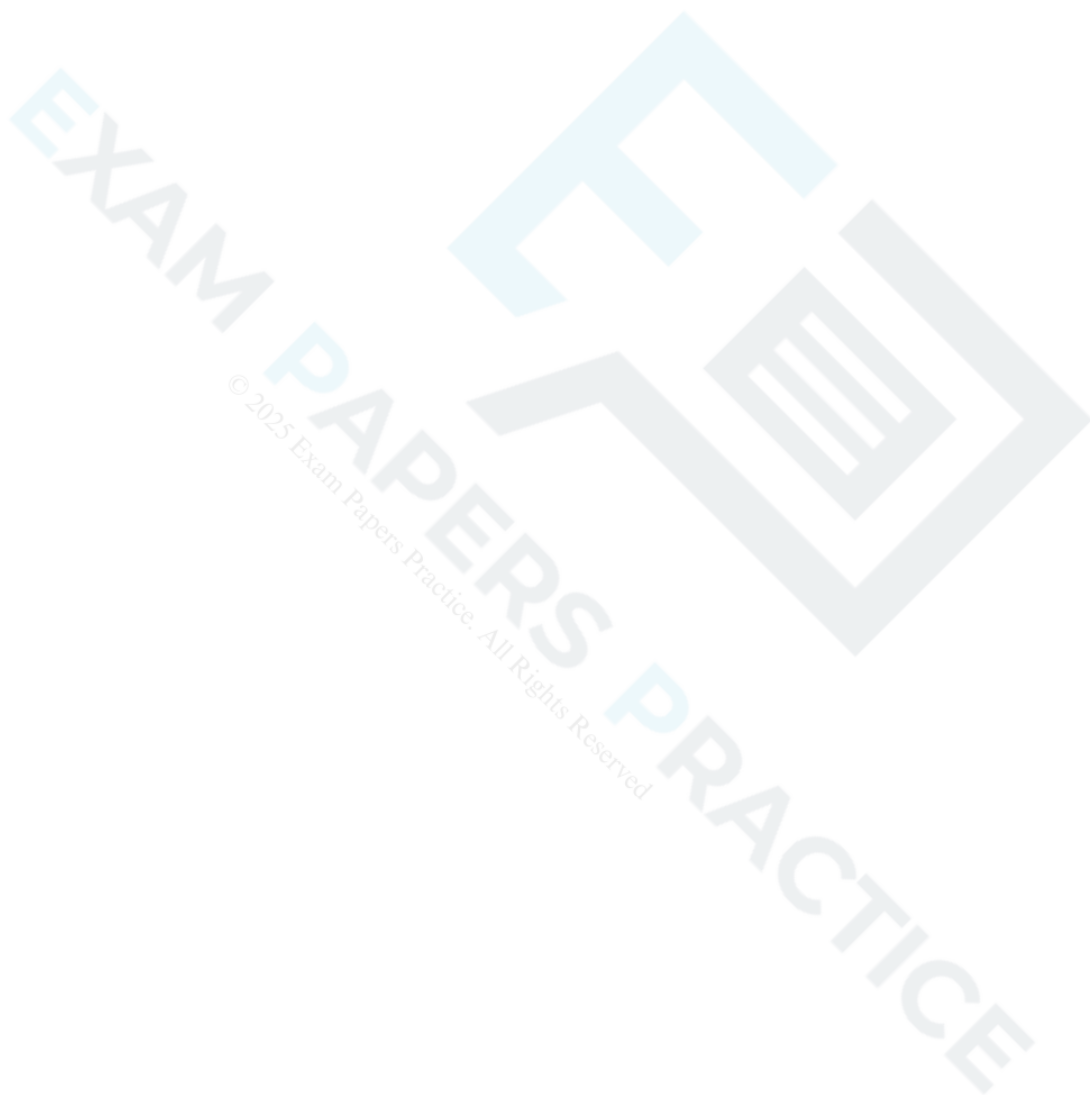
The tangent to the curve $y = x^2 + \frac{1}{x^2}$ at the point $(2, \frac{17}{4})$ meets the x-axis at A and meets the y-axis at B. O is the origin.

Use calculus to prove that the complete curve has two minimum points and no maximum point.

[6]

(b) Find the exact area of the triangle OAB.

[6]



12(a) In this question you must show detailed reasoning.

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A curve has equation $y = 4x^3 - 6x^2 - 9x + 4$.

Sketch the gradient function for this curve, clearly indicating the points where the gradient is zero.

[4]

(b) Find the set of values of x for which the gradient function is decreasing. Give your answer using set notation. [2]

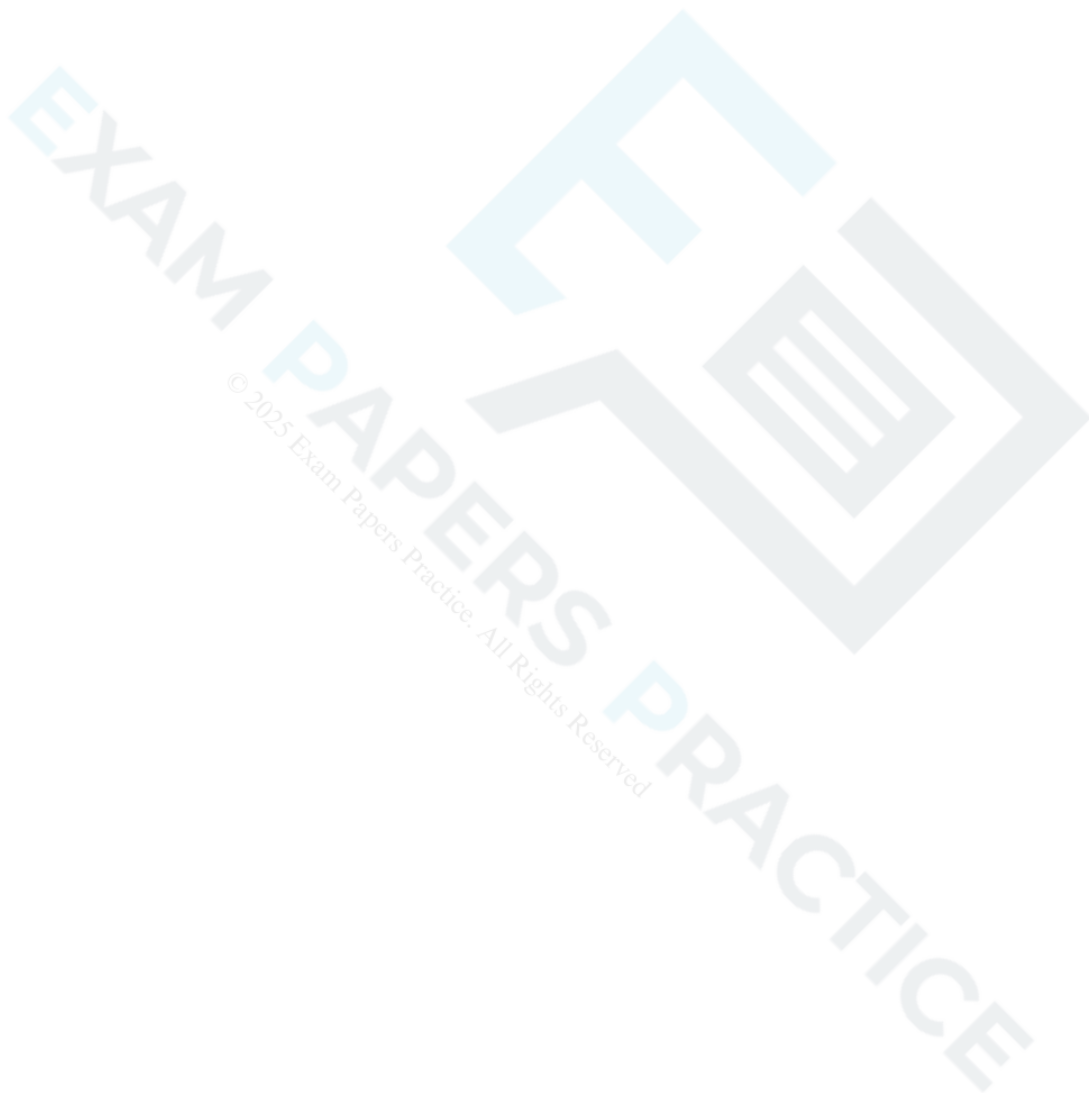
- 13 A function is defined by $f(x) = x^3 - x$.

Show that the curve $y = f(x)$ has a single point of inflection which is not a stationary point.

[3]

- 14 Find the second derivative of $(x^2 + 5)^4$, giving your answer in factorised form.

[5]



15(a) To answer this question, please refer to the Insert for H640-03 June 2019.

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Hence show that $V = \frac{2}{3}\pi r^2 h$, as given in line 41.

[3]

(b)

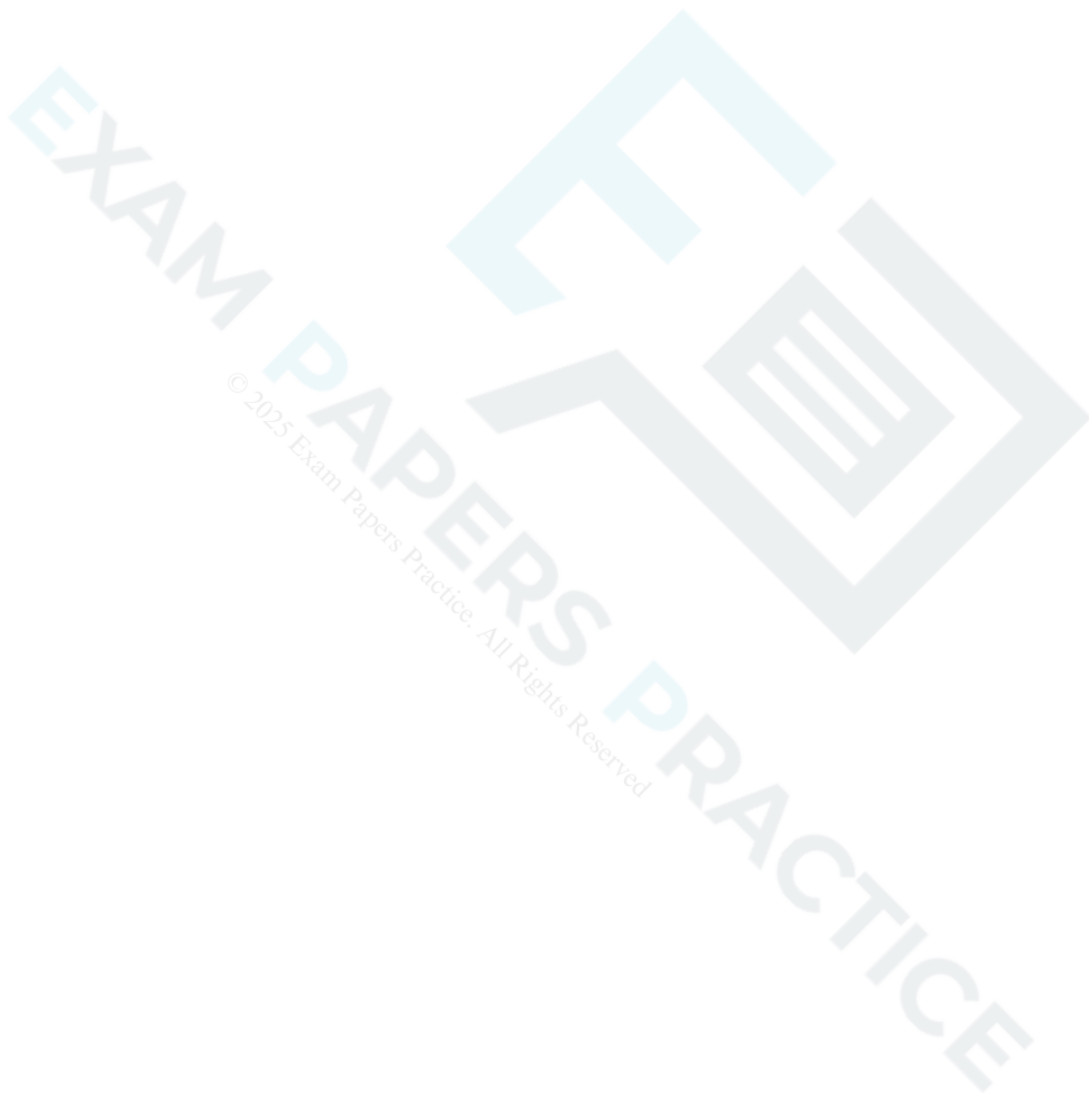
Express $\lim_{\delta y \rightarrow 0} \sum_0^h (h^2 - y^2) \delta y$ as an integral.

[1]

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By using the substitution $u = 1 + \sqrt{x}$, find $\int \frac{x}{1 + \sqrt{x}} dx$.

[6]





Show that $f(x) = \frac{e^x}{1+e^x}$ is an increasing function for all values of x .

[4]

- 18 In this question you must show detailed reasoning.

The curve $xy+y^2 = 8$ is shown in Fig. 9.

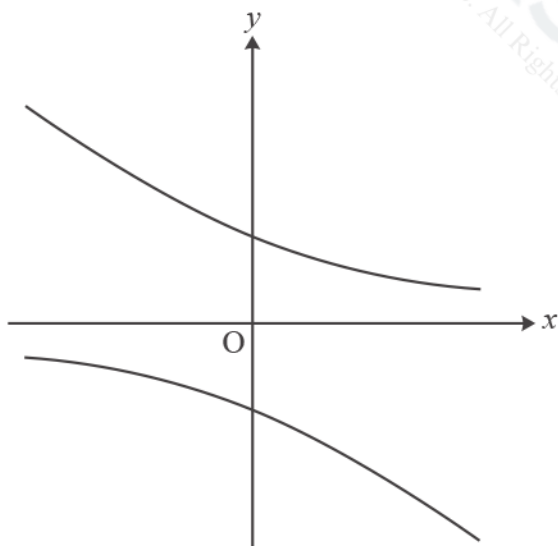
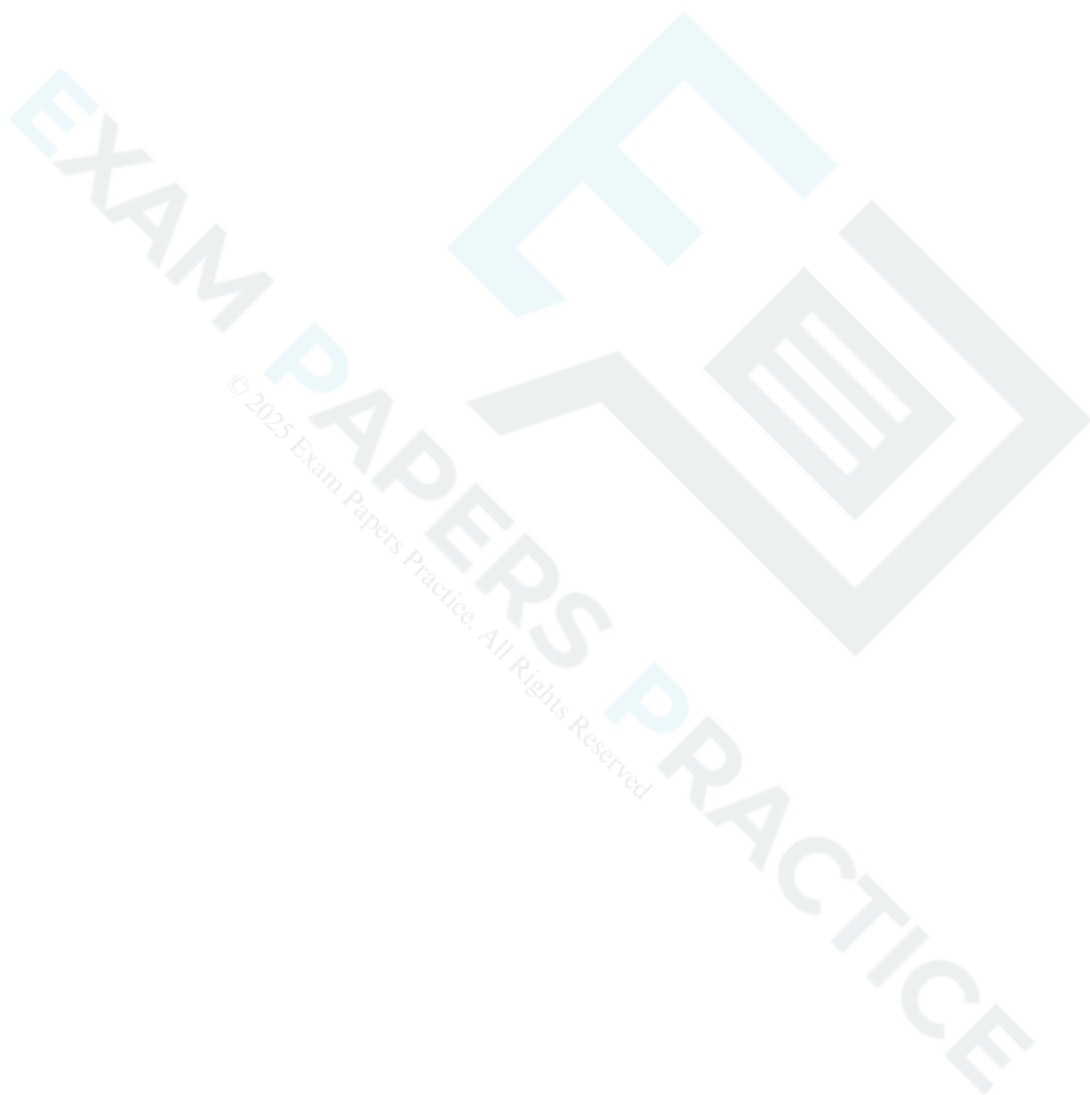


Fig. 9

Find the coordinates of the points on the curve at which the normal has gradient 2.

[6]

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A is the point (1, 0), B is the point (1, 1) and D is the point where the tangent to the curve $y = x^3$ at B crosses the x -axis, as shown in Fig. 8. The tangent meets the y -axis at E.

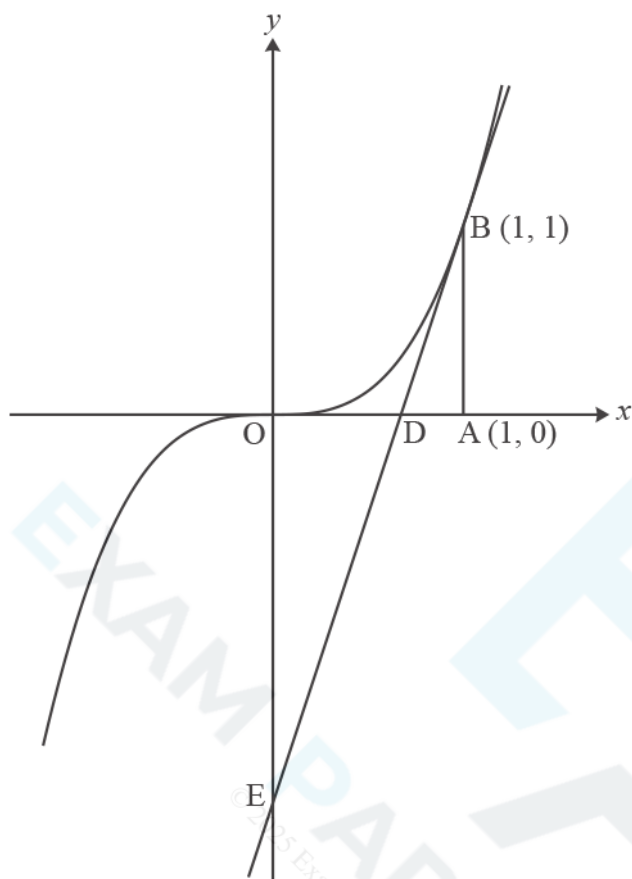


Fig. 8

Find the area of the region bounded by the curve $y = x^3$, the tangent at B and the y -axis.

[4]



Find $\int \frac{1}{(x+2)(x+3)} dx$ in the form $\ln(f(x)) + c$, where c is the constant of integration and $f(x)$ is a function to be determined. [3]

21 The function $f(x)$ is defined for all real x by

$$f(x) = 3x - 2.$$

Find the set of values of x for which $f(x) > f^{-1}(x)$. [2]

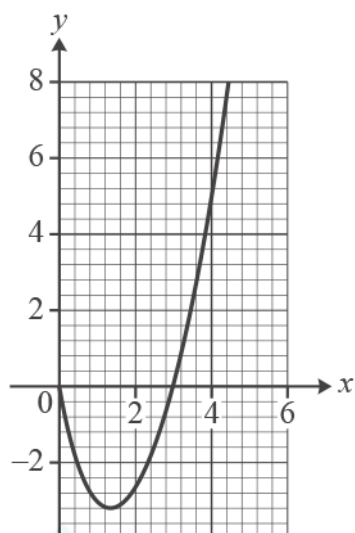


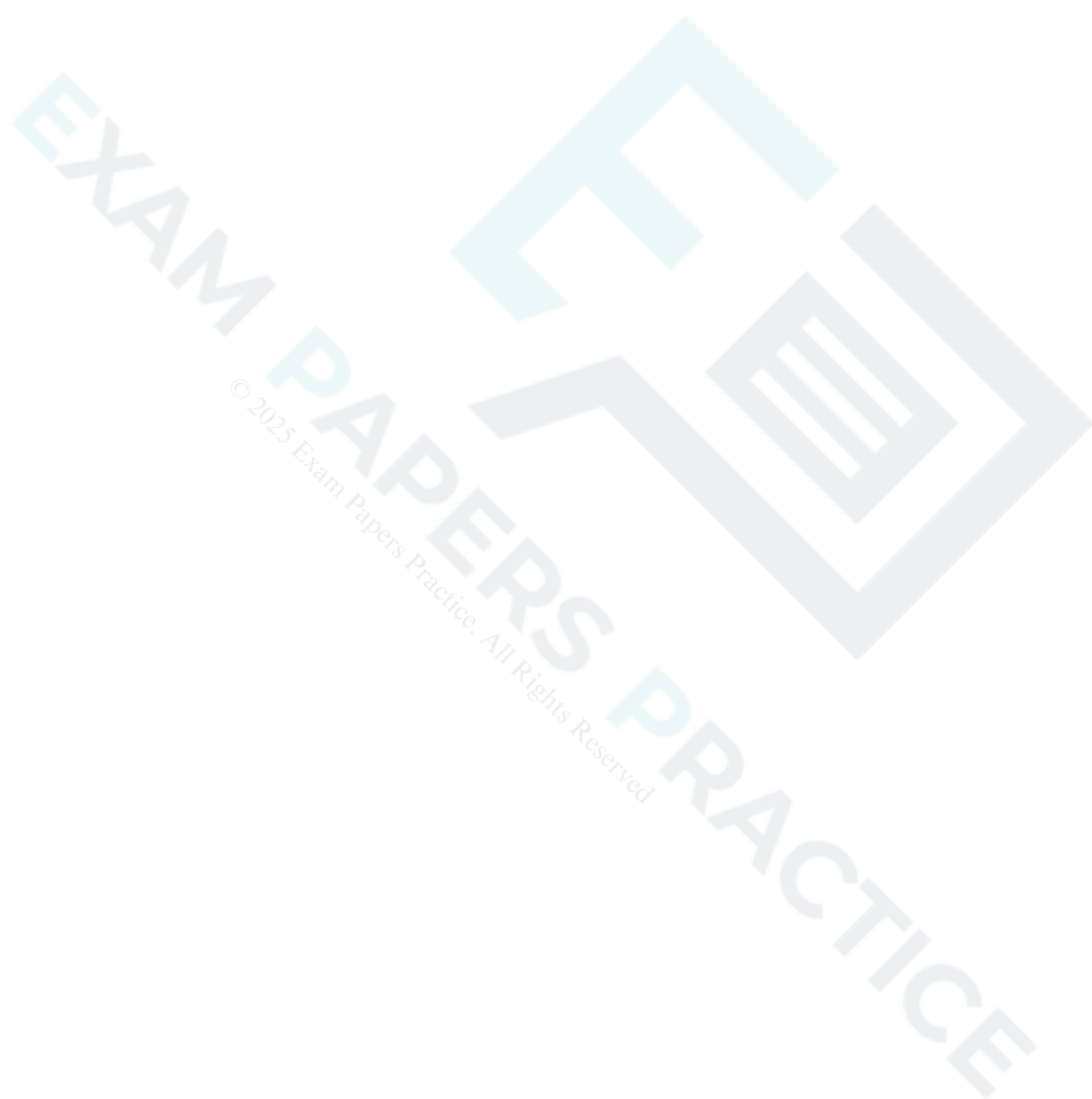
Fig. 11

Show that the x -coordinate of the stationary point on the curve may be found from the equation $2x - 3 + \ln x = 0$.

[4]

Find $\int \frac{32}{x^5} \ln x \, dx$.

[4]



24(a)

Fig. 4 shows the graph of $y = \sqrt{1+x^3}$.

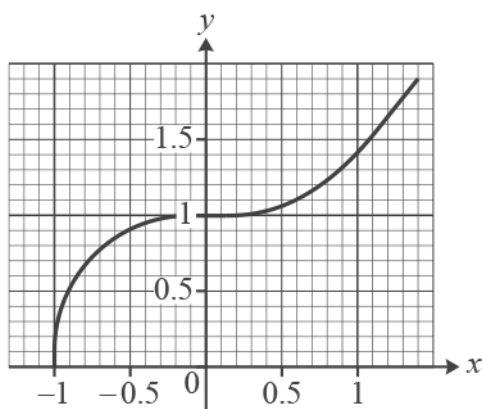


Fig. 4

Use the trapezium rule with $h = 0.5$ to find an estimate of $\int_{-1}^0 \sqrt{1+x^3} dx$, giving your answer correct to 6 decimal places. [3]

(b) State whether your answer to the above part is an under-estimate or an over-estimate, justifying your answer. [1]

25(a) Given that $y = (x^2 + 5)^{12}$,

Hence find $\int 48x (x^2 + 5)^{11} dx$.

[2]

(b)

Find $\frac{dy}{dx}$.

[2]

26(a) A model for the motion of a small object falling through a thick fluid can be expressed using the differential equation

$$\frac{dv}{dt} = 9.8 - kv,$$

where $v \text{ ms}^{-1}$ is the velocity after $t \text{ s}$ and k is a positive constant.

Given that $v = 0$ when $t = 0$, solve the differential equation to find v in terms of t and k .

[7]

(b) Sketch the graph of v against t .

[2]

- (c) Experiments show that for large values of t , the velocity tends to 7ms^{-1} .

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Find the value of k .

[2]

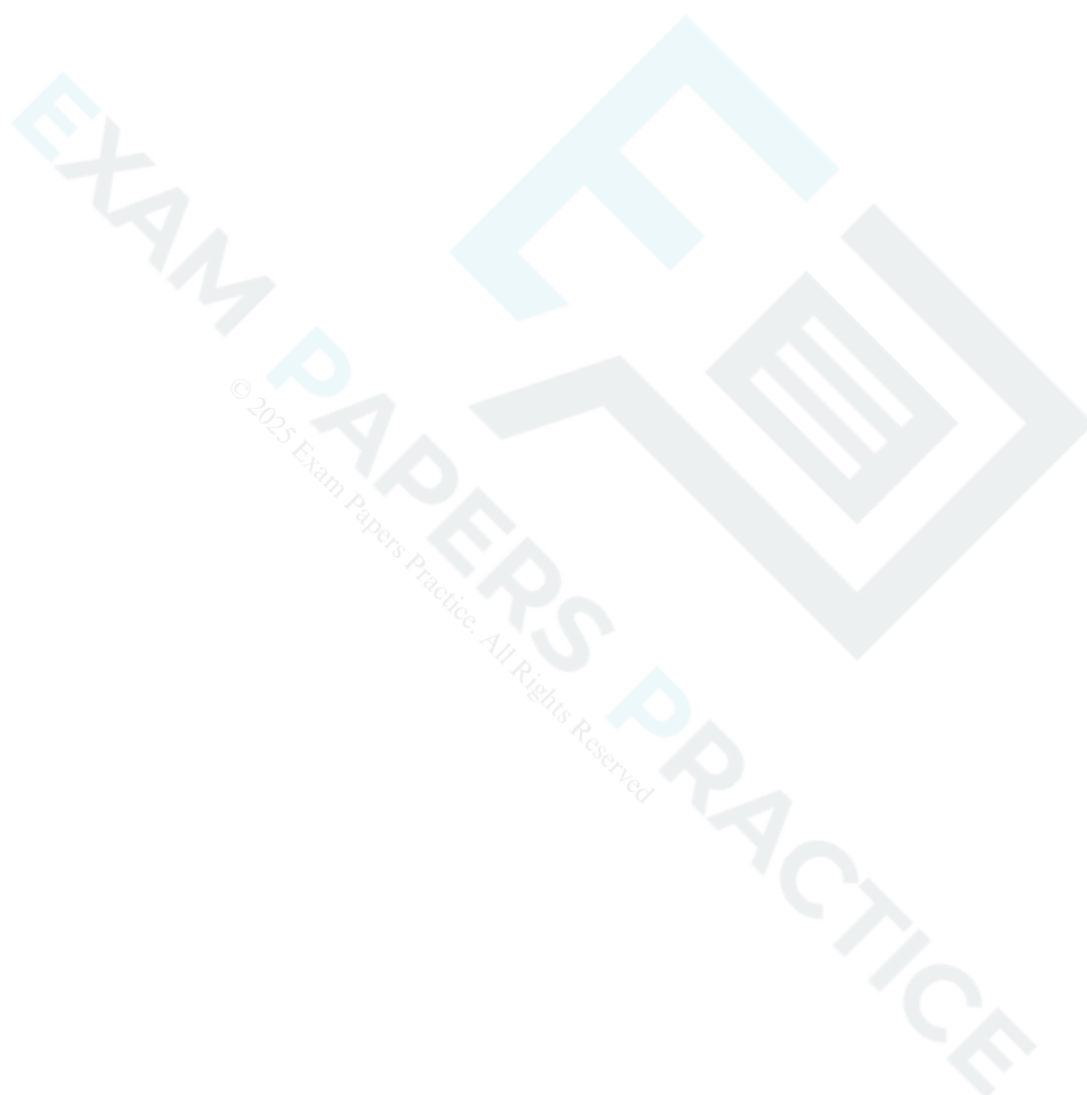
- (d) Find the value of t for which $v = 3.5$.

[1]

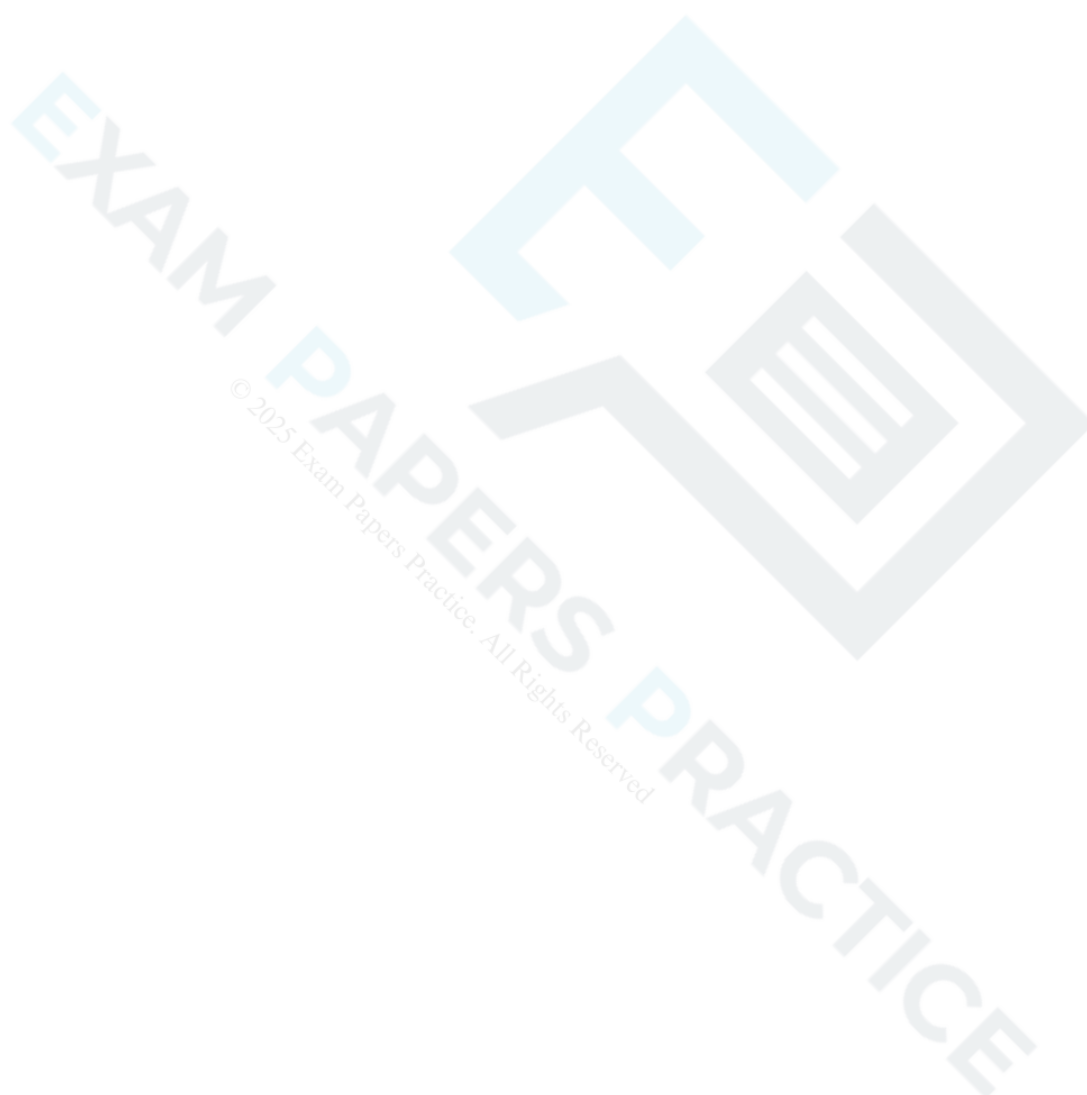
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Show that $\int_4^9 (2x + \sqrt{x}) dx = \frac{233}{3}$.

[3]



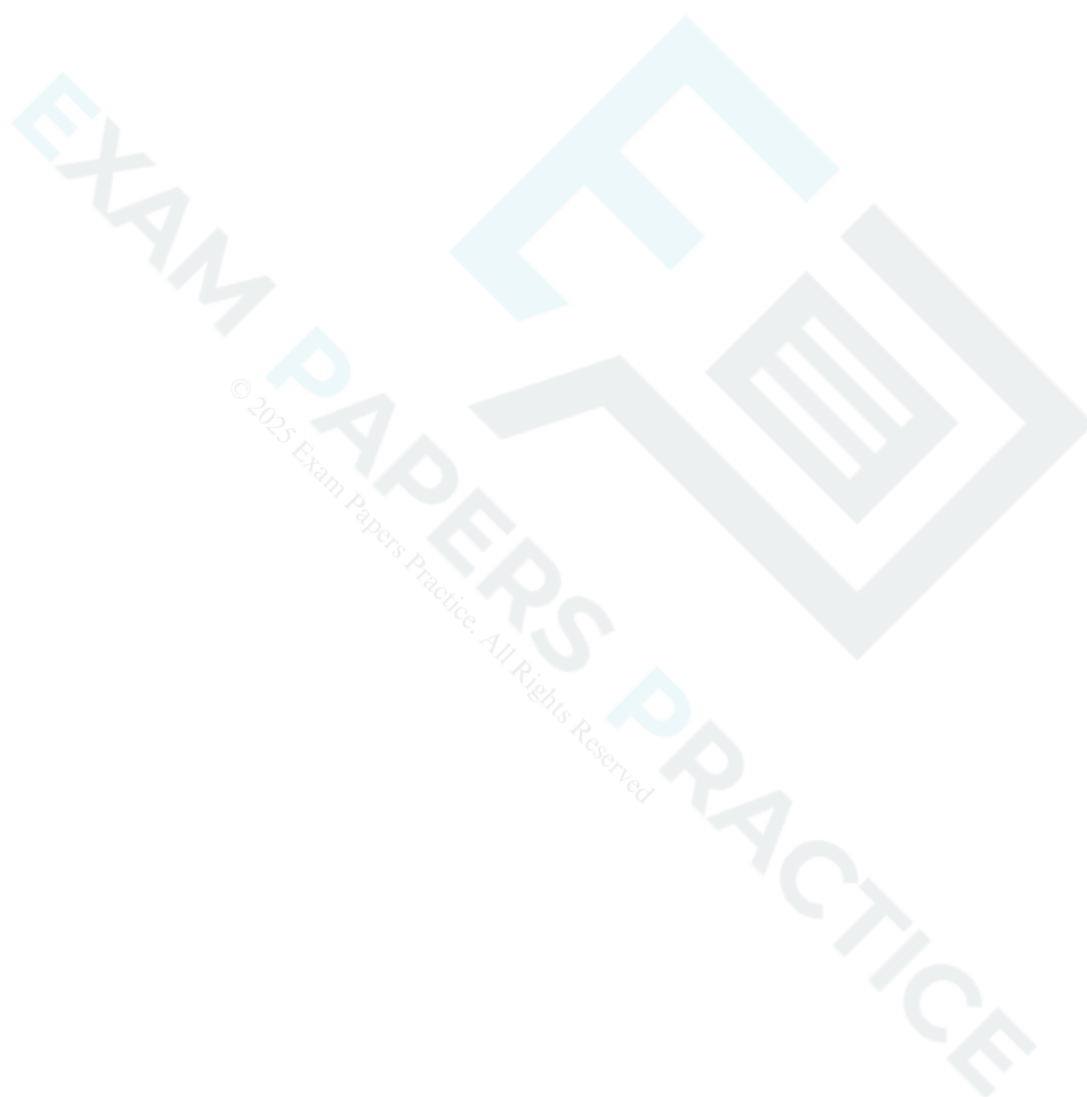
Show that $\int_0^8 x^3 \left(15x + \frac{11}{\sqrt[3]{x}} \right) dx = a \times 2^{11}$, where a is a positive integer to be determined.



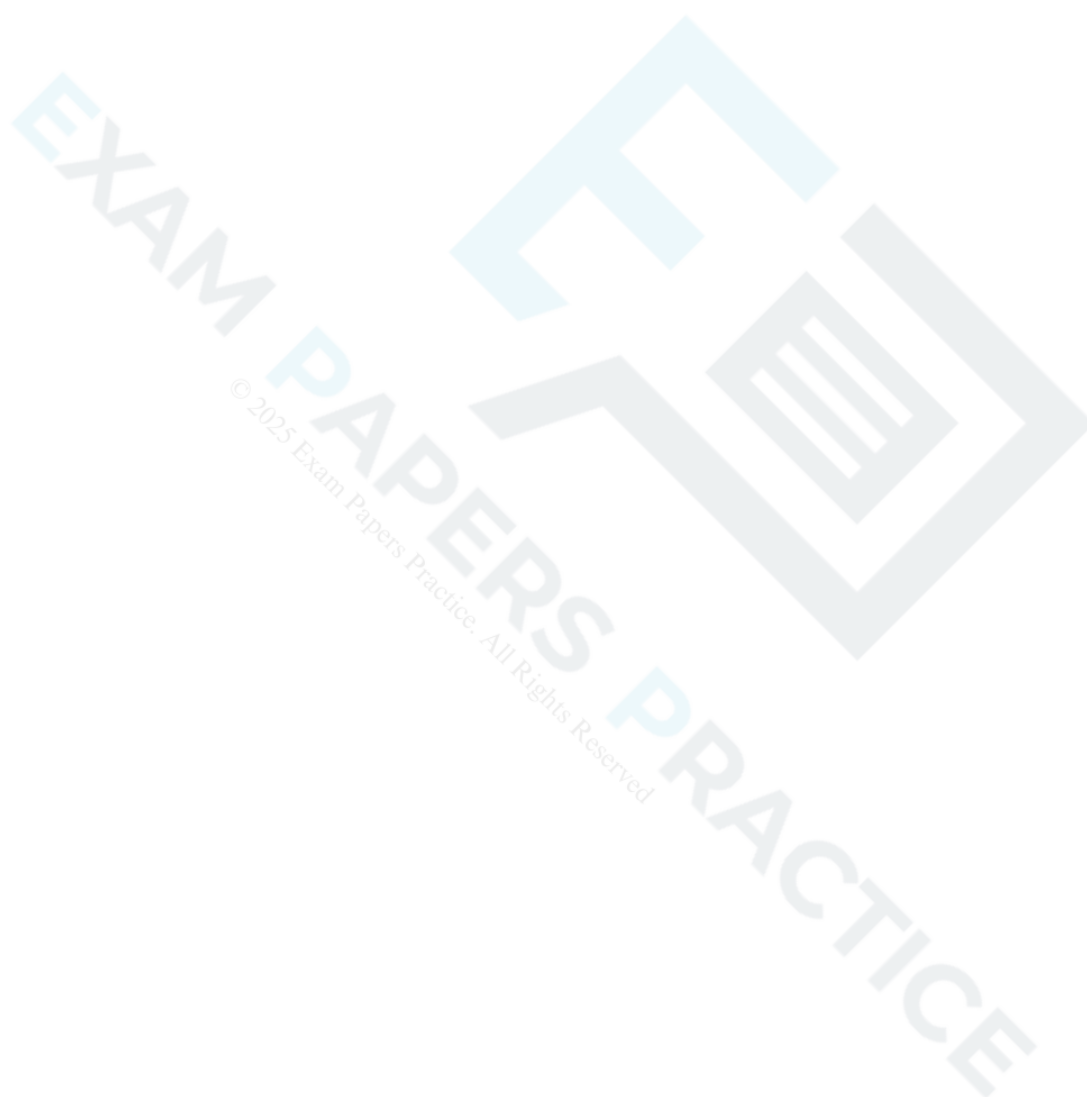
The equation of a curve is $y = \frac{x^2}{4} + \frac{2}{x} + 1$. A tangent and a normal to the curve are drawn at the point where $x = 2$.

Calculate the area bounded by the tangent, the normal and the x-axis.

[10]



Find $\int x^3 \left(15x + \frac{11}{\sqrt[3]{x}} \right) dx$.



31(a) In this question you must show detailed reasoning.

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Determine the set of values of x for which $24x - 3x^2 - x^3$ is decreasing.

[2]

(b) Sketch the gradient function for the curve $y = 24x - 3x^2 - x^3$.

[5]



32 In this question you must show detailed reasoning.

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Show that the only stationary point on the graph of $y = x^2 - 4\sqrt{x}$ is a minimum point at $(1, -3)$.

[7]

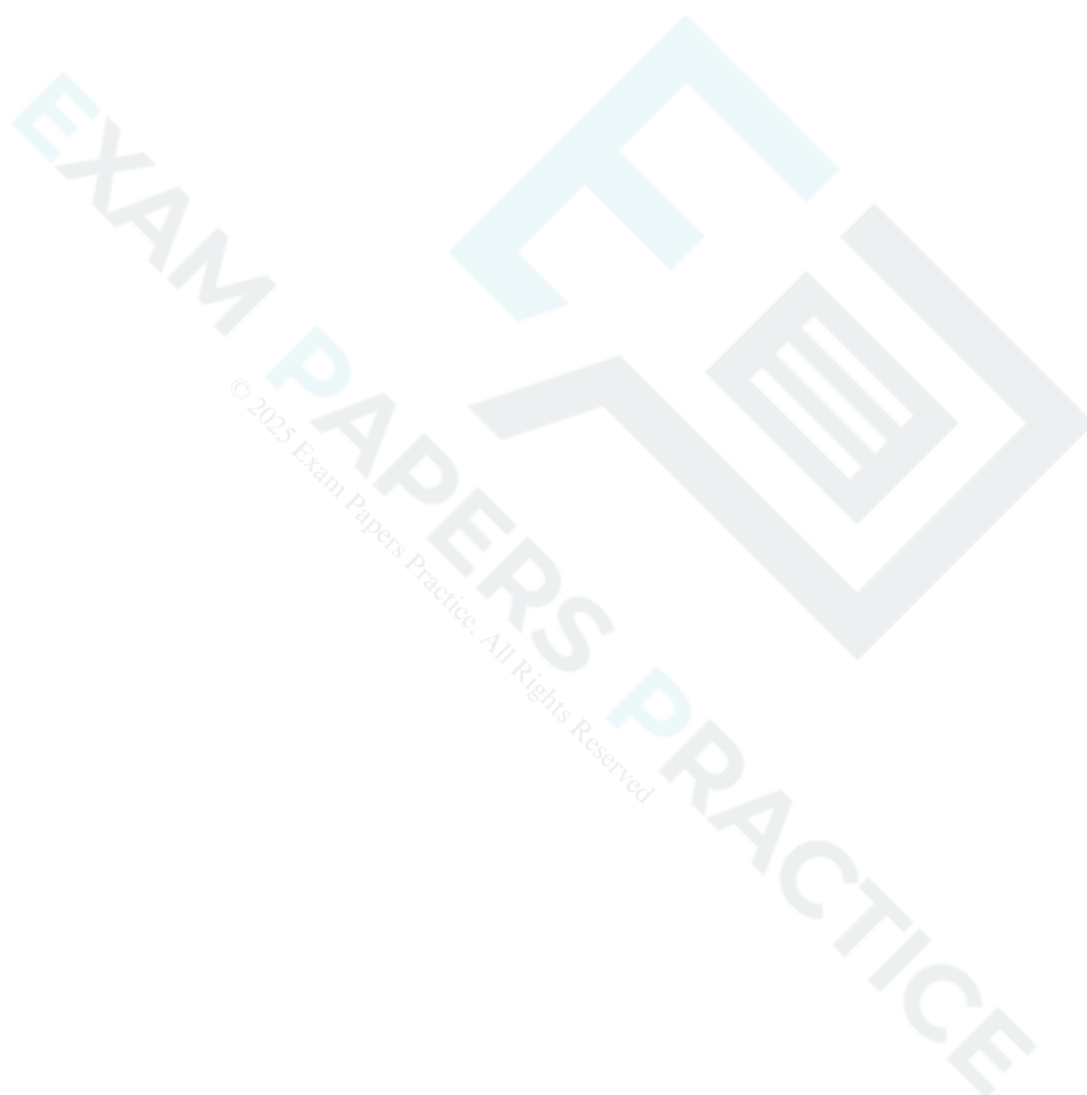


Fig. 9 shows the line $y = x$ and the curve $y = \frac{4}{3}x^3 + a$ where a is a constant. The line is a tangent to the curve, touching the curve in the first quadrant.

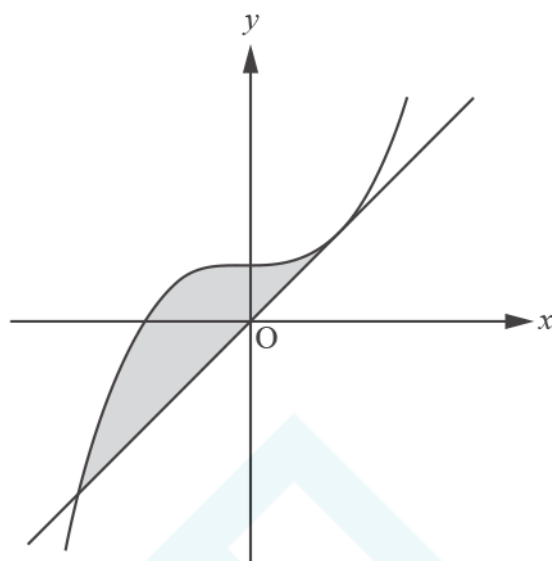


Fig.9

- (a) (i) Show that, at the point of contact between the line and the curve, $4x^2 = 1$.

[2]

- (ii) Hence find the value of a .

[4]

(b) Find the area of the shaded region between the curve and the line.

[7]

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The function $h(x) = \sin\left(\frac{1}{x}\right)$ is defined for the domain $x > \frac{2}{\pi}$.

(a) Differentiate $h(x)$ with respect to x .

[2]

(b) Find the values between which $\frac{1}{x}$ lies for $x > \frac{2}{\pi}$.

[2]

(c) Show that $h(x)$ is a decreasing function.

[3]

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35 In this question you must show detailed reasoning.

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- (a) Find the exact coordinates of the stationary point of the curve $y = x \ln x$.

[5]

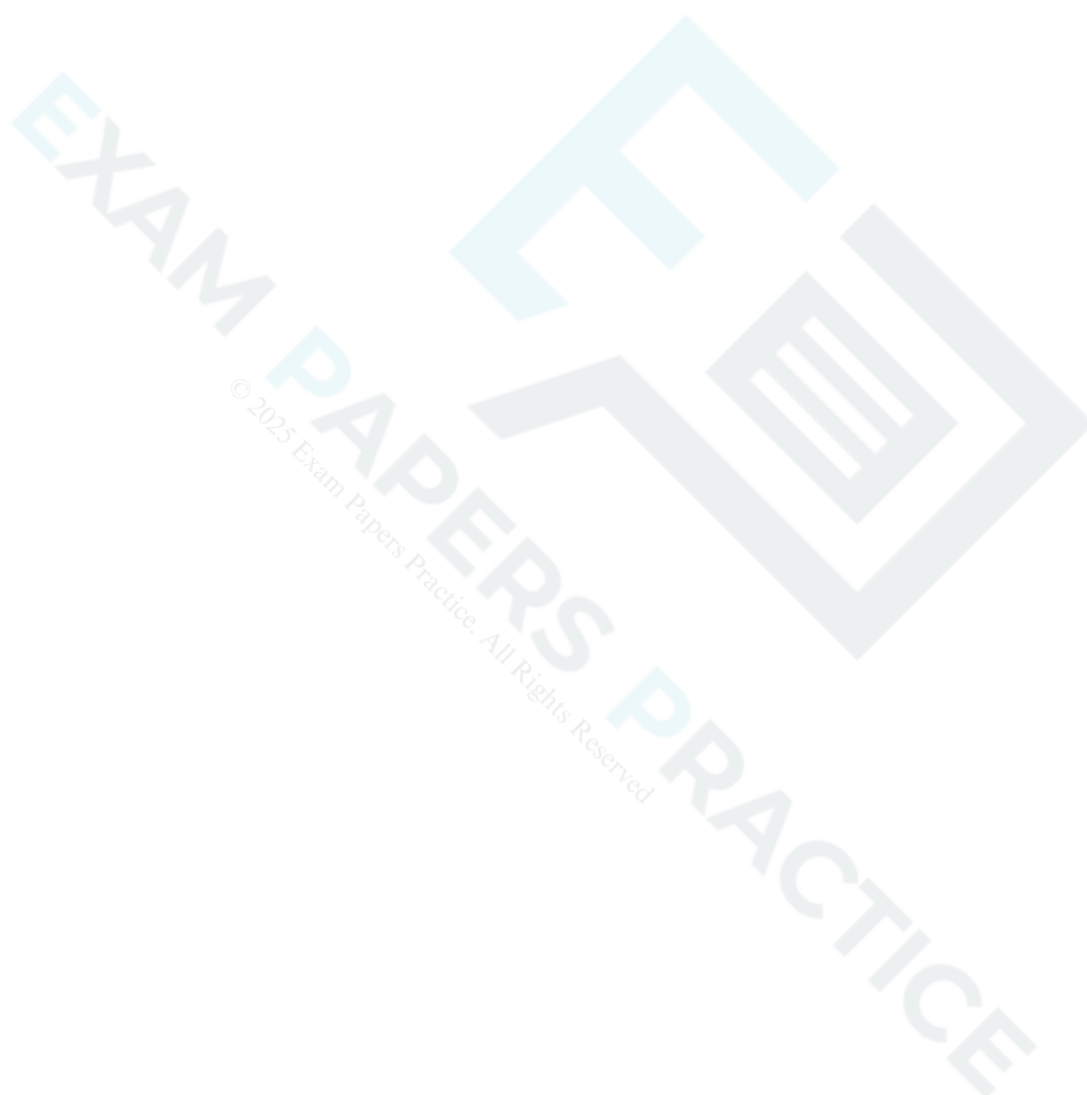
- (b) Show that the stationary point is a minimum turning point.

[2]



Using a suitable substitution, or otherwise, show that $\int_3^8 \frac{2x}{2x-1} dx = 5 + \frac{1}{2} \ln 3$.

[6]



37 In this question you must show detailed reasoning.

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The curve $y = \ln x$ passes through the point (a, b) , where $a > 1$.

The area A is bounded by the x -axis, the line $x = a$ and the curve $y = \ln x$.

The area B is bounded by the x -axis, the y -axis, the line $y = b$ and the curve $y = \ln x$.

The area A is equal in magnitude to the area B .

- (a) Show that a satisfies the equation $pa \ln a + qa + r = 0$, where p , q and r are constants to be determined.

[7]

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The value of a is found using the Newton-Raphson method on a spreadsheet. The output is shown in Fig. 15.

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r	x_r
0	4
1	5.177399
2	4.931531
3	4.921571
4	4.921554

Fig. 15

Heidi states that the value of a is 4.921554 correct to 6 decimal places.

(b) Determine whether she is correct.

[2]

38 In this question you must show detailed reasoning.

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The equation of a curve is $y = \frac{\ln(x+2)}{\cos x}$.

Find the equation of the tangent to the curve at the point where $x = 0$.

[5]





The function $f(x)$ for the domain $-\frac{1}{2}\pi \leq x \leq \frac{1}{2}\pi$ is defined by $f(x) = e^{kx} \sin 2x$, where k is a non-zero constant. Fig. 13 shows the graph of $y = f(x)$.

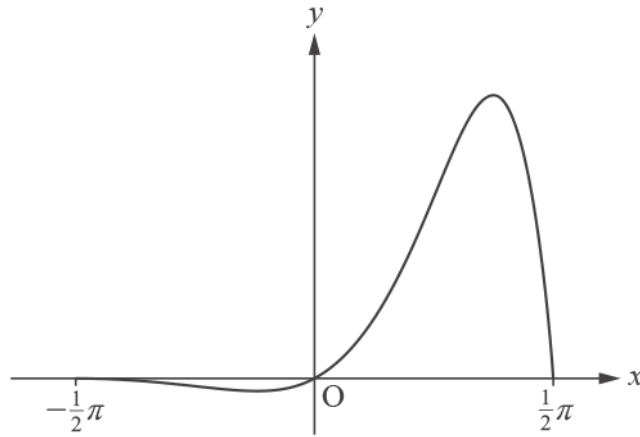


Fig. 13

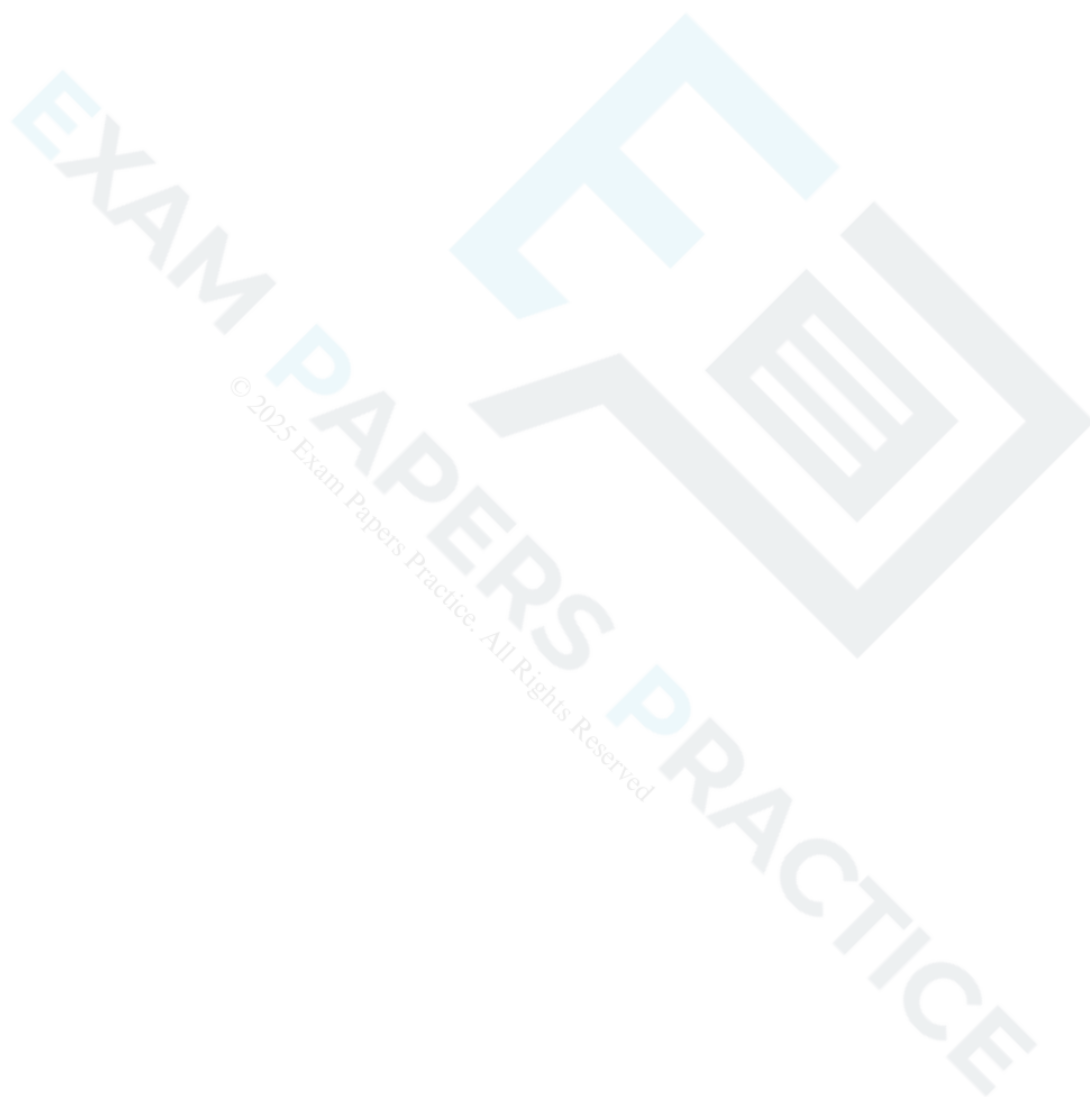
Given that the maximum value of $f(x)$ occurs when $x = \frac{3}{8}\pi$, find the exact minimum value of $f(x)$.

[8]



Solve the differential equation $5x \frac{dy}{dx} = y^2 - y - 6$ given that $y = 8$ when $x = 1$. Give your answer in the form $y = f(x)$.

[10]



41 In this question you must show detailed reasoning.

Fig. 15 shows the graph of $y = 5xe^{-2x}$. Find the exact value of the area of the shaded region between the curve, the x-axis and the line parallel to the y-axis through the maximum point on the curve.

[9]

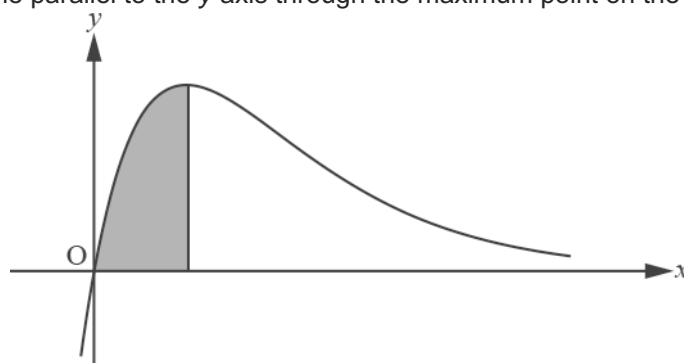


Fig. 15

42 Rebecca is looking for the root of the equation



$\sin 2x^2 - \cos 5x = 0$

that lies between 0.2 and 0.3.

She uses the standard small angle approximations for $\sin \theta$ and $\cos \theta$ to find an estimate.

(a) Show that Rebecca's method gives an estimate of 0.26 when rounded to 2 decimal places.

[3]

(b) Use a change of sign method to determine whether this value gives an estimate of the root which is correct to 2 decimal places.

[2]

Rebecca then uses the Newton-Raphson method to find another estimate for the root.

(c) Show that using $x_0 = 0.2$ gives the value $x_1 = 0.291\,989\,3$ correct to 7 decimal places.

[5]

(d) Continue this method, showing the result of each iteration, to find the root correct to 3 significant figures.

[3]

- 43 In a laboratory experiment, the motion of a small object moving in a straight line is being studied. A model for its velocity $v \text{ m s}^{-1}$ at time $t \text{ s}$ is given by

$$v = at^4 + bt^3,$$

where a and b are constants and $t > 0$.

The velocity has maximum value of 0.1 m s^{-1} when $t = 1$.

- (a) Determine the values of a and b .

[5]

(b) Find the time at which the particle changes direction.

[2]

(c) Explain why the model would not be suitable for very large values of t .

[1]

44 Fig. 10 shows the graph of $x^3 + y^3 = xy$.

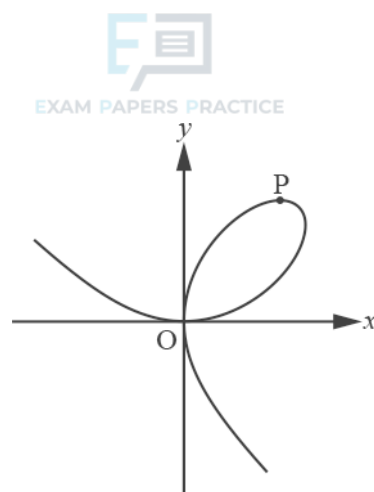


Fig. 10

- (a) Find an expression for $\frac{dy}{dx}$ in terms of x and y .

[4]

- (b) Explain why the expression cannot be used directly to find the gradient of the tangents to the curve at the origin.

[1]

- (c) P is the maximum point on the curve. The parabola $y = kx^2$ intersects the curve at P. Find the value of the constant k .

[2]

- 45 Sanjiv is investigating points on the curve $y = f(x)$. P is the point $(x, f(x))$ and Q is the point $(x + h, f(x + h))$. He has produced the spreadsheet shown in Fig. 2.

	A	B	C	D	E	F
1	x	f(x)	h	(x+h)	f(x+h)	
2	2	24	0.1	2.1	26.901	29.01
3	2	24	0.01	2.01	24.281001	28.1001
4	2	24	0.001	2.001	24.028010001	28.010001
5	2	24	0.0001	2.0001	24.002800100001	28.00100001
6						

Fig. 2

Column F shows the value of $\frac{f(x+h) - f(x)}{h}$ for different values of h .

- (a) Explain what the values in column F represent.

[1]

- (b) Explain what the spreadsheet suggests about the curve at the point P.

[1]

- 46 When a container is filled with water to a depth of y cm, the volume of water, V cm³, in the container is modelled by the formula

$$V = 20y + 3y^2 - 0.1y^3.$$

When the container is filled to its maximum possible depth of 10 cm, the volume of water in the container is 400 cm³.

Water is poured into the container after t seconds at a rate which is modelled by the differential equation

$$\frac{dV}{dt} = \frac{k}{\sqrt{t+1}},$$

where k is a positive constant. Initially the container is empty and after 3 seconds it is full.

- (a) Express V in terms of t .

[5]

(b) Calculate the time T taken, according to the model, until the container is filled to a depth of 5 cm. [2]

(c) Calculate the rate at which the depth of water in the container is increasing at this time T . [3]

A student states that, according to the model, the water will overflow.

(d) Determine whether the student's statement is correct. [2]

Find $\int \left(\frac{x}{1 + \sqrt{x}} \right) dx$. You may use the substitution $u = 1 + \sqrt{x}$.

(a)

[7]

(b) Hence show that $\int_0^1 \left(\frac{x}{1 + \sqrt{x}} \right) dx = A - \ln B$ where A and B are constants to be determined.

[2]

$$f(x) = x^4 - x, \quad x > 1.$$

The graphs of $y = f(x)$, $y = x$ and $y = f^{-1}(x)$ are shown in Fig. 9.

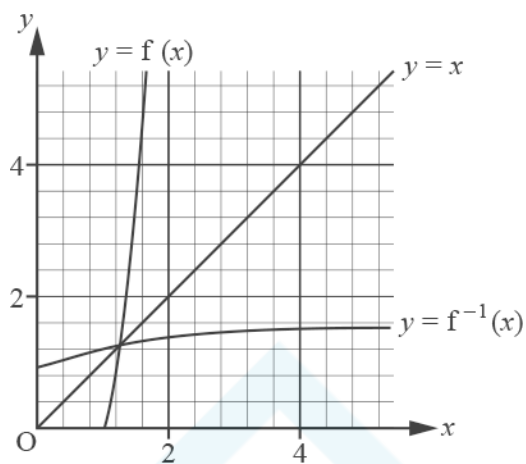


Fig. 9

- (a) Show that the inverse function, $f^{-1}(x)$, passes through the point (14, 2).

[1]

- (b) Find the gradient of $f^{-1}(x)$ at the point (14, 2).

[3]



- (a) (see insert for practice3 H640/03) The differential equation $\frac{dP}{dQ} = \frac{1}{k} \frac{P}{Q}$ is given on line 42. Find the general solution, giving Q as a function of P . [3]

- (b) (see insert for practice3 H640/03) Hence show that, when the PED is constant, a 5% increase in price results in the demand changing by a percentage which is independent of the original price, as stated in lines 43–44. [3]

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50 (see insert for practice3 H640/03) This question is about a straight line demand curve with equation $P = mQ + c$, where $m < 0$ and $c > 0$.

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(a) Find an expression for the revenue in terms of Q , m and c .

[1]

(b) Hence show that, in this case, the maximum revenue occurs where the PED is -1 , as stated on line 35.

[5]

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$$x = \tan \theta, y = 1 + \cos 2\theta, \text{ for } 0 \leq \theta \leq \pi \text{ with } \theta \neq \frac{1}{2}\pi.$$

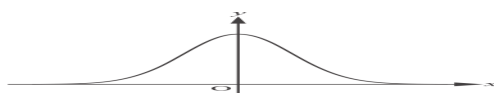


Fig. 9

- (a) Explain why $\theta \neq \frac{1}{2}\pi$ [1]

- (b) Show that the maximum value of y for points on the curve is 2. [1]

- (c) Show that the cartesian equation of the curve is $y = \frac{2}{1+x^2}$. [3]

- (d) In this question you must show detailed reasoning.

The point P in the first quadrant lies on the curve. Find the coordinates of P given that OP is the normal to the curve at P. [7]



Use the substitution $u = x + 1$ to find $\int (5x + 2)\sqrt{x + 1} \, dx$. Give your answer in the form $kx(x + 1)^p + c$ where k , p and c are constants.

[7]



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Fig. 5.1 shows the curve $y = e^{1-x^2}$. Fig. 5.2 shows a spreadsheet used to calculate an estimate of $\int_0^2 e^{1-x^2} dx$ using the trapezium rule with four strips.

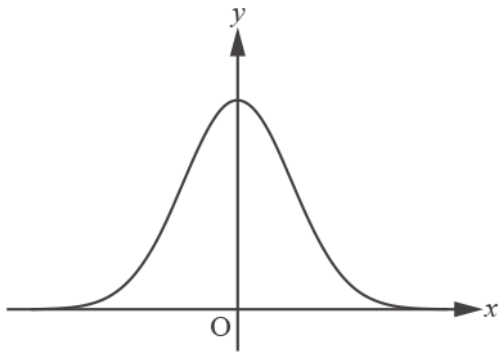


Fig. 5.1

	A	B	C
1	x		y
2	0	1	2.718282
3	0.5	0.75	2.117
4	1	0	1
5	1.5	-1.25	0.286505
6	2	-3	0.049787
7			

Fig. 5.2

(a) Show how the value in cell B3 is calculated.

[1]

(b) Complete the calculation to estimate $\int_0^2 e^{1-x^2} dx$, giving the answer correct to 3 significant figures.

[2]

(c) Show that the only stationary point on the curve is at $(0, e)$.

[2]

- (a) Show that $x - 3$ is a factor of $4x^3 - 12x^2 - x + 3$.

[1]

- (b) Fig. 4 shows the curve $y = 4x^3 - 12x^2 - x + 3$. Find the coordinates of the points where it crosses the x -axis.

[4]

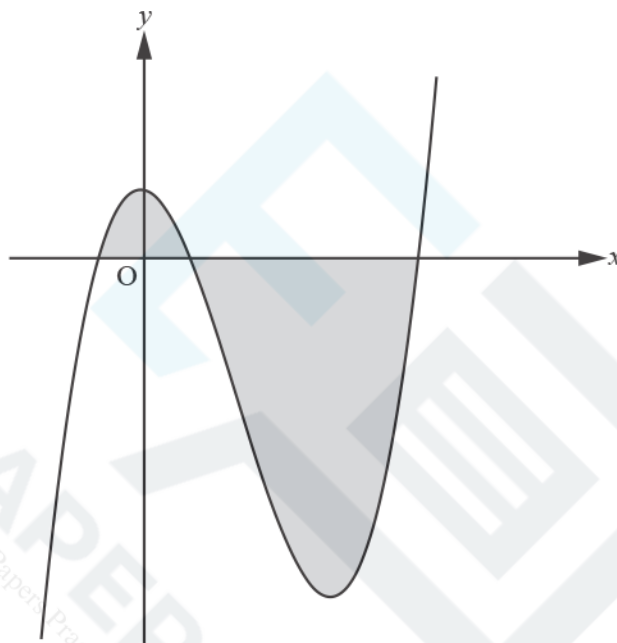


Fig. 4

- (c) The two regions bounded by the curve $y = 4x^3 - 12x^2 - x + 3$ and the x -axis are shaded in Fig. 4. Determine the total area of the shaded regions.

[5]

- 55 The intensity of the sun's radiation, y watts per square metre, and the average distance from the sun, x astronomical units, are shown in Fig. 11 for the planets Mercury and Jupiter.

	x	y
Mercury	0.3075	14400
Jupiter	4.950	55.8

Fig. 11

The intensity y is proportional to a power of the distance x .

- (a) Write down an equation for y in terms of x and two constants.

[1]

(b) Show that the equation can be written in the form $\ln y = a + b \ln x$.

[2]

(c) Complete the table for $\ln x$ and $\ln y$ correct to 4 significant figures.

[2]

	$\ln x$	$\ln y$
Mercury		
Jupiter		

(d) Use the values from part (c) to find a and b .



EXAM PAPERS PRACTICE

[3]

(e) Hence rewrite your equation from part (a) for y in terms of x , using suitable numerical values for the constants.

[2]

(f) Sketch a graph of the equation found in part (e).

[2]

EXAM PAPERS PRACTICE

(g) Earth is 1 astronomical unit from the sun. Find the intensity of the sun's radiation for Earth.

[1]



The curve $y = (x - 1)^2$ maps onto the curve C_1 following a stretch scale factor $\frac{1}{2}$ in the x -direction.

- (a) Show that the equation of C_1 can be written as $y = 4x^2 - 4x + 1$.

[2]

The curve C_2 is a translation of $y = 4.25x - x^2$ by $\begin{pmatrix} 0 \\ -3 \end{pmatrix}$.

- (b) Show that the normal to the curve C_1 at the point $(0, 1)$ is a tangent to the curve C_2 .

[7]

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57 In this question you must show detailed reasoning.

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Fig. 8 shows the graph of a quadratic function. The graph crosses the axes at the points $(-1, 0)$, $(0, -4)$ and $(2, 0)$.

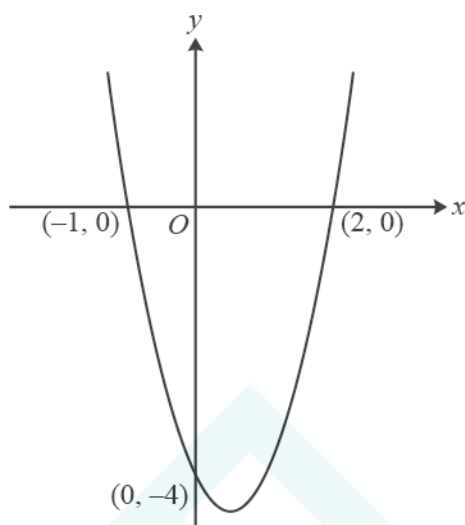


Fig. 8

Find the area of the finite region bounded by the curve and the x-axis.

[8]

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58 (a) Sketch the graphs of $y = 4 \cos x$ and $y = 2 \sin x$ for $0^\circ \leq x \leq 180^\circ$ on the same axes.

[2]

(b) Find the exact coordinates of the point of intersection of these graphs, giving your answer in the form $(\arctan a, k\sqrt{b})$, where a and b are integers and k is rational.

[4]

(c) A student argues that without the condition $0^\circ \leq x \leq 180^\circ$ all the points of intersection of the graphs would occur at intervals of 360° because both $\sin x$ and $\cos x$ are periodic functions with this period. Comment on the validity of the student's argument.

[1]



A curve has parametric equations $x = \frac{t}{1+t^3}$, $y = \frac{t^2}{1+t^3}$ where $t \neq -1$.

(a) In this question you must show detailed reasoning.

Determine the gradient of the curve at the point where $t = 1$.

[5]



(b) Verify that the cartesian equation of the curve is $x^3 + y^3 = xy$.

[3]

60 In this question you must show detailed reasoning.

A curve has equation $y = x - 5 + \frac{1}{x-2}$. The curve is shown in Fig. 4.

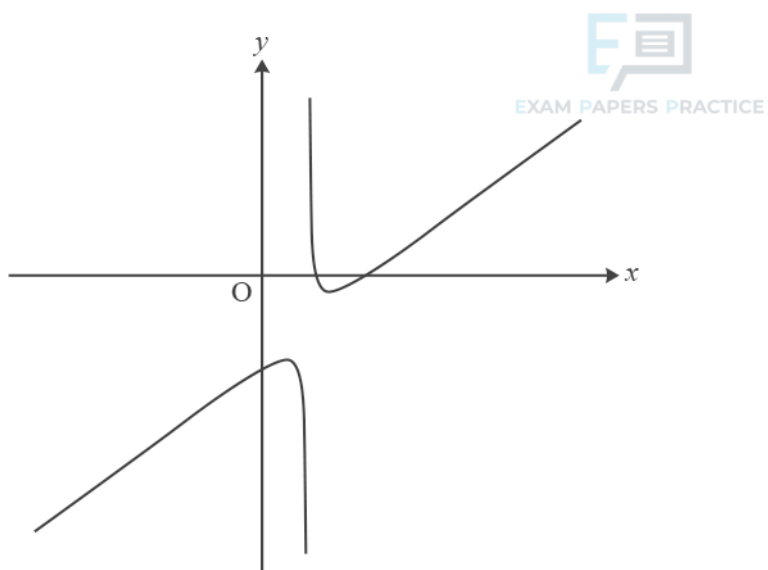


Fig. 4

- (a) Determine the coordinates of the stationary points on the curve.

[5]

- (b) Determine the nature of each stationary point.

[3]

(c) Write down the equation of the vertical asymptote.

[1]

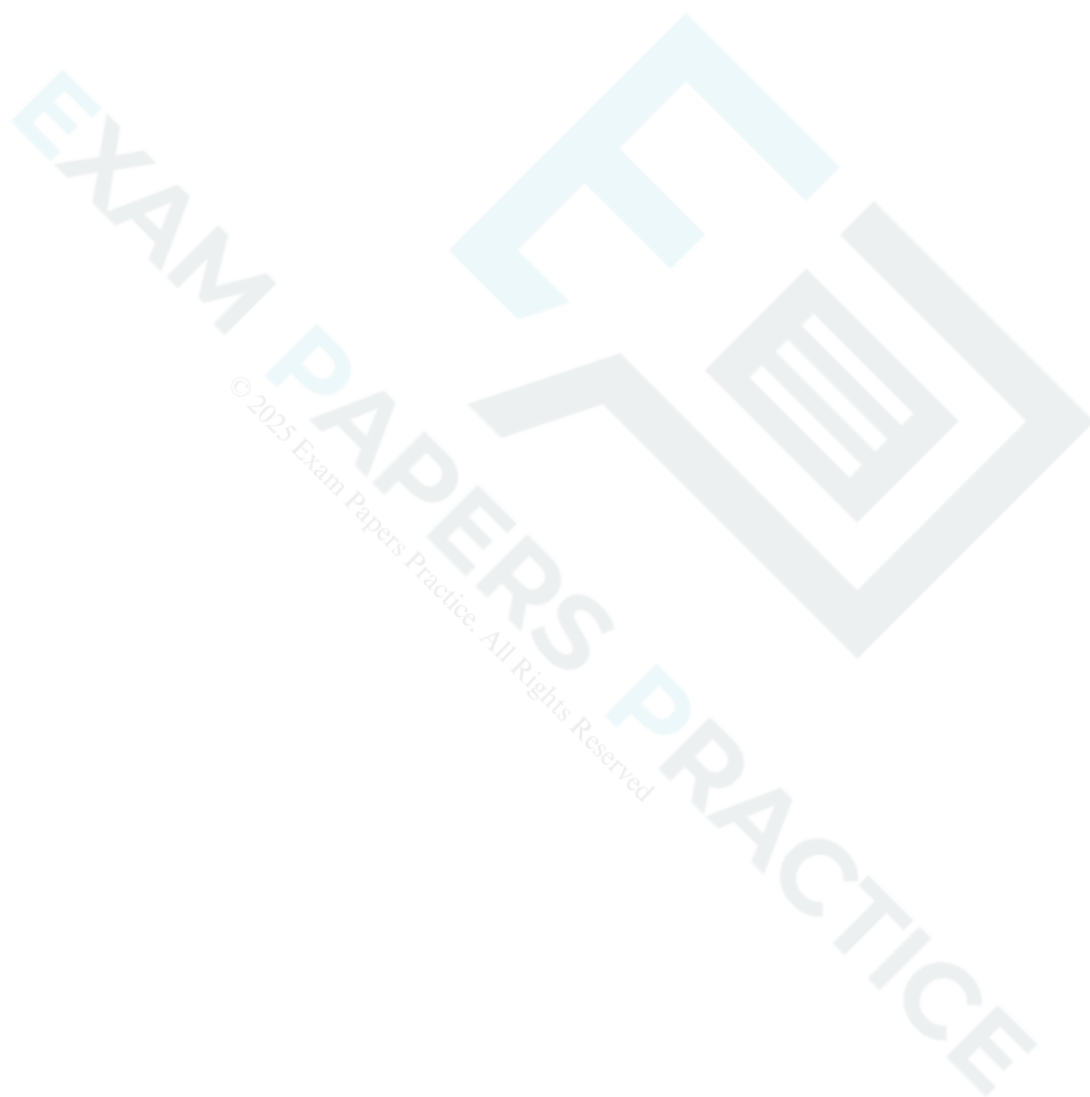
(d) Deduce the set of values of x for which the curve is concave upwards.

[1]



The curve $y = x^3 - 2x$ is translated by the vector $\begin{pmatrix} 1 \\ -4 \end{pmatrix}$. Write down the equation of the translated curve.

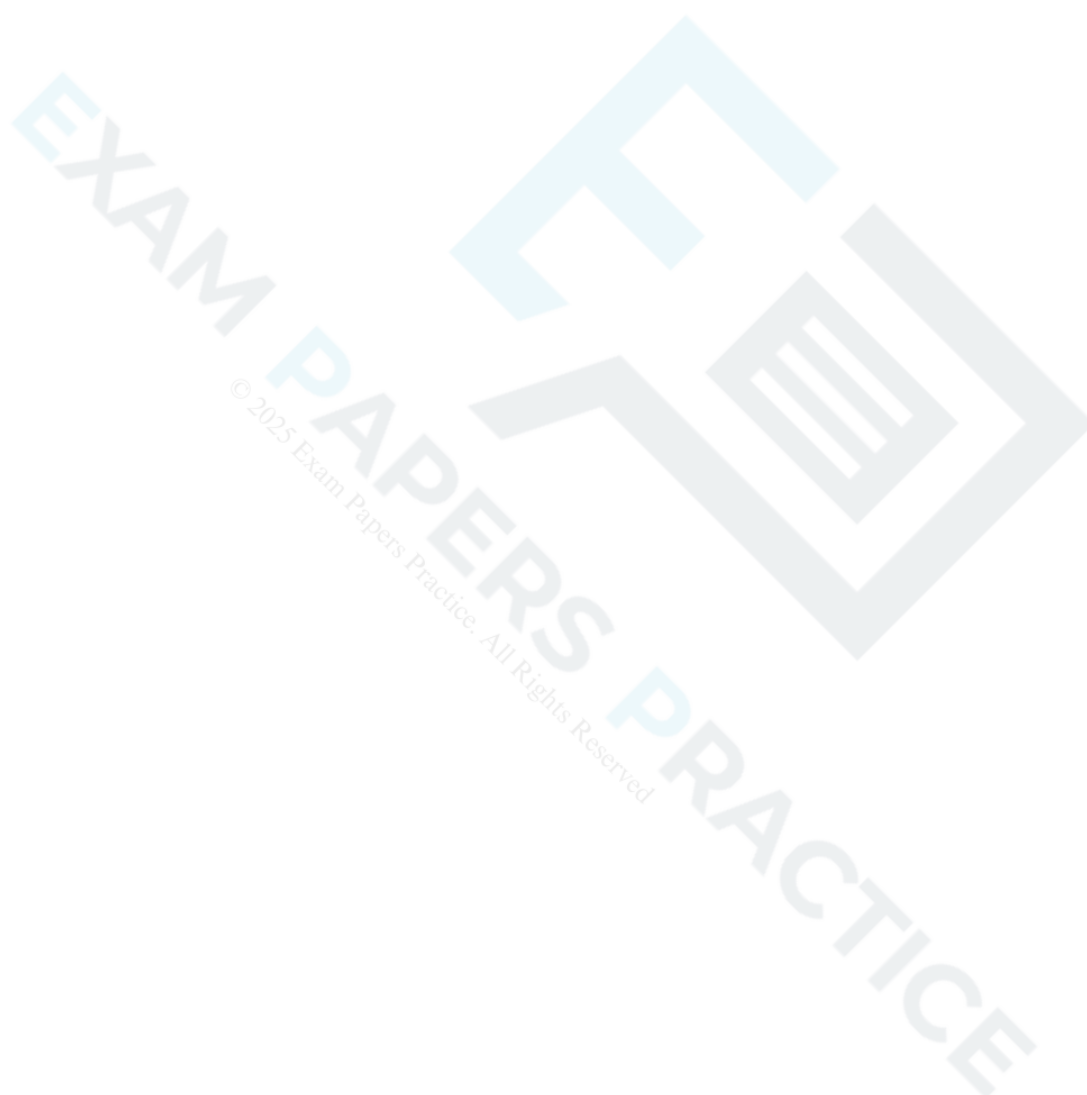
[2]





- (a) Express $\frac{(x^2 - 8x + 9)}{(x+1)(x-2)^2}$ in partial fractions.

[5]



(b) Express y in terms of x given that



EXAM PAPERS PRACTICE

$$\frac{dy}{dx} = \frac{y(x^2 - 8x + 9)}{(x+1)(x-2)^2} \text{ and } y = 16 \text{ when } x = 3.$$

[7]



63 You must show detailed reasoning in this question.

The equation of a curve is

$$y^3 - xy + 4\sqrt{x} = 4.$$

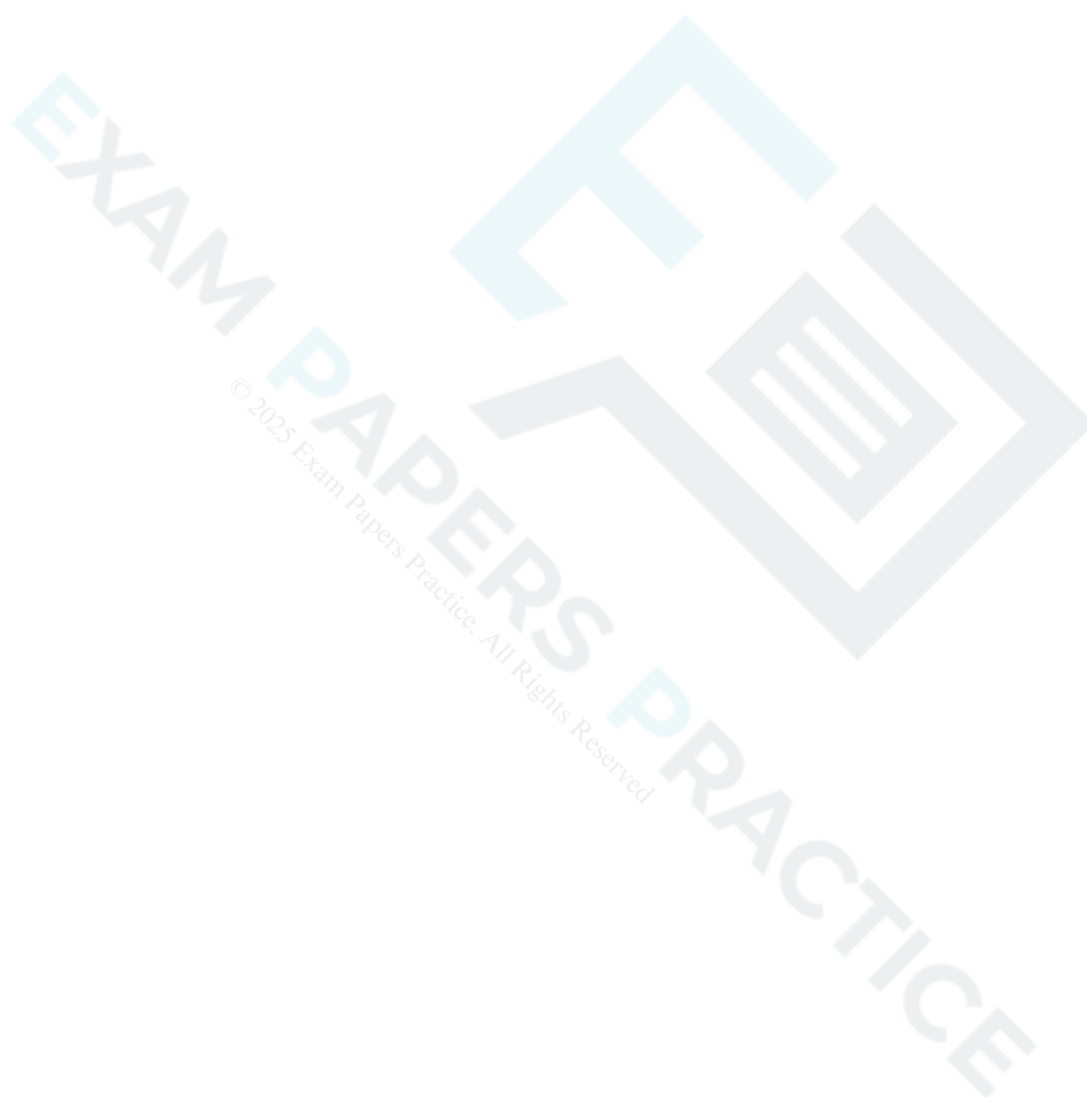
Find the gradient of the curve at each of the points where $y = 1$.

[9]



Find $\int \left(4\sqrt{x} - \frac{6}{x^3} \right) dx$.

[4]



65 (a) (i) Sketch the graph of $y = 3^x$.

[1]

(ii) Give the coordinates of any intercepts.

[1]

The curve $y = f(x)$ is the reflection of the curve $y = 3^x$ in the line $y = x$.

(b) Find $f(x)$.

[1]



The function $f(x)$ is defined by $f(x) = \sqrt[3]{27-8x^3}$. Jenny uses her scientific calculator to create a table of values for $f(x)$ and $f'(x)$.

x	$f(x)$	$f'(x)$
0	3	0
0.25	2.9954	-0.056
0.5	2.9625	-0.228
0.75	2.8694	-0.547
1	2.6684	-1.124
1.25	2.2490	-1.977
1.5	0	ERROR

- (a) Use calculus to find an expression for $f'(x)$ and hence explain why the calculator gives an error for $f'(1.5)$.

[3]

- (b) Find the first three terms of the binomial expansion of $f(x)$.

[3]

- (c) Jenny integrates the first three terms of the binomial expansion of $f(x)$ to estimate the value of $\int_0^1 \sqrt[3]{27-8x^3} \, dx$. Explain why Jenny's method is valid in this case. (You do not need to evaluate Jenny's approximation.)

[2]

- (d) Use the trapezium rule with 4 strips to obtain an estimate for $\int_0^1 \sqrt[3]{27-8x^3} \, dx$.

[3]

The calculator gives 2.921 174 38 for $\int_0^1 \sqrt[3]{27-8x^3} dx$. The graph of $y = f(x)$ is shown in Fig. 13.

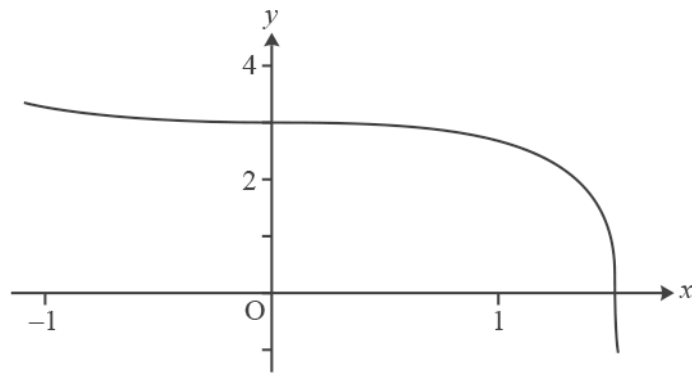


Fig. 13

(e) Explain why the trapezium rule gives an underestimate.

[1]

67 Fig. 10 shows the graph of $y = (k - x)\ln x$ where k is a constant ($k > 1$).

EXAM PAPERS PRACTICE

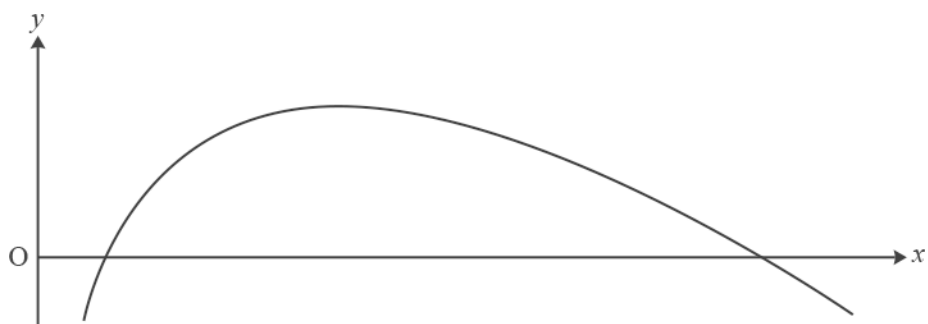


Fig. 10

Find, in terms of k , the area of the finite region between the curve and the x -axis.

[8]

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68 (a) Show that $8 \sin^2 x \cos^2 x$ can be written as $1 - \cos 4x$.

[3]

EXAM PAPERS PRACTICE

(b) Hence find $\int \sin^2 x \cos^2 x \, dx$.

[3]

Show that $\int_0^9 (3 + 4\sqrt{x}) \, dx = 99$.

[4]



70 (See Insert for Practice2 64003.) When solving the equation $x^3 + bx^2 + cx + d = 0$ by depressing the cubic and using Vieta's substitution, it can be shown that the resulting quadratic in y^3 must have real roots if $3c > b^2$.

(a) Show that $y = x^3 + bx^2 + cx + d$ has no stationary points if $3c > b^2$.

[3]

(b) Deduce that $x^3 + bx^2 + cx + d = 0$ has exactly one real root if $3c > b^2$.

[1]

Fig. 6 shows the curves $y = \sin x$ and $y = \cos x$ for $-\pi \leq x \leq \pi$.

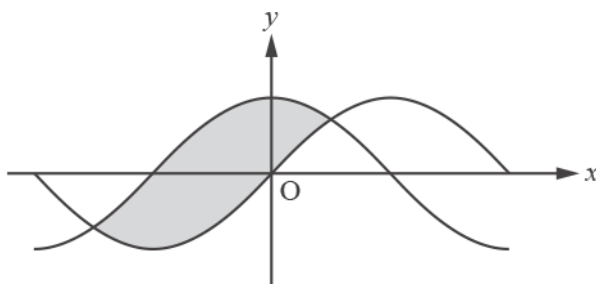
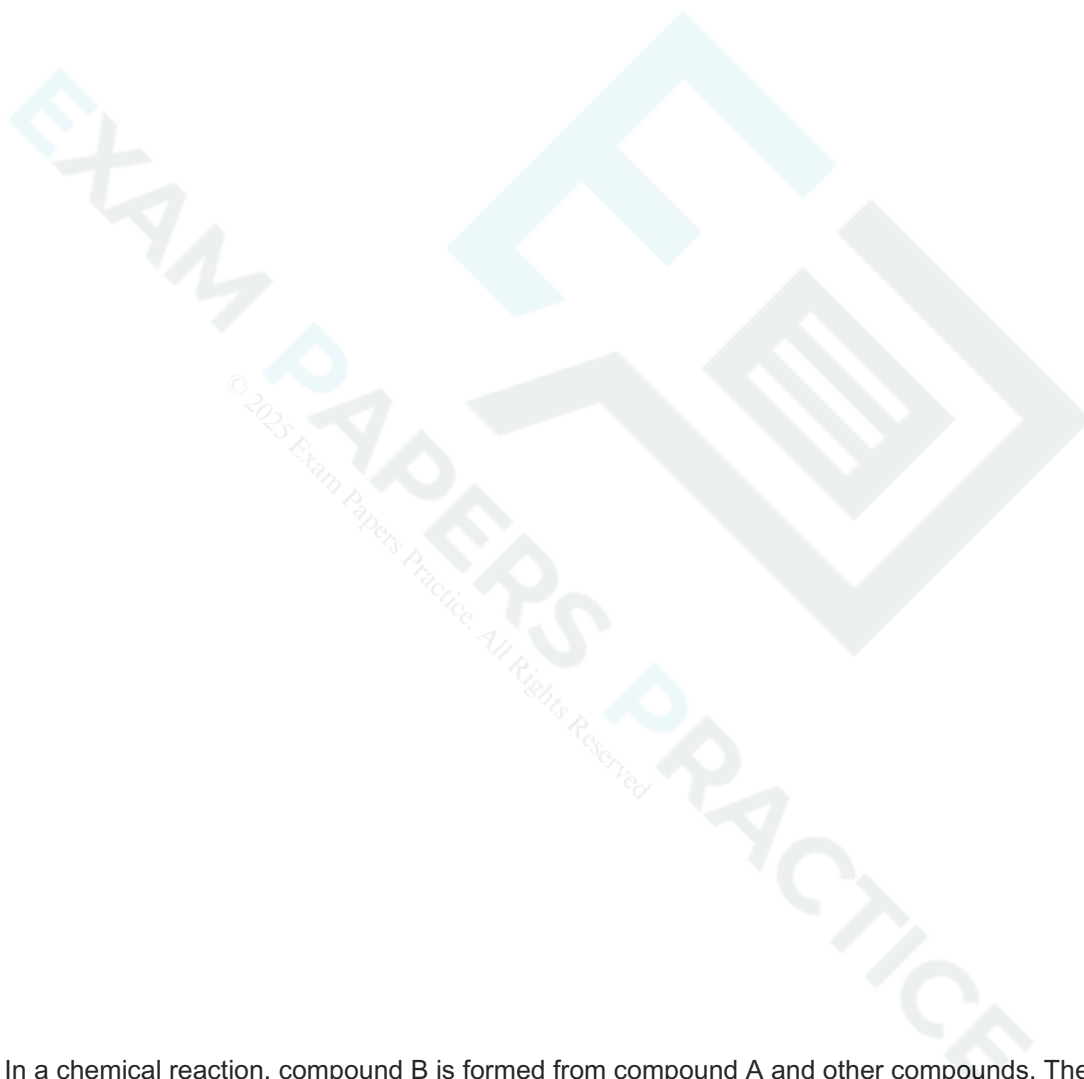


Fig. 6

- (a) Show that the area of the region enclosed between the curves is given by $\int_a^b (\cos x - \sin x) dx$, where the values of a and b are to be determined. [4]

(b) Find the exact value of the area of this region.

[3]



- 72 In a chemical reaction, compound B is formed from compound A and other compounds. The mass of B at time t minutes is x kg. The total mass of A and B is always 1 kg. Sadiq formulates a simple model for the reaction in which the rate at which the mass of B increases is proportional to the product of the masses of A and B.

Show that the model can be written as

(a)

[1]

Initially, the mass of B is 0.2 kg.

(b) Solve the differential equation, expressing x in terms of k and t .

[7]

After 15 minutes, the mass of B is measured to be 0.9 kg.

(c) Find the value of k , correct to 3 significant figures.

[2]

EXAM PAPERS PRACTICE

(d) Find the mass of B after 30 minutes.

[1]

(e) Explain what the model predicts for the mass of A remaining for large values of t .

[1]

73 The function $f(x)$ is defined by $f(x) = x^4 - 3x^3 + 3x^2$ for $x \in \mathbb{R}$.

EXAM PAPERS PRACTICE

(a) Show that the only stationary point on the curve $y = f(x)$ is a minimum point at the origin.

[7]



(b) Explain why $f(x)$ does not have an inverse function.

[1]

The function $g(x)$ is defined by $g(x) = x^4 - 3x^3 + 3x^2$ for $0 \leq x \leq 2$.

(c) State the domain and range of the inverse function $g^{-1}(x)$.

[2]

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74 Using the substitution $x = e^u$, find $\int (\ln x)^2 dx$.



EXAM PAPERS PRACTICE

[6]

75

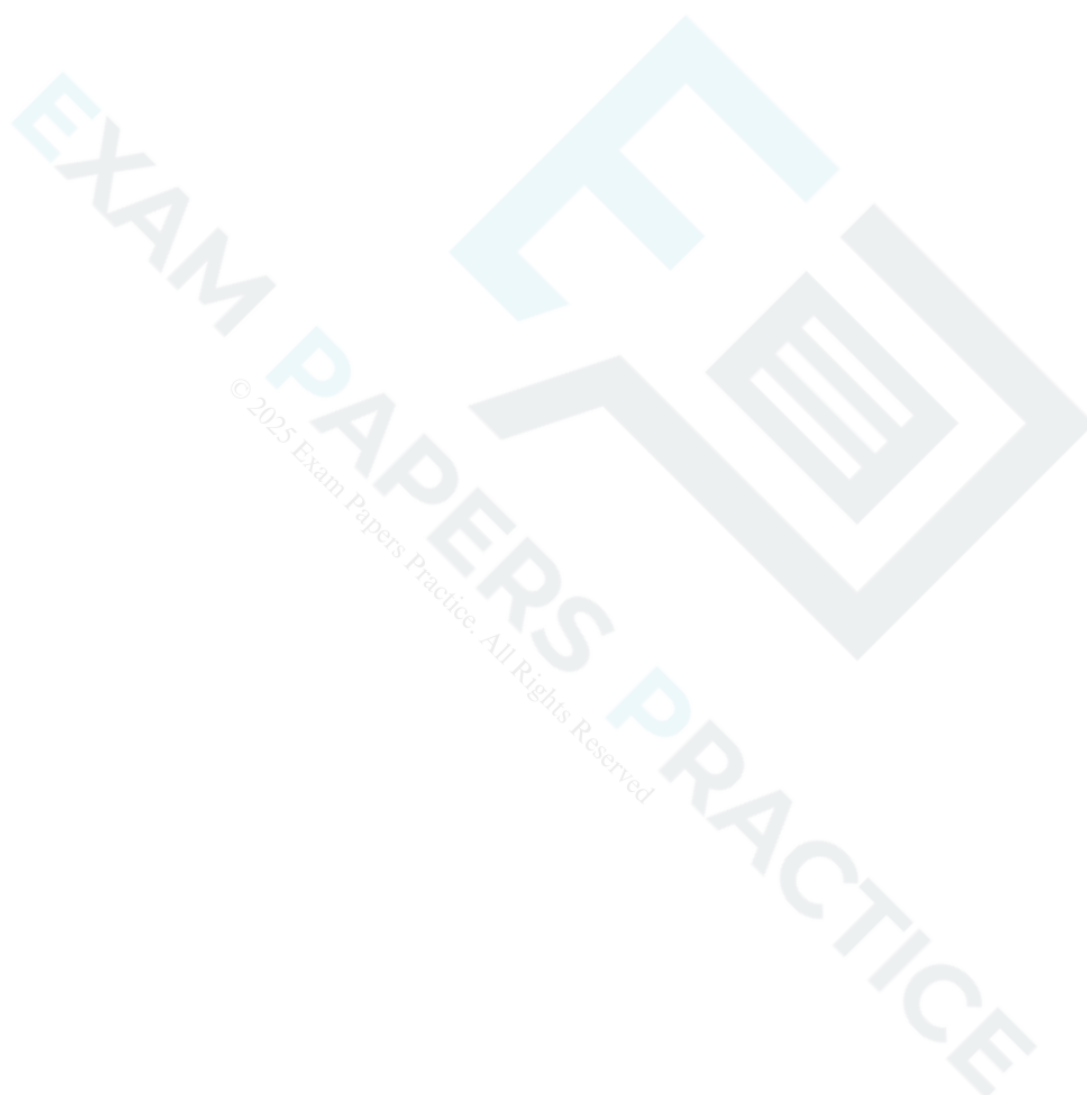
Find $\frac{dy}{dx}$ given that $y = 3x^2 \sin 2x$.

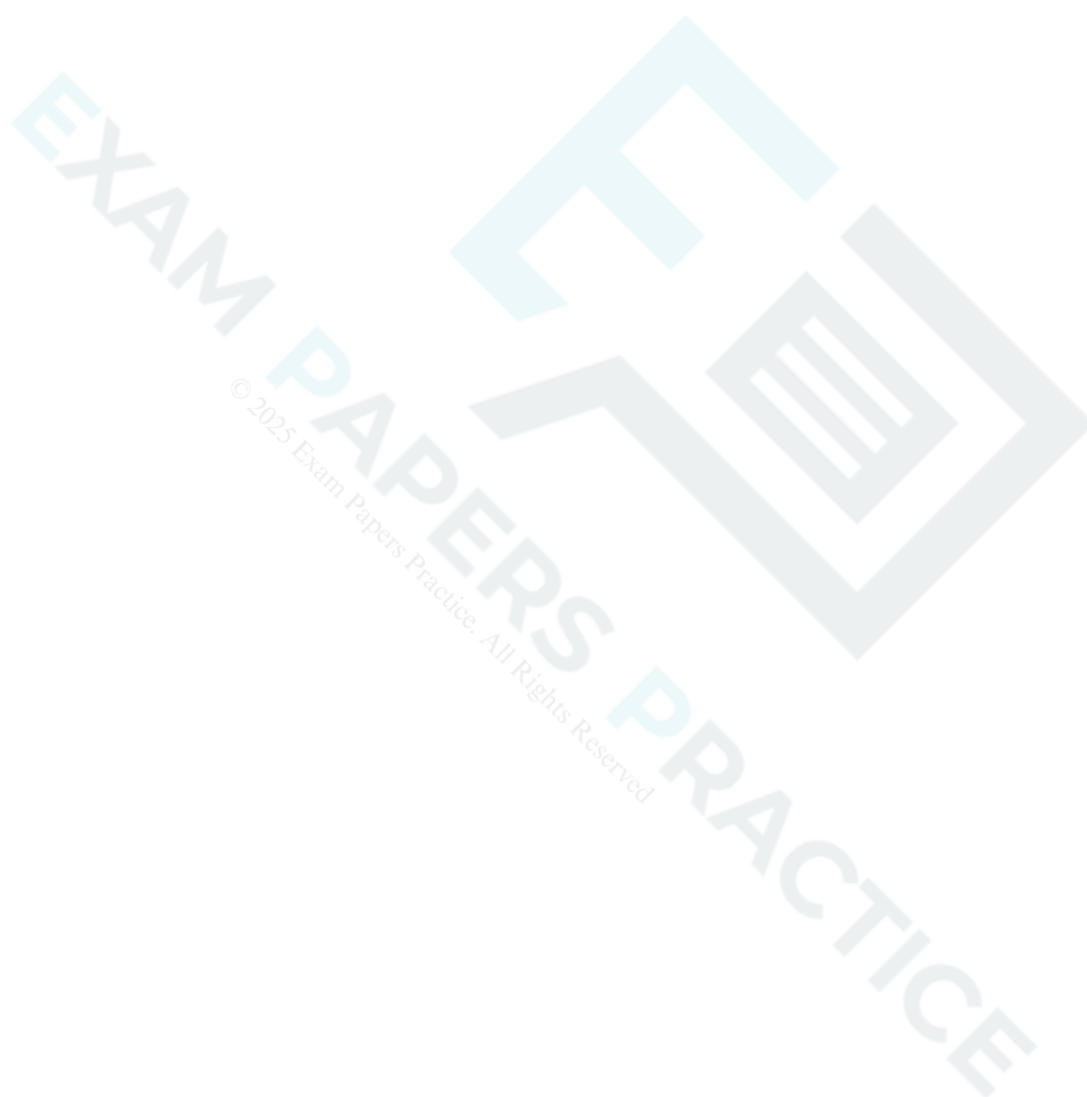
[3]

76 In this question you must show detailed reasoning.

EXAM PAPERS PRACTICE

The equation of a curve is $y = \frac{2x}{\ln x}$. Show that the only stationary point on this curve is a minimum point at $(e, 2e)$. [9]





- 78 Initially a car is travelling along a stretch of motorway at a steady speed of 22ms^{-1} . The driver changes lane and accelerates in order to overtake a lorry. The speed of the car, $V\text{ms}^{-1}$, is modelled by the formula

$$V = A + B(1 - e^{-0.2t}),$$

where A and B are constants to be determined and t is the time in seconds after the car has started to accelerate.

2 seconds after the car starts to accelerate it is travelling at 25.3ms^{-1} .

- (a) Find the values of A and B , giving the value of B correct to 2 significant figures.

[3]



5 seconds after the car starts to accelerate it is travelling at 28.4ms^{-1} .

(b) Determine whether the model is a good fit for the data.

[2]

(c) According to the model, what is the acceleration of the car at $t = 5$?

[3]

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On this stretch of motorway the speed limit is 110 km h^{-1} . The level of penalty given to a driver caught speeding above 110 km h^{-1} depends on the percentage the speed is above the speed limit. The driver continues to accelerate until reaching the maximum speed predicted by the model.

(d) Determine whether the driver will exceed the speed limit by 10%.

[4]



79 The number of bacteria in a population at time t is denoted by P . The rate of increase of P is proportional to the square root of P .

EXAM PAPERS PRACTICE

(i) Write down a differential equation relating P , the time t , and a constant of proportionality k .

[1]

(ii) Verify that $P = (A + Bt)^2$, where A and B are constants, satisfies the differential equation, and find k in terms of B .

[3]

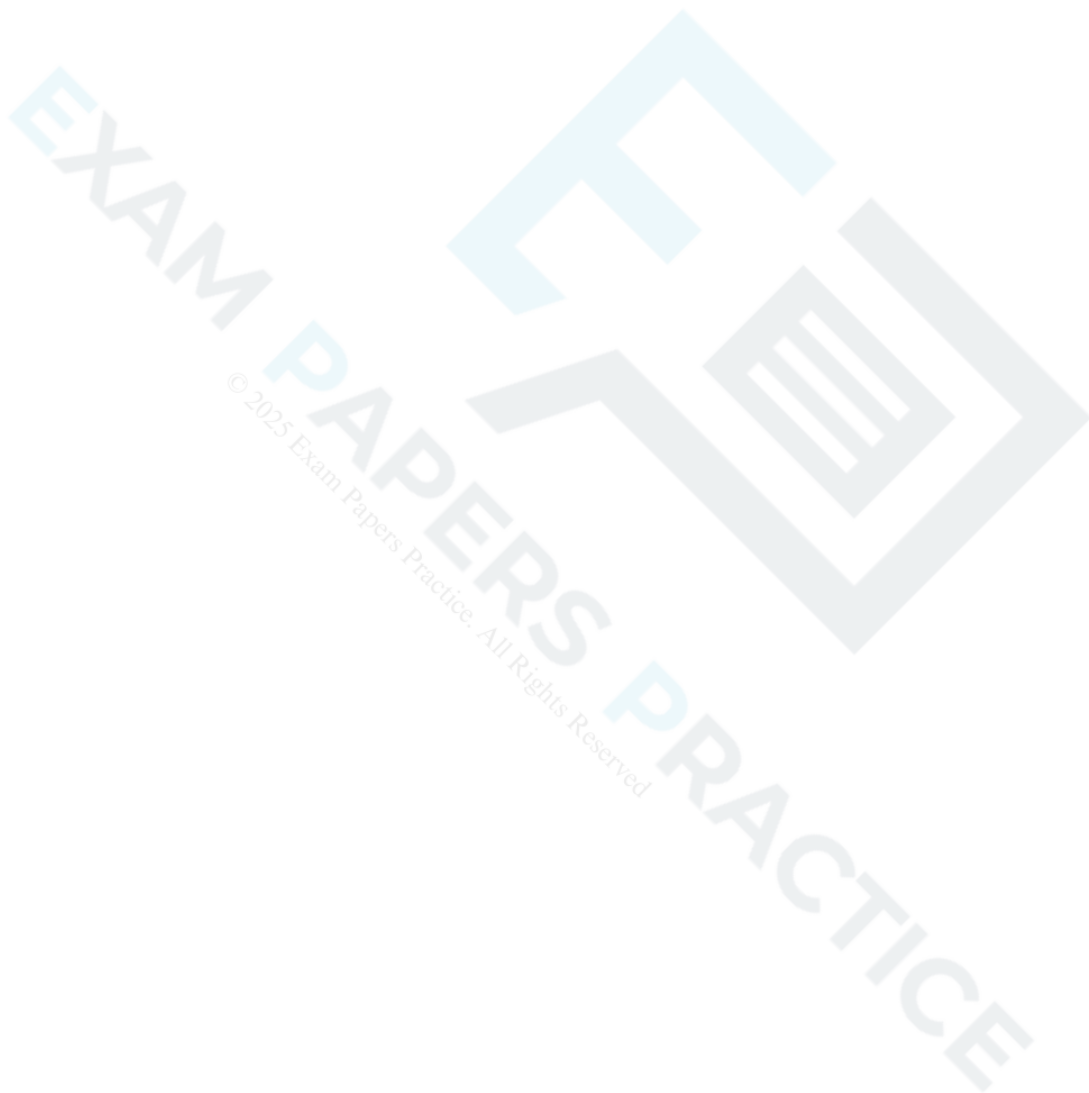
EXAM PAPERS PRACTICE
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- (i) Show that $f(x)$ is an odd function, and interpret this result in terms of the graph of the curve $y = f(x)$. [3]



- (ii) Find the coordinates of the stationary points of the curve. Give answers correct to 2 decimal places where appropriate.

[7]



(iii) Sketch the curve for $-2 \leq x \leq 2$.

[2]

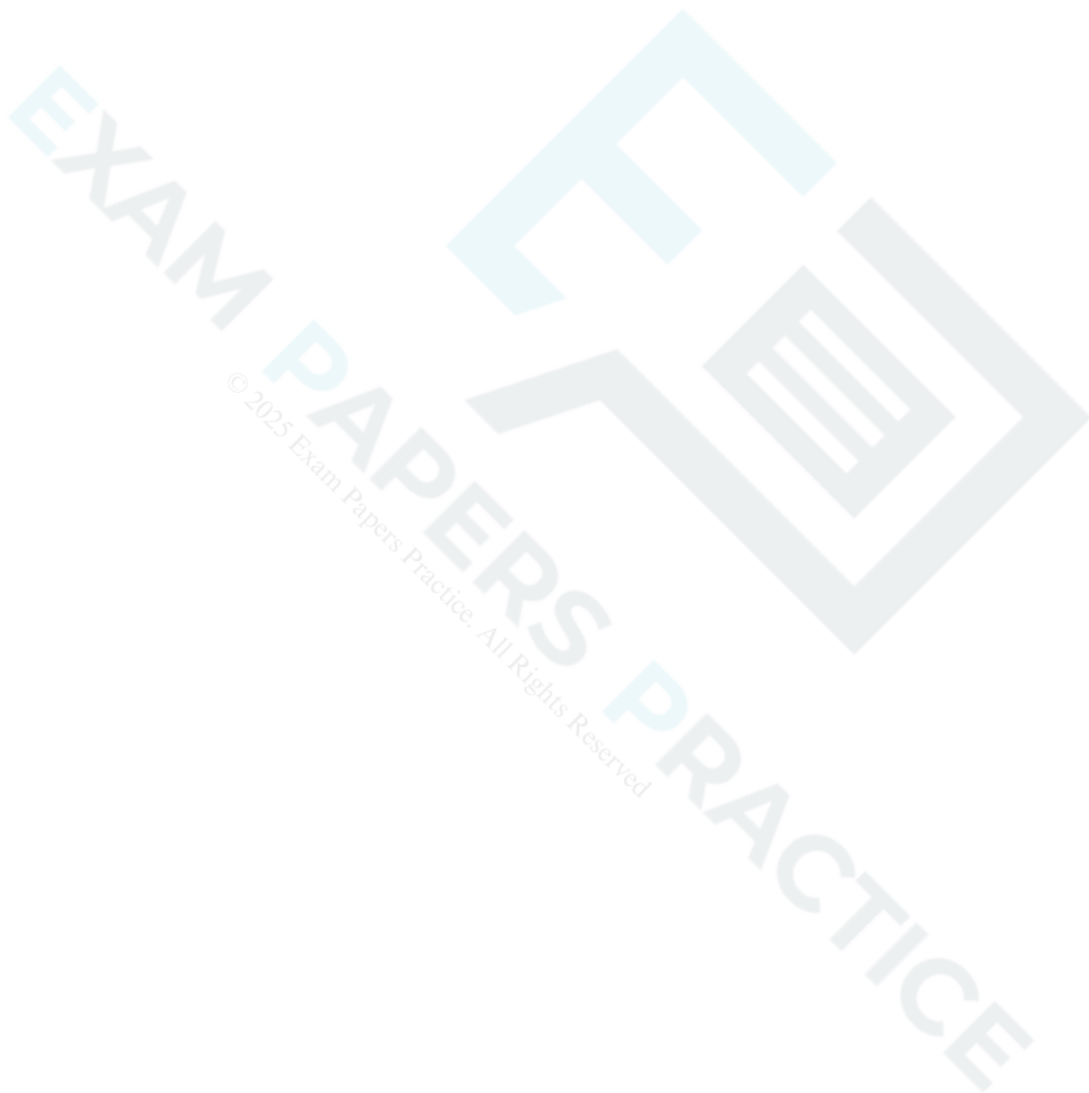
(iv) (A) Show, using the substitution $t = x^2$, that $\int f(x)dx$ may be expressed as $\int kt e^{-t} dt$, where k is a constant to be determined.

[2]



(B) Hence find the exact area of the region enclosed by the curve $y = f(x)$, the positive x -axis and the line $x = 2$.

[4]



- 81 Fig. 8 shows part of the curve $y = \frac{\cos x}{2 - \sin x}$. The curve intersects the x - and y -axes at A and C respectively, and has a turning point at B.

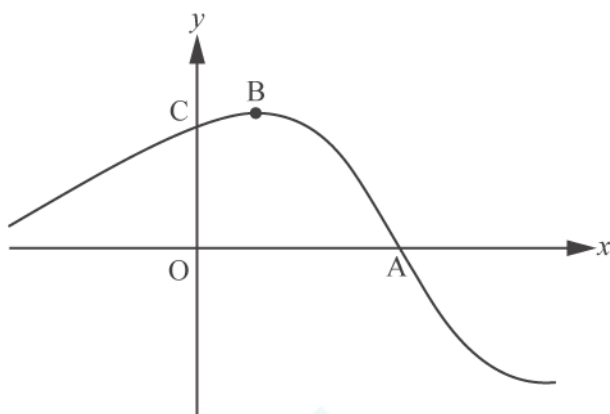


Fig. 8

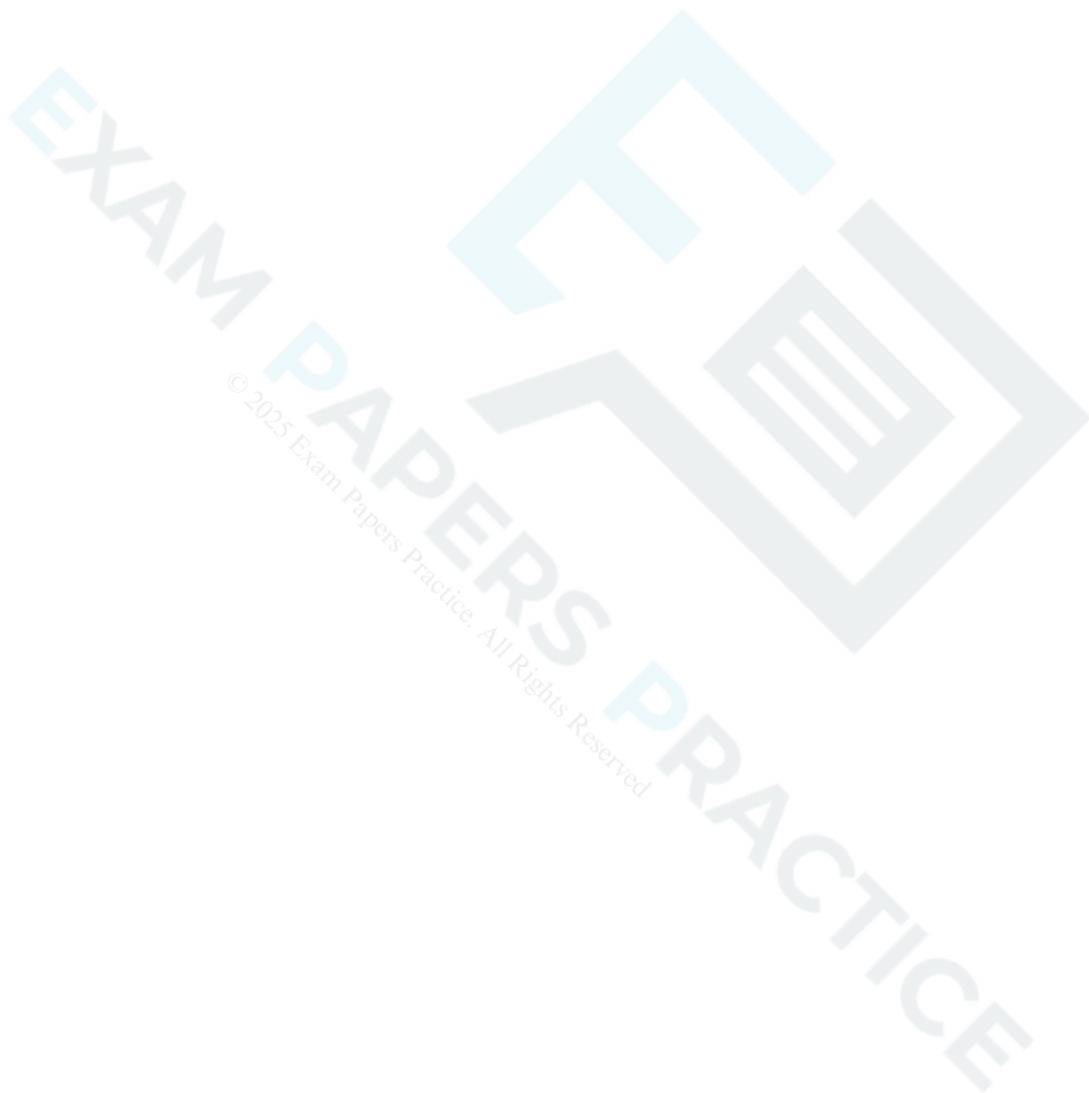
- (i) Write down the coordinates of A and C.

[2]



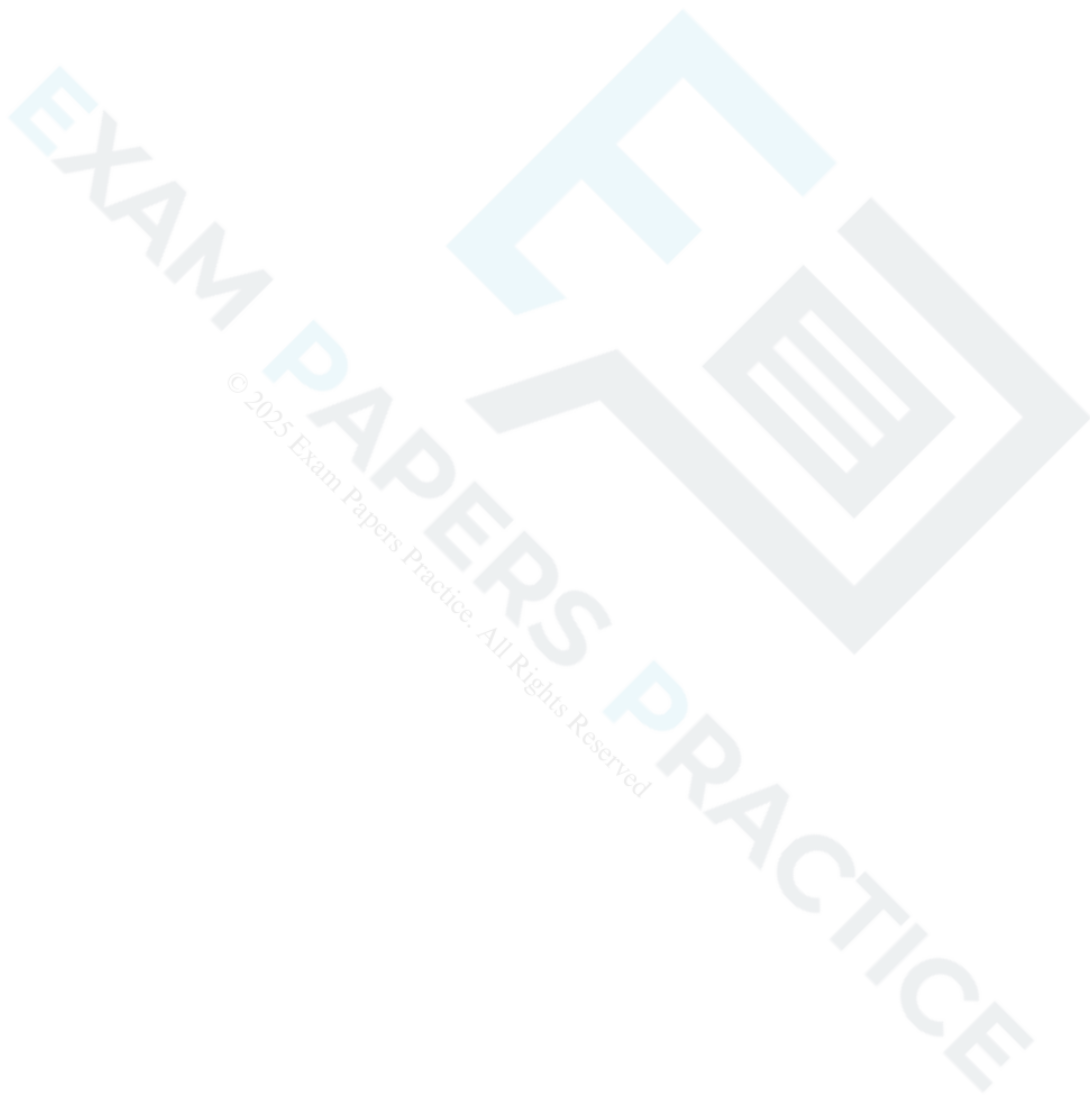
(ii) Find $\frac{dy}{dx}$ and the exact coordinates of B.

[7]



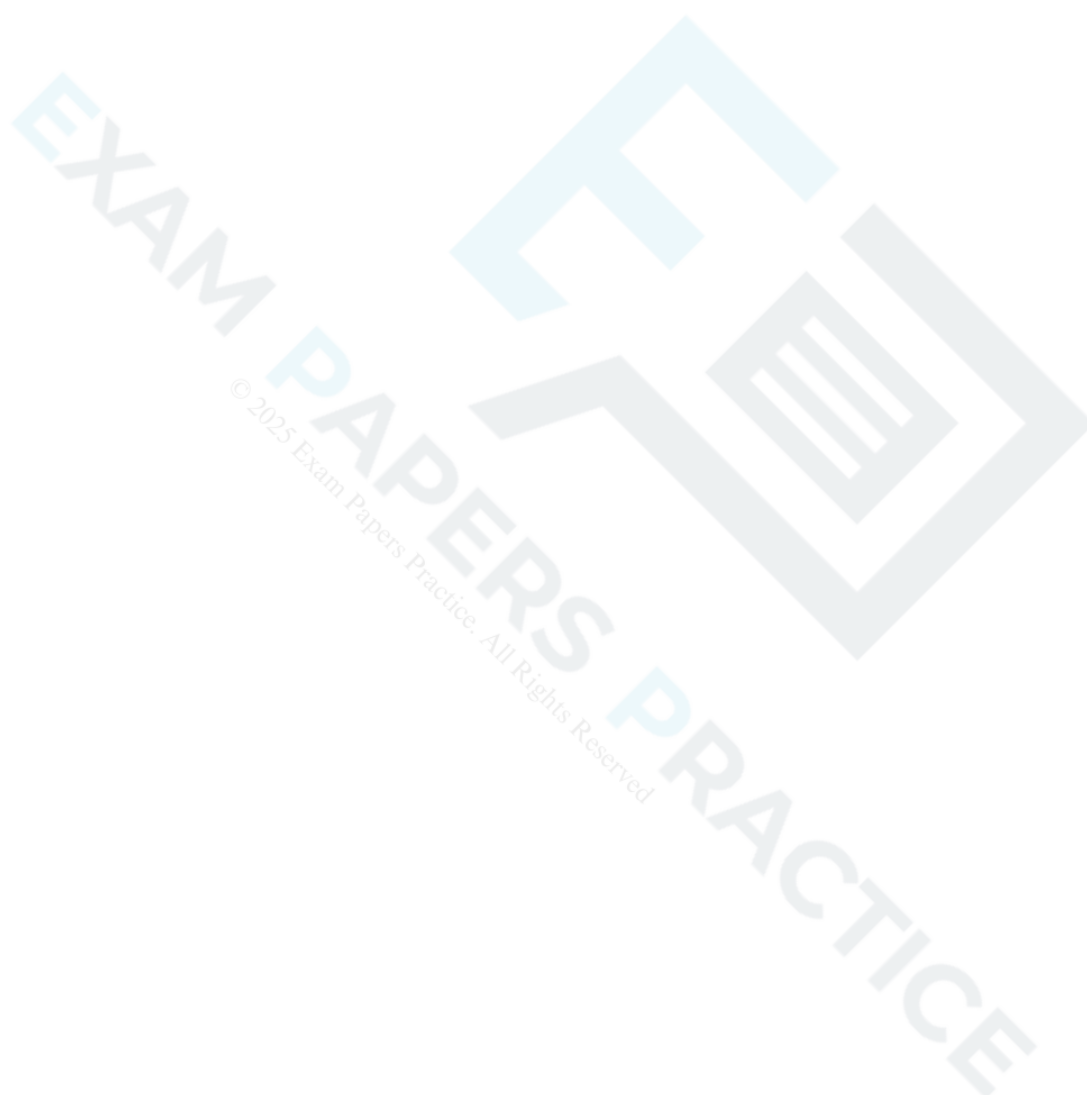
(iii) (A) Using integration by substitution, or otherwise, find the exact area of the region enclosed by the curve, the y -axis and the positive x -axis. PAPERS PRACTICE

[4]



(B) The line $x = k$ divides this region into two parts of equal area. Show that $k = \arcsin(2 - \sqrt{2})$.

[5]



- (i) Express each of x and $\frac{dx}{dy}$ in terms of y .

[2]

- (ii) A point is moving on the curve, and has coordinates (x, y) at time t . When $x = 1$, the value of $\frac{dx}{dt}$ is 2.

Find the exact value of $\frac{dy}{dt}$ at this instant.

[5]

83 A curve has equation $3x^{\frac{2}{3}} + 2y^{\frac{1}{3}} = 7$.

(i) By differentiating implicitly, find $\frac{dy}{dx}$ in terms of x and y .

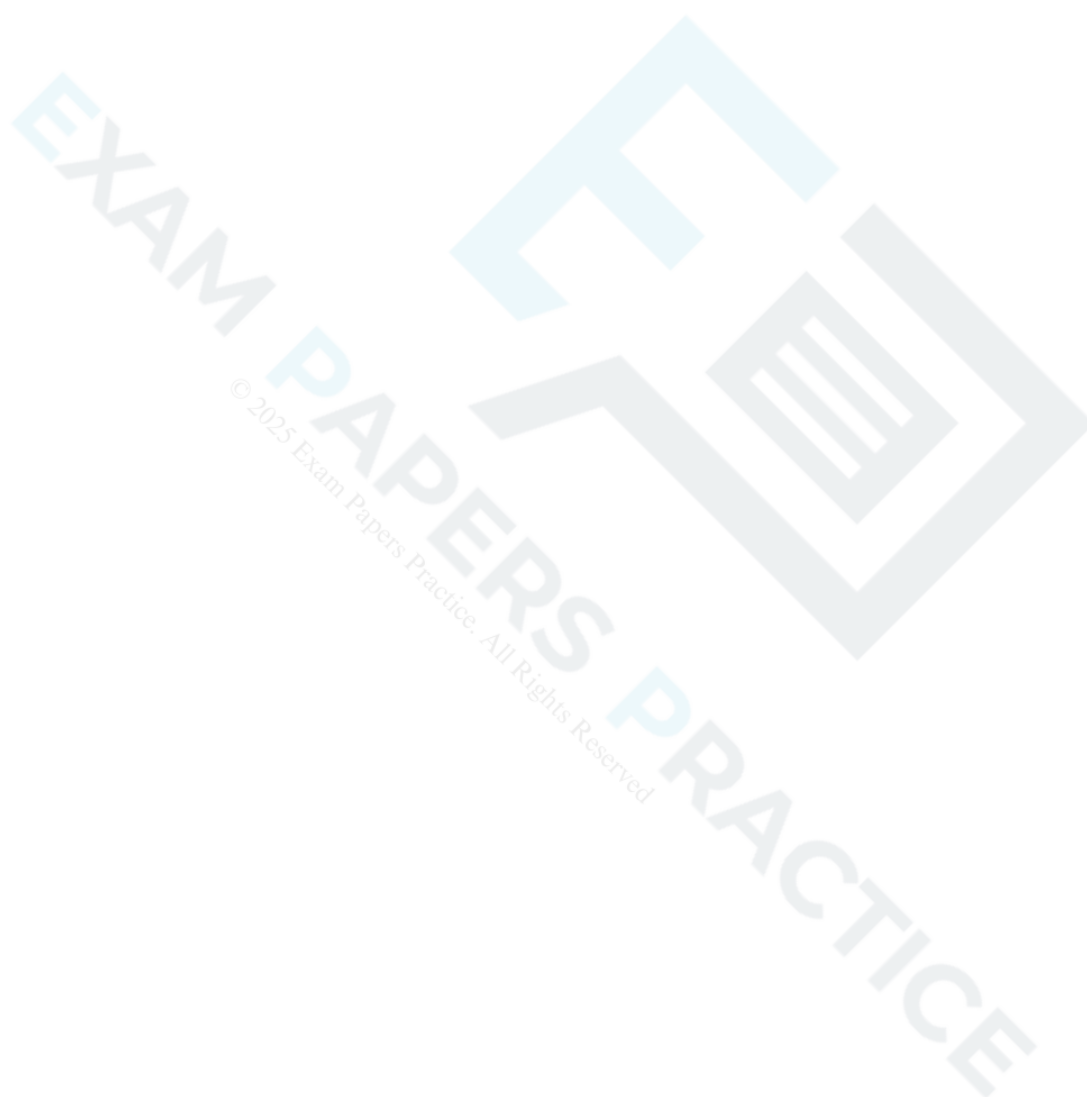
[3]

(ii) Hence find the gradient of the curve at the point with coordinates $(1, 8)$.

[2]

Differentiate $\frac{1}{(5-2x^3)^2}$.

[3]



85 The standard formulae for the volume V and total surface area A of a solid cylinder of radius r and height h are

EXAM PAPERS PRACTICE

$$V = \pi r^2 h \text{ and } A = 2\pi r^2 + 2\pi rh.$$

You are given that $V = 400$.

(i) Show that $A = 2\pi r^2 + \frac{800}{r}$.

[2]

(ii) Find $\frac{dA}{dr}$ and $\frac{d^2A}{dr^2}$.

[4]

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- (iii) Hence find the value of r which gives the minimum surface area. Find also the value of the surface area in this case.

[4]



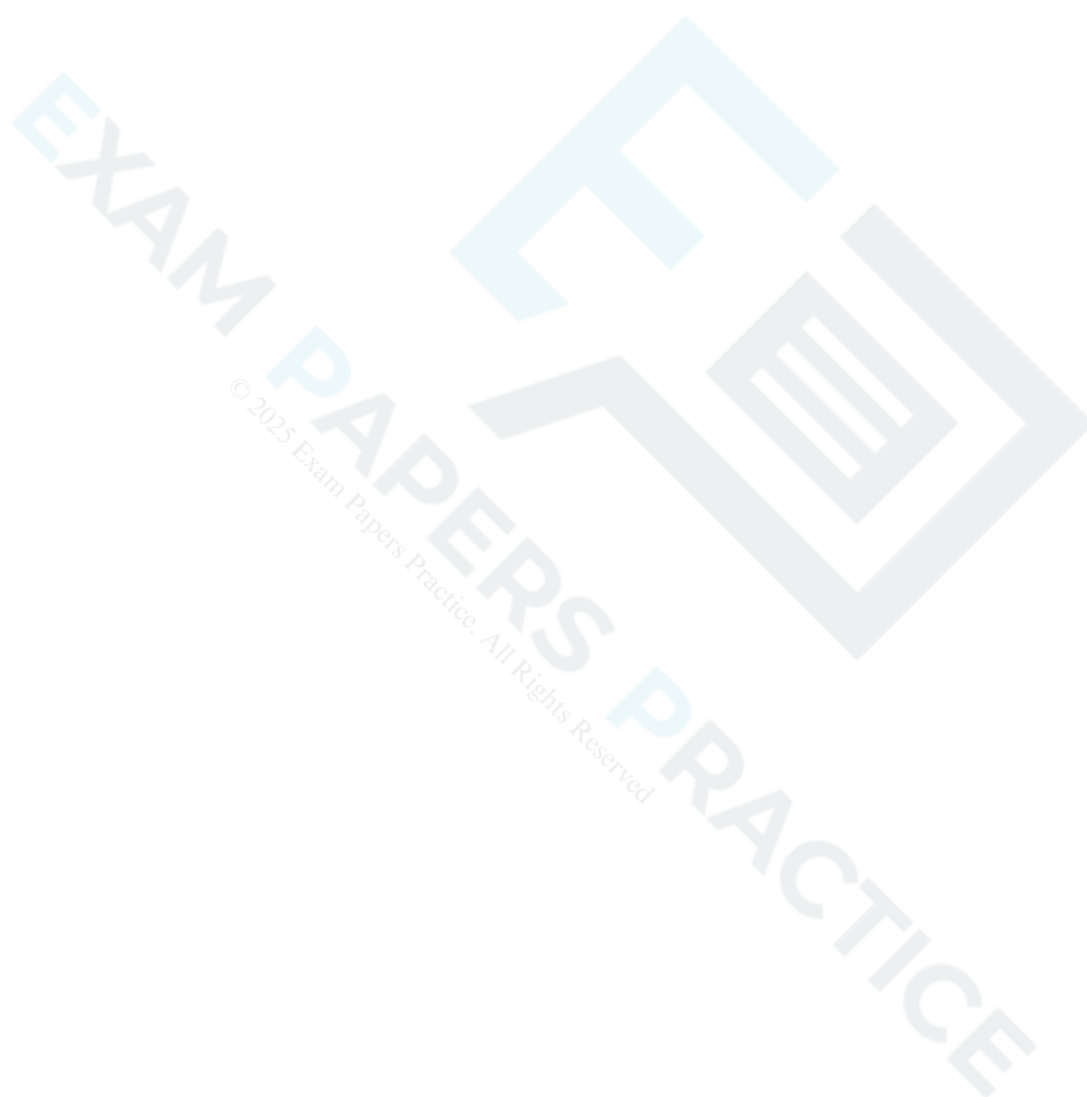


86

A curve passes through the point (2, 10) and has gradient $\frac{dy}{dx} = 12x^3 - 7$. Find the equation of the curve. [5]



- 87 Find the equation of the normal to the curve $y = 2x^3$ at the point on the curve where $x = 2$. Give your answer in the form $ax + by = c$. [5]



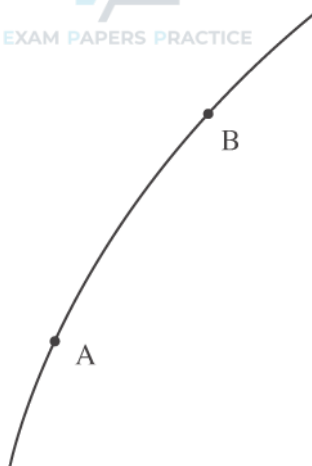


Fig. 3

Fig. 3 shows two points A and B on the curve $y = \log_{10} x$. At A, $x = 0.1$ and at B, $x = 0.2$.

- (i) Calculate the gradient of the chord AB.

[2]

- (ii) The gradient of the chord AB gives an estimate for the gradient of the curve at A. On the copy of Fig. 3 below, mark a point C on the curve such that the gradient of the chord AC would give a better estimate. [1]

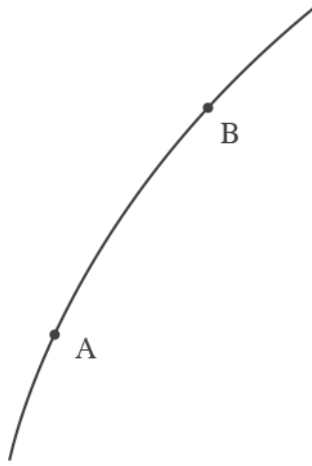


Fig. 3

(i) Find $\int_1^5 4x \, dx$

(ii) Find $\int 6x^{\frac{1}{2}} \, dx$

END OF QUESTION PAPER

$$3 - y - 6 \Rightarrow 2x \frac{dx}{dy} = 3y^2 - 10$$

Question			Answer/Indicative content	Marks	Guidance
					were unable to make meaningful progress in part (c), although they may have earned credit for a correct domain in part (a) and a FT calculation in part (b).  Misconception A significant minority of candidates appeared unclear how composite functions are formed and thought $fg(x) = f(x) \times g(x)$. There was also a surprising number of candidates that expressed $\sqrt{x^3 - x - 6}$ as $\sqrt{x^3} - \sqrt{x} - \sqrt{6}$
			Total	7	

Question			Answer/Indicative content	Marks	Guidance	
2			$\frac{1}{2} - \sin 2x \cos x = \sin x \cos 2x$ $\sin 3x = \frac{1}{2}$ oe $x = \frac{\pi}{18}$ and $x = \frac{5\pi}{18}$ $\pm \int \left(\sin x \cos 2x - \left(\frac{1}{2} - \sin 2x \cos x \right) \right) dx$ oe $F[x] = -\frac{x}{2} - \frac{\cos 3x}{3}$ $F\left[\frac{5\pi}{18}\right] - F\left[\frac{\pi}{18}\right]$ $\frac{\sqrt{3}}{3} - \frac{\pi}{9}$ or $\frac{3\sqrt{3}-\pi}{9}$ cao	M1 (AO3.1a) M1 (AO2.1) A1 (AO3.2a) A1 (AO1.1) M1 (AO1.1) A1 (AO1.1) M1 (AO1.1) A1 (AO3.2a) [8]	from compound angle formula allow sign errors only A1 for each ignore limits allow the positive of this F[x] must be one of the correct forms	or $4\sin^3 x - 3\sin x + \frac{1}{2} = 0$ oe 0.17453... A1 0.87266... A1 to 2 or more sf $\pm \left(-\frac{4}{3}\cos^3 x + \cos x - \frac{x}{2} \right)$ oe or $\pm \left(-\frac{1}{3}\cos 2x \cos x + \frac{1}{3}\sin x \sin 2x - \frac{1}{2}x \right)$ oe for A1 F[0.87266] – F[0.17453] for M1
Total				8	<u>Examiner's Comments</u> Candidates who did well either spotted the compound angle application or were able to derive the correct cubic in sin x when finding the points of intersection of the curve. They recognised how to apply integration and some were able to do this correctly. Those who did less well were unable to make progress with the integration or slipped up in forming the cubic in sin x. Candidates who did not do well were unable to progress from setting the two equations equal or from stating the correct integration needed.	

Examiner's Comments

Candidates who did well either spotted the compound angle application or were able to derive the correct cubic in $\sin x$ when finding the points of intersection of the curve. They recognised how to apply integration and some were able to do this correctly. Those who did less well were unable to make progress with the integration or slipped up in forming the cubic in $\sin x$. Candidates who did not do well were unable to progress from setting the two equations equal or from stating the correct integration needed.

Question		Answer/Indicative content	Marks	Guidance
3	a	$\sin 2x \approx 2x$ or $\sin x \approx x$ used $\int \left(\frac{1}{x}\right) dx$ or $\int \left(\frac{1}{x} - x\right) dx$ obtained oe nfww $F[x] = \ln x$ oe or $F[x] = \ln x - \frac{1}{2} x^2$ oe $\ln(0.05) - \ln(0.01) = \ln 5$ oe or $\ln(0.05) - \ln(0.01) + 0.0012 \approx \ln 5$ oe	M1 (AO3.1a) A1 (AO1.1) A1 (AO1.1) A1 (AO3.2a) [4]	may see $\cos x \approx 1 - \frac{x^2}{2}$ intermediate step needed from here to earn final mark
	b	differentiation of <i>their</i> $\frac{1}{x}$ substitution of 0.01 and $-10\,000$ correctly obtained	M1 (AO2.1) A1 (AO1.1) [2]	or differentiation of y using quotient rule and use of small angle approximation from $-\frac{1}{x^2}$ or $-\frac{1}{x^2} - 1$ oe

Question			Answer/Indicative content	Marks	Guidance
	c		4.54066×10^{-6} or 0.00000454066 cao (no sign change for 6 dp), but sign change for 5 dp or last two iterates agree to 5dp 0.94775	B1 (AO2.5) E1 (AO3.1a) B1 (AO3.2a) [3]	allow sign change between 0.947745 and 0.9477475 <u>Examiner's Comments</u> Candidates who did well in this question made efficient use of the small angle approximation for $\sin x$ and sensibly used $\sin 2x \approx 2x$ to obtain $\int \frac{1}{x} dx$, or used the double angle formula and then the small angle approximation for x . Full marks in parts (a) and (b) was commonly achieved. Those who did less well did not simplify their expression for $\frac{1}{x}$ fully before integrating and went on to make a slip (often integrating by parts) and similarly used the product or quotient rule in part (b) to differentiate $\frac{x}{x^2}$ for example. Candidates who did well in part (c) were clearly familiar with spreadsheets and were able to give a clear and correct response to both requests. Candidates who did less well either didn't understand the spreadsheet notation, or didn't appreciate the significance of the change of sign in the table.
			Total	9	

Question			Answer/Indicative content	Marks	Guidance
4	a		$4 + 8\cos 8x$	M1* (AO1.1) A1 (AO1.1) [2]	differentiation with either term correct all correct
	b		attempt to solve their $4 + 8\cos 8x = 0$ $\frac{\pi}{12}$ isw cao	M1dep* (AO1.1) A1 (AO1.1) [2]	one intermediate step seen Examiner's Comments Candidates who did well in Question 3 differentiated successfully in part (a) and set their derivative equal to 0 in part (b). They understood that the answer should be presented in radians. Those who did less well made a slip in part (a)– usually omitting an 8 – or worked in degrees in part (b). The very few candidates who did not do well were unable to differentiate either term successfully in part (a).
			Total	4	

Question			Answer/Indicative content	Marks	Guidance	
5			$\int_a^b (x^3 - 5x^2 + 7x - 3) dx$ $a = 1 \text{ and } b = 3$ $F[x] = \frac{x^4}{4} - \frac{5x^3}{3} + \frac{7x^2}{2} - 3x$ $F[3] - F[1] \text{ evaluated}$ $= -\frac{4}{3} \text{ so required area is } \left -\frac{4}{3} \right = \frac{4}{3} \text{ AG}$	M1 (AO2.1) A1 (AO1.1) B1 (AO2.2a) M1 (AO1.1) M1 (AO1.1) A1 (AO2.4) [6]	All three brackets expanded. Allow sign errors and one coefficient error All correct integration of their expanded brackets; condone +c Allow sign errors and one coefficient error Subst seen, allow sign or num slip $\left[\frac{81}{4} - 45 + \frac{63}{2} - 9\right] - \left[\frac{1}{4} - \frac{5}{3} + \frac{7}{2} - 3\right]$ Sign change must be seen; no reliance on decimals	Must be 4 term cubic Later M marks both dep on first M1 $\left[-\frac{9}{4}\right] - \left[-\frac{11}{12}\right]$
Total				6		

Examiner's Comments

Almost all candidates started off by expanding the given expression to find a four-term cubic, which was usually correct. Most were able to integrate their expression correctly. Only a minority of candidates realised that this had to be evaluated between the limits of 1 and 3, and only a very small number of candidates correctly justified the change of sign involved in getting to the final area given in the question.

Question			Answer/Indicative content	Marks	Guidance	
6	a		$\frac{dy}{dx} = \frac{x - \ln x}{x^2}$ $\frac{1 - \ln x}{x^2} = 0 \Rightarrow \ln x = 1 \Rightarrow x = e$	M1 (AO1.1a) A1 (AO1.1) E1 (AO2.2a) [3]	M1 for attempt to use quotient rule (allow one error) Convincing completion (AG)	Subbing $x = e$ gets M0
	b		$\frac{dy}{dx} = \frac{1}{x^2} - \frac{\ln x}{x^2}$ $\frac{d^2y}{dx^2} = -\frac{2}{x^3} - \frac{\frac{x^2}{x} - 2x \ln x}{x^4}$ $= -\frac{2}{x^3} - \frac{x - 2x \ln x}{x^4} \text{ or } \frac{x^2(-\frac{1}{x}) - (1 - \ln x)2x}{x^4}$ $= \frac{-3 + 2 \ln x}{x^3}$ When $x = e$, $\frac{d^2y}{dx^2} = -\frac{1}{e^3} < 0$ hence maximum	M1 (AO1.1a) A1 (AO1.1) E1 (AO2.1) [3]	Attempt to differentiate again (allow one error) Correct second derivative Correct conclusion from correct working	OR M1 subst (must be seen) values either side of e into derivative A1 correct conclusion about sign of gradient E1 correct conclusion from correct working regarding maximum (-0.05)
	c		$\frac{\ln e}{e} > \frac{\ln a}{a} \Rightarrow \frac{1}{e} > \frac{\ln a}{a} \text{ or } a \ln e > e \ln a$ $e^{\frac{a}{e}} > a \text{ hence } e^a > a^e \text{ or } \ln e^a > \ln a^e$ $e^a > a^e$	M1 (AO3.1a) A1 (AO2.4) [2]	Convincing completion (AG)	
			Total	8		
7			$\frac{dy}{dx} = e^x$ Hence $\frac{dy}{dx} > 0$ for all x so e^x is an increasing function for all x	M1 (AO1.1a) E1 (AO2.4) [2]	Convincing completion (AG)	
			Total	2		

Question			Answer/Indicative content	Marks	Guidance	
8			DR $\int_e^{\pi} \frac{1}{x} dx = [\ln x]_e^{\pi}$ $\ln \pi - \ln e = \ln \pi - 1$	M1 (AO1.1a) A1 (AO2.2a) [2]	May have $\ln x $ Convincing completion inc at least 1 intermediate line of working (AG)	Don't allow if '+ c' or if no [] or limits but condone no dx
			Total	2		

Question		Answer/Indicative content	Marks	Guidance	
9	a	$\frac{dx}{d\theta} = \sec^2 \theta$ $\int_{x=-1}^{x=1} \frac{\sec^2 \theta}{(1 + \tan^2 \theta)^2} d\theta$ $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\sec^2 \theta}{(1 + \tan^2 \theta)^2} d\theta$ $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{\sec^2 \theta} d\theta$ $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \cos^2 \theta d\theta$ $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{2} (\cos 2\theta + 1) d\theta$ $\frac{1}{2} \left[\frac{1}{2} \sin 2\theta + \theta \right]_{-\frac{\pi}{4}}^{\frac{\pi}{4}}$ $\frac{1}{2} \left(1 + \frac{\pi}{2} \right) = \frac{1}{2} + \frac{\pi}{4}$	M1 (AO1.1a) M1 (AO1.1) M1 (AO1.1) M1 (AO2.2a) M1 (AO1.1) M1 (AO3.1a) M1 (AO1.1) A1 (AO2.1) [8]	o.e. Substitution for either x or dx (limits may be missing) Limits (may be done at any point) Use of $1 + \tan^2 \theta = \sec^2 \theta$ Use of $\sec \theta = \frac{1}{\cos \theta}$ Use of $\cos 2\theta = 2 \cos^2 \theta - 1$ Integration. Must include at least one trig term. Limits may be wrong or missing Exact form	Condone $\sec^2 x$

Question			Answer/Indicative content	Marks	Guidance	
	b	i	$\frac{dy}{dx} = \frac{-4x}{(x^2 + 1)^3}$ $\frac{d^2y}{dx^2} = \frac{-4(x^2+1)^3 + 4x \cdot 2x \cdot 3(x^2+1)^2}{(x^2+1)^6}$ $\frac{d^2y}{dx^2} = \frac{-4(x^2 + 1) + 24x^2}{(x^2 + 1)^4}$ $\Rightarrow \frac{d^2y}{dx^2} = \frac{20x^2 - 4}{(x^2 + 1)^4}$	M1 (AO1.1a) A1 (AO1.1) M1 (AO1.1a) A1 (AO1.1) A1 (AO2.1) [5]	Attempt to differentiate Correct first derivative Attempt to use quotient rule to find second derivative Any correct expression for second derivative Simplifying expression for second derivative by cancelling $(x^2 + 1)$ and correct completion (AG)	chain or quotient or product rule Don't have to simplify For each M1 allow one error

Examiner's Comments

In this question many candidates just substituted for x . In Question 11 many did not realise that to show it is an increasing function they need to show that the gradient function is >0 .



AfL

When doing integration by substitution there are three aspects to consider:

- Substitute for x .
- Substitute for dx (by differentiating the substitution and rearranging).
- Deal with the limits (either change them to limits of u or change the answer in u back into x).

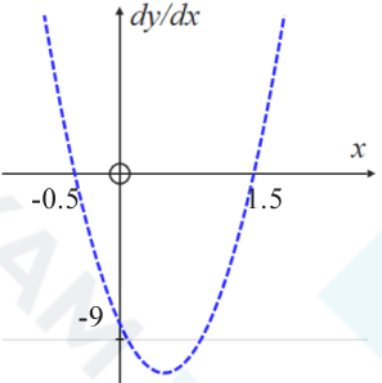
Question			Answer/Indicative content	Marks	Guidance
		ii	DR For concave downwards, $\frac{20x^2-4}{(x^2+1)^4} < 0$ so $20x^2 - 4 < 0$ $5x^2 < 1$ so $x^2 < \frac{1}{5}$ $-\frac{1}{\sqrt{5}} < x < \frac{1}{\sqrt{5}}$	M1 (AO2.2a) M1 (AO1.1) A1 (AO2.5) [3]	Or $x = \pm \frac{1}{\sqrt{5}}$ Correct solution correctly expressed Condone $x^2 > \frac{1}{5}$ following $y'' > 0$ for M1 Allow decimals (± 0.447)
			Total	16	

Question			Answer/Indicative content	Marks	Guidance	
10	a		$\frac{A}{x(A-x)}$	B1 (AO1.1) [1]	Denominator may be multiplied out Examiner's Comments Only the strongest candidates appreciated the connection between this question parts (a) and (b) and many attempted, unnecessarily, a partial fraction decomposition with variable degrees of success. Not all candidates appreciated the need to give the number of fish in the lake as an integer.  AfL The front cover instructions state that candidates should 'Give your final answer to a degree of accuracy that is appropriate to the context'. This may will be 3 significant figures in most cases, but it is always good practice to check back with the question before stating the final numerical answer.	
	b		DR $\int \frac{400}{x(400-x)} dx = \int dt$ $\int \left(\frac{1}{x} + \frac{1}{400-x} \right) dx = \int dt$ $\ln x - \ln(400-x) = t + c$ $\ln \left(\frac{x}{400-x} \right) = t + c$ When $t = 0$, $x = 100$ so $c = \ln \left(\frac{1}{3} \right)$ $t = \ln \left(\frac{3x}{400-x} \right)$ When $t = 10$, $10 = \ln \left(\frac{3x}{400-x} \right)$	M1 (AO1.1a) M1 (AO3.1a) M1 (AO1.1) A1 (AO1.1) M1 (AO1.1) M1 (AO3.4) M1 (AO2.1) A1 (AO3.2a) M1 M1 M1 A1	Separates variables (inc dx and dt) Use of result from (a) or method for partial fractions Attempt to integrate (at least one term correct) All integration correct Or $\ln \left \frac{Ax}{400-x} \right = t$ Finding constant (or $A = 3$) Using log laws to	400 either side

Question			Answer/Indicative content	Marks	Guidance
			$\frac{3x}{400-x} = e^{10} \Rightarrow 3x = e^{10}(400-x)$ 400 OR DR $\int_{100}^X \frac{400}{x(400-x)} dx = \int_0^{10} dt$ $\int_{100}^X \left(\frac{1}{x} + \frac{1}{400-x} \right) dx = \int_0^{10} dt$ $[\ln x - \ln(400-x)]_{100}^X = [t]_0^{10}$ $\left[\ln \frac{x}{(400-x)} \right]_{100}^X = [t]_0^{10}$ $\ln \frac{X}{(400-X)} - \ln \frac{100}{300} = 10 - 0$ $\ln \frac{X}{(400-X)} - \ln \frac{1}{3} = 10$ $\ln \frac{3X}{(400-X)} = 10$ $\frac{3X}{(400-X)} = e^{10}$ $3X = e^{10}(400-X)$ $X = 400$	M1 M1 M1 A1 [8]	combine the c value Attempt to rearrange to find x Must be rounded to nearest whole number Separates variables Use result from (a) or method for partial fractions Attempt to integrate (at least one term correct) All integration correct Limits applied (condone one error) Combining log terms together Attempt to rearrange to find X Must be rounded to nearest whole number
			Total	9	

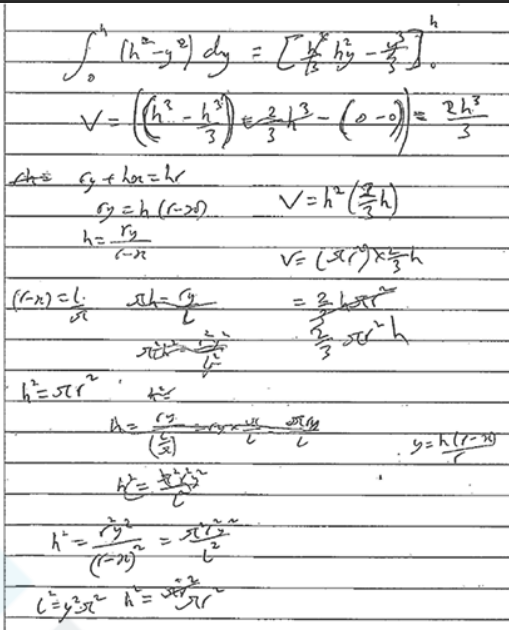
Question		Answer/Indicative content	Marks	Guidance	
11	a	At a stationary point $\frac{dy}{dx} = 2x - 2x^{-3} = 0$ $x^4 = 1$ giving $x = \pm 1$ so there are only two stationary points $\frac{d^2y}{dx^2} = 2 + 6x^{-4}$ When $x = 1$, $\frac{d^2y}{dx^2} = 2 + 6 \times 1^4 [= 8] > 0$ so minimum point When $x = -1$, $\frac{d^2y}{dx^2} = 2 + 6 \times (-1)^4 [= 8] > 0$ so also minimum point The two stationary points are both minimum points so there is no maximum point.	M1(AO1.1a) A1(AO1.1) M1(AO1.1a) M1(AO2.1) A1(AO2.1) E1(AO2.2a) [6]	Equating their derivative to zero and attempting to solve Finding both roots and no others Differentiating their $\frac{dy}{dx}$ Evaluating at (at least one of) their stationary point(s) Argues that both points are minimum from correct working. Allow arguing that second derivative is positive everywhere or a symmetry argument Deduces that there are no maximum points.	Allow for evaluating gradient at appropriate points either side. Arguing point is minimum from their gradients [incl sketch]

Question		Answer/Indicative content	Marks	Guidance	
	b	$\frac{dy}{dx} = 2x - 2x^{-3}$ $\text{At } \left(2, \frac{17}{4}\right) \text{ gradient} = 2 \times 2 - 2 \times 2^{-3} = \frac{15}{4}$ $\text{Equation of the tangent } y - \frac{17}{4} = \frac{15}{4}(x - 2)$ $y = \frac{15}{4}x - \frac{13}{4}$ $\text{Crosses x-axis when } y = 0, x = \frac{13}{15}$ $\text{Crosses y-axis when } x = 0, y = -\frac{13}{4}$ $\text{Area of triangle } \frac{1}{2} \times \frac{13}{15} \times \frac{13}{4} = \frac{169}{120} \text{ [below the axis]}$	M1(AO3.1a) A1(AO1.1) M1(AO3.1a) A1(AO1.1) A1(AO1.1) A1(AO1.1) [6]	Attempt to differentiate Correct value in any form Using their gradient to find the equation of the tangent Allow 0.867 or better any form FT their values but must be exact. Accept positive or negative value	Note Area 1.41 gets 5/6 marks
		Total	3		

Question		Answer/Indicative content	Marks	Guidance	
12	a	$\frac{dy}{dx} = 12x^2 - 12x - 9$ When $\frac{dy}{dx} = 12x^2 - 12x - 9 = 0$ $3(2x + 1)(2x - 3) = 0 \text{ so } x = -0.5, 1.5$ 	M1(AO1.1a) M1(dep)(AO1.1a) A1(AO1.1a) B1(AO1.1) [4]	DR Attempt to differentiate seen Attempt to solve their $\frac{dy}{dx} = 0$ Both values seen – may be indicated on the graph Correct shape through (0, -9)	SC For cubic graph of the function drawn with M0M0A0 allow SC1 for correct shape with minimum when $x = 1.5$, and maximum when $x = -0.5$
	b	Min point of gradient function when $\frac{d^2y}{dx^2} = 24x - 12 = 0 \text{ so } x = \frac{1}{2}$ Gradient is decreasing for $\left\{x : x < \frac{1}{2}\right\}$	M1(AO3.1a) A1(AO2.5) [2]	DR Attempt to find the vertex (including completing the square or symmetry argument) Inequality correctly formed and expressed as a set. Allow either $<$ or $\leq$	
Total			9		

Question			Answer/Indicative content	Marks	Guidance	
13			Point of inflection when $f''(x) = 0 \quad f''(x) = 6x = 0 \text{ has only one root } x = 0$ When $x = 0$, $f'(x) = -1 \neq 0$ so the point of inflection is not a stationary point.	M1(AO2.1) A1(AO2.1) E1(AO2.1) [3]	Equating their second derivative to zero Must explain that this is the only point of inflection Must prove that the point is not stationary from correct value for $f'(0)$	Also allow if shown that the stationary points are at $\left(\pm \frac{1}{\sqrt{3}}, 0\right)$
			Total	3		
14			$\frac{dy}{dx} = 4(x^2 + 5)^3 \times 2x = 8x(x^2 + 5)^3$ Using product rule with $u = 8x$ $v = (x^2 + 5)^3$ $\frac{d^2y}{dx^2} = 8(x^2 + 5)^3 + 8x \times 3(x^2 + 5)^2 \times 2x$ $= 8(x^2 + 5)^3 + 48x^2(x^2 + 5)^2$ $= 8(x^2 + 5)^2(7x^2 + 5)$	M1(AO1.1a) A1(AO1.1) M1(AO1.1a) A1(1.1) [5]	Attempting to use the chain rule Any form Need not be written explicitly FT their $\frac{dy}{dx}$ of correct form Any form Must be factorised – allow for $(x^2 + 5)^2$ ($56x^2 + 40$)	For candidates who have fully expanded the brackets and differentiate a polynomial without factorising, give SC1 for each correct term of the second derivative $56x^6 + 600x^4 + 1800x^2 + 1000$
			Total	5		

Question		Answer/Indicative content	Marks	Guidance
15	a	$V = \frac{\pi r^2}{h^2} \int_0^h (h^2 - y^2) dy$ $\frac{\pi r^2}{h^2} \left[h^2 y - \frac{y^3}{3} \right]_0^h$ $\frac{\pi r^2}{h^2} \left(h^3 - \frac{h^3}{3} \right)$ $= \frac{2}{3} \pi r^2 h$	M1 (AO2.2a) M1 (AO1.1a) A1 (AO2.1) [3]	Multiplying their integral from 14a $\frac{\pi r^2}{h^2}$ May be at a later stage Integration (one term correct condone omission of $\frac{\pi r^2}{h^2}$) Convincing completion (answer given) Examiner's Comments Only the better candidates made it through this part. Errors included integrating h^2 as $h^3/3$, forgetting to include the multiplying term or just getting into a mess with the algebra. Exemplar 8

Question			Answer/Indicative content	Marks	Guidance
					 This candidate made a good start on this part with correct integration earning M1. Notice that they avoided the potential pitfall of integrating h^2 to get $\frac{h^3}{3}$. However, they did not multiply their answer $\frac{\pi r^2}{h^2}$ missing out on the second method mark.
	b		$\int_0^h (h^2 - y^2) dy$	B1 (AO1.1) [1]	Condone dy missing or as δy but nothing else Condone integration started provided seen or used in the below part Examiner's Comments Only few candidates answered this part correctly, recognising that the given summation was the 'definition' of integration. Some did not change δy to dy, some put both and others omitted the limits.
			Total	4	

Question	Answer/Indicative content	Marks	Guidance
16	Let $u = 1 + \sqrt{x}$ $\Rightarrow dx = 2(u - 1)du$ or $dx = \frac{du}{\frac{1}{2}x^{-\frac{1}{2}}}$ oe or $2\sqrt{x}du$ $\int \frac{x}{1 + \sqrt{x}} dx = \int \frac{2(u - 1)^3}{u} du$ oe $\int \frac{2(u^3 - 3u^2 + 3u - 1)}{u} du$ $\int 2\left(u^2 - 3u + 3 - \frac{1}{u}\right) du$ $\frac{2u^3}{3} - 3u^2 + 6u - 2\ln u + c$ $\frac{2(1 + \sqrt{x})^3}{3} - 3(1 + \sqrt{x})^2 + 6(1 + \sqrt{x}) - 2\ln 1 + \sqrt{x} + c$ oe	M1 (AO1.1) A1 (AO1.1) M1 (AO1.1) M1 (AO3.1a) M1 (AO1.1) A1 (AO2.5) [6]	In terms of u For expanding cubic (allow one error) Dividing by u 3 of <i>their</i> terms correctly integrated to inc $\ln u$. Constant term may be missing at this stage Correct answer in terms of x with constant term Examiner's Comments The more able candidates often produced fully correct work but some lost the factor of 2 after integrating. The expansion of $(u - 1)^3$ was often inaccurately done. Some candidates wasted time expanding brackets and attempting to factorise at the end – no particular form was requested so candidates could leave their answer unsimplified. Some candidates attempted to use integration by parts rather than simplifying the integral by expanding brackets and dividing by u. Condone omission of modulus (since $1 + \sqrt{x}$, and hence u, must be positive)
	Total	6	

Question		Answer/Indicative content	Marks	Guidance
17		$f'(x) = \frac{(1+e^x)e^x - e^xe^x}{(1+e^x)^2}$ or $f'(x) = \frac{e^x}{1+e^x} - \frac{e^{2x}}{(1+e^x)^2}$ from product rule $f'(x) = \frac{e^x}{(1+e^x)^2}$ $f'(x) > 0 \Rightarrow f(x)$ is an increasing function for all x	M1 (AO1.1a) A1 (AO1.1) M1 (AO2.1) E1 (AO2.4) [4]	Use of quotient rule or product rule (allow one error) Correct derivative Simplifying derivative to a form which shows it is positive All correct and clear completion <u>Examiner's Comments</u> Most candidates realised that they needed to find dy/dx . Those who used the quotient rule were generally more successful than those who re-wrote the function as $(e^x)(1+e^x)^{-1}$ and used the product rule. In the latter case candidates often missed the e^x which was the 'differential' of the bracket and gave $dy/dx = (e^x)(-1)(1+e^x)^{-2} + (e^x)(1+e^x)^{-1}$. Those using the quotient rule occasionally forgot to square the denominator or wrote $(e^x)^2$ as the denominator. Simplifying the numerator was usually successful but some candidates wrote e^{x^2} when calculating $(e^x)(e^x)$. By far the most common fault was in completion where candidates merely wrote $e^x > 0$ as justification for dy/dx being positive and made no mention of the denominator.

Question			Answer/Indicative content	Marks	Guidance
					Exemplar 7 $f(x) = \frac{e^x}{1+e^{2x}} \quad \begin{matrix} u = e^x & v = 1+e^{2x} \\ u' = e^x & v' = 2e^{2x} \end{matrix}$ $f'(x) = \frac{e^x(1+e^{2x}) - e^x(e^{2x})}{(1+e^{2x})^2}$ $f'(x) = \frac{e^x + e^{2x} - e^{2x}}{(1+e^{2x})^2} \Rightarrow \frac{e^x}{(1+e^{2x})^2}$ $\frac{e^x}{(1+e^{2x})^2}$ must always be positive because denominator is a square number (+ve) & numerator & value (e^x) is always +ve. $\therefore f'(x)$ is always positive $\therefore$ gradient never never neg also always increasing. This is a good example of a candidate who clearly explains how they know that both the numerator and denominator of their final fraction for their derivative are positive.
			Total	4	

Question	Answer/Indicative content	Marks	Guidance
18	DR Gradient of normal = 2 $\Rightarrow$ gradient of tangent = -0.5 $2y \frac{dy}{dx} + y + x \frac{dy}{dx} = 0$ eg $-y + y - 0.5x = 0 \Rightarrow x = 0$ $y^2 = 8$ $(0, \sqrt{8})$ and $(0, -\sqrt{8})$ Alternative method $\frac{dx}{dy} = \frac{-8 - y^2}{y^2} \text{ oe}$ $\frac{8 + y^2}{y^2} = 2 \text{ oe}$ $y^2 = 8$ $(0, \sqrt{8})$ and $(0, -\sqrt{8})$	M1 (AO3.1a) M1 (AO1.1) A1 (AO1.1) A1 (AO2.2a) M1 (AO1.1) A1 (AO1.1) M1 M1 A1 M1 M1 A1 [6]	$\frac{-y}{x + 2y} = -\frac{1}{2}$ Attempt at implicit differentiation with one algebraic term differentiated correctly Correct implicit expression Implied by at least one y-value Rearranging to $x = \frac{8 - y^2}{y}$ and attempting differentiation Differentiation by dividing first or quotient rule – allow one error Correct differentiation Using negative reciprocal Implied by at least one y-value Examiner's Comments There seemed to be a good understanding of the need for implicit differentiation, and many fully correct solutions were seen. The marking of this question allowed for those with the correct concepts, but poor algebraic manipulative skills to achieve a

Question			Answer/Indicative content	Marks	Guidance
					reasonable number of marks. A few candidate tried to rearrange to get a function of the form $y = g(x)$, but this then lead to greater difficulty in completing it correctly. Many got through the whole question but did not get the last mark for only giving one solution to $y^2 = 8$.
			Total	6	

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Question	Answer/Indicative content	Marks	Guidance
19	DR Area $= \int_0^1 (x^3 - (3x - 2)) \, dx$ $\left[\frac{x^4}{4} - \frac{3x^2}{2} + 2x \right]_0^1$ $\frac{3}{4}$ Alternative method Area between curve and x-axis = $\int_0^1 x^3 \, dx = \left[\frac{x^4}{4} \right]_0^1 = \left[\frac{1}{4} \right]$ Area triangle ABD = $\frac{1}{2} \times \frac{1}{3} \times 1 = \frac{1}{6}$ Area = area between curve and x-axis + triangle ODE – triangle ABD $\frac{3}{4}$	M1 (AO3.1a) M1 (AO1.1) M1 (AO1.1) A1 (AO2.2a) M1 M1 M1 A1 [4]	Integrand. FT their tangent Limits (may be seen later) Allow one error Using <i>their</i> values Examiner's Comments A reasonable number of complete answers, but many candidates did not know what area they were trying to find and ended up with only part marks. Some tried integrating between the curve and the y-axis but only very few were successful doing it this way. Exemplar 6 Handwritten work for Exemplar 6: $\int_0^1 x^3 \, dx \Rightarrow \left[\frac{1}{4} x^4 \right]_0^1 \Rightarrow \frac{1}{4} - 0 \text{ units}^2$ $1 - \frac{2}{3} = \frac{1}{3}$ $\frac{1}{2} \text{ base} \times \text{height} = \frac{1}{2} \times \frac{1}{3} \times 1 = \frac{1}{6}$ $\frac{2}{3} + \frac{1}{4} - \frac{1}{6} = \frac{3}{4} \text{ units}^2$ Answer At first glance, with the correct answer, $\frac{3}{4}$, it would appear that this would score full marks. However examiners need to make sure that the answer does not come from wrong working. Notice that their initial

Question			Answer/Indicative content	Marks	Guidance
					integration is wrong so they do not get the first M1. They do get the M1 for the triangle $\frac{1}{6}$ area of $\frac{1}{6}$ and also the M1 for knowing how to combine their areas. Since their initial integration was not accurate they do not score the final accuracy mark (i.e. overall they get M0M1M1A0.)
			Total	4	
20			$\int \left(\frac{1}{(x+2)} - \frac{1}{(x+3)} \right) dx$ $= \ln x+2 - \ln x+3 [+c]$ $\ln \left \frac{x+2}{x+3} \right + c$	M2 (AO1.1a) (AO1.1) A1 (AO2.2a) [3]	For both terms correct FT their fractions M1 for one of their terms correct A0 if modulus sign missing or if further work eg to try and find c Need brackets or better (eg single fraction) Ignore 'f(x) = ' Examiner's Comments In (b) most could integrate the reciprocals correctly to get natural logarithms but only the strongest realised that modulus should be used as negatives are not part of the domain for log functions Exemplar 3 $\int \frac{1}{x+2} dx - \int \frac{1}{x+3} dx = \int \frac{1}{(x+2)(x+3)} dx$ $= \ln(x+2) - \ln(x+3) + c \quad \ln a - \ln b = \ln \frac{a}{b}$ $= \ln \left(\frac{x+2}{x+3} \right) + c$ This was probably the most common response we saw to this part. The candidate has correctly integrated each fraction to get ln and also uses a log law correctly. If they had used modulus bars instead of round brackets they would have earned the last mark as well,
			Total	3	

Question			Answer/Indicative content	Marks	Guidance
21			$3x - 2 = x$ when $x = 1$ $x > 1$	M1 (AO3.1a) A1 (AO2.2a) [2]	Attempt to find when $f(x) = x$ or attempt to solve $3x - 2 = \frac{x+2}{3}$ or Using <i>their</i> inverse $3x - 2 > \frac{x+2}{3}$ soi by $x = 1$ Examiner's Comments A good sketch made it easier for candidates to answer this part although most could handle the necessary algebra.
			Total	2	
22			differentiate to obtain $2x - 4$ $+ 1 \times \ln x + x \times \frac{1}{x}$ oe derivative = 0 oe seen and terms combined $2x - 3 + \ln x = 0$ www isw AG	B1 (AO3.1a) M1 (AO2.1) A1 (AO1.1) A1 (AO2.4) [4]	use of Product Rule all correct allow one error Examiner's Comments Candidates who did well showed how $\frac{dy}{dx}$ was found and explained why it was set equal to zero. Candidates who did less well made slips in applying the Product rule and/or forgot to set their derivative equal to zero.
			Total	4	

Question			Answer/Indicative content	Marks	Guidance
23			$k \frac{x^{-4}}{-4} \ln x - \int k \frac{x^{-4}}{-4} \times \frac{1}{x} dx$ oe $[32] \frac{x^{-4}}{-4} \ln x - \int [32] \frac{x^{-4}}{-4} \times \frac{1}{x} dx$ oe $-8x^{-4} \ln x - 2x^{-4} + c$ oe isw	M1 (AO3.1a) A1 (AO1.1) A1 (AO1.1) A1 (AO1.1) [4]	allow sign errors only all correct two of three elements correct all three elements correct <u>Examiner's Comments</u> Candidates who did well in this question were able to apply the method of integration by parts successfully and remembered to include the constant of integration in their final answer. Candidates who did less well made sign errors or omitted the constant in their final answer. A significant minority of candidates either didn't recognise that integration by parts was the correct approach or were unable to apply it successfully.
			Total	4	

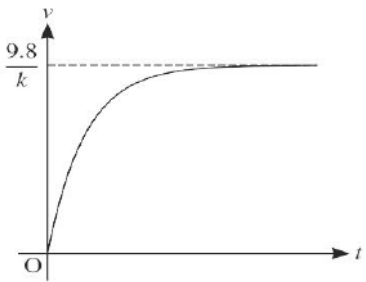
Question		Answer/Indicative content	Marks	Guidance	
24	a	$\frac{0.5}{2} \left[\sqrt{1+(-1)^3} + 2\sqrt{1+(-0.5)^3} + \sqrt{1+0^3} \right]$ oe $\sqrt{1+(-0.5)^3} \text{ soi}$ 0.717707 cao	M1 (AO1.1) B1 (AO1.1) A1 (AO1.1) [3]	condone omission of brackets NB $\frac{\sqrt{14}}{4} = 0.9(35414346693)$ NB $\frac{2 + \sqrt{14}}{8}$ unsupported implies M1B1	must be three terms in the bracket if unsupported allow SC3 for 0.717707 and SC2 for 0.717707173347 unsupported to 7 or more dp Examiner's Comments Candidates who did well in this question knew how to apply the trapezium rule and gave full details of their calculation. Candidates who did less well either ignored the request for an answer to 6 decimal places or made slips in the arithmetic. A surprising number of candidates were unable to make any progress with this question. Common errors were to substitute $h = 0.5$ in the formula but to then work with 4 strips, to use $x = 0.5$ instead of $x = -0.5$ and a few candidates substituted x-values in the formula instead of y-values.

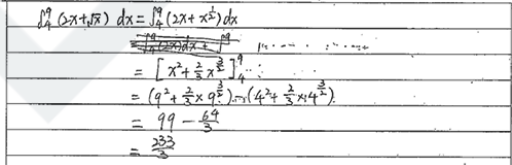
Question			Answer/Indicative content	Marks	Guidance
	b		under-estimate since curve is concave down / convex up oe	B1 (AO2.4) [1]	or eg the slant lines of both trapezia are entirely below the curve allow annotated diagram with at least one trapezium condone eg trapezium below curve allow integral is 0.841309 BC so my answer is an underestimate Examiner's Comments Candidates who did well understood how their calculated value related to the value of the integral and were able to explain this either by referencing that the curve is concave down, the behaviour of the gradient of the interval or by supplying an annotated diagram. A surprising number of candidates were unable to explain clearly why their value was an underestimate – or they explained why it was an over-estimate with a correct reason for an underestimate.
			Total	4	
25	a		$a(x^2 + 5)^{12}$ $2(x^2 + 5)^{12}(+c)$	M1 (AO1.1) A1 (AO1.1) [2]	condone omission of + c A1 FT their $kx(x^2 + 5)^{11}$ from part (a) Examiner's Comments Candidates who did well were familiar with the Chain rule and recognised the connection between these parts. Candidates who did less well made arithmetic slips in this part, or attempted this part using integration by parts.
	b		$k(x^2 + 5)^{11}$ seen $24x(x^2 + 5)^{11}$	M1 (AO1.1a) A1 (AO1.1) [2]	Examiner's Comments Candidates who did well on this part were familiar with the Chain rule. Arithmetical slips were seen.
			Total	4	

Question		Answer/Indicative content	Marks	Guidance	
26	a	$\int \frac{1}{9.8 - kv} dv = \int 1 dt$ $-\frac{1}{k} \ln(9.8 - kv) = t + c$ $9.8 - kv = Ae^{-kt}$ When $t = 0$, $v = 0$ giving $0 = \frac{1}{k} (9.8 - Ae^{-0})$ $A = 9.8$ Hence $v = \frac{9.8}{k} (1 - e^{-kt})$ Alternative using a Further Maths method $\frac{dv}{dt} + kv = 9.8$ Integrating factor $e^{\int P(t) dt} = e^{kt}$ $ve^{kt} = \int (9.8e^{kt}) dt = \frac{9.8}{k} e^{kt} + c$ $v = \frac{9.8}{k} + ce^{-kt}$ When $t = 0$, $v = 0$ giving $c = -\frac{9.8}{k}$	M1 (AO 3.1a) A1 (AO 1.1b) A1 (AO 1.1b) M1 (AO 1.1a) M1 (AO 3.1a) A1 (AO 1.1b) A1 (AO 1.1b) [7] M1 A1 A1 M1 M1 A1	Attempt to separate variables and integrate Correct rearrangement Correctly integrated – must include +c somewhere Attempt to write in the form $v =$ (Clearing natural logs; may be done after evaluation of the constant) Using initial conditions leading to a value for the constant; may be done before clearing logs Correct value for A (or $c = -\frac{1}{k} \ln 9.8$) Must be in the form $v = \dots$ Writing in the correct form and using integrating factor Correct integrating factor Correct equation Attempt to write in the form $v =$ Using initial conditions leading to a value for the constant Correct value for c Must be in the form $v = \dots$	Allow up to 3M marks for an attempt with an incorrect rearrangement.



Question	Answer/Indicative content	Marks	Guidance
	Hence $v = \frac{9.8}{k}(1 - e^{-kt})$	A1	Examiner's Comments Most candidates did attempt to separate the variables and the first mark was given even when the rearrangement was clearly the result of poor algebra and was not in the required form for integrating. $\frac{dt}{dv} = \frac{1}{9.8 - kv}$ Candidates who wrote $\frac{dt}{dv} = \frac{1}{9.8 - kv}$ were much more likely to integrate correctly. Many other candidates did not have the equation in the correct form so lost all the accuracy marks, however further method marks were available for using the boundary conditions and for rewriting into the form '$v =$'. Some candidates with knowledge of Further Maths techniques were usually successful, when using for example an integrating factor. Exemplar 5 $v = 9.8t - kv t$ $v + kv t = 9.8t$ $v(1 + kt) = 9.8t$ $v = \frac{9.8t}{1 + kt}$ This candidate has not separated the variables here and treats v as a constant value. However, the solution is written in the form '$v = \dots$' so the third method mark was given (it was not dependent on the first M mark). This is an example of an incorrect function that has a non-zero limiting value for large values of t, so all

Question			Answer/Indicative content	Marks	Guidance
					marks were available in parts (b) and (c).
	b			B1 (AO 1.1b) B1 (AO 1.1b) [2]	Graph through (0, 0) with the correct general shape or correct for their v. Horizontal asymptote labelled $\frac{9.8}{k}$ Examiner's Comments The shape of the graph was often drawn well by candidates who used the context rather than their solution, with the zero initial velocity and the terminal velocity using the zero value for acceleration directly from the differential equation. One follow-through mark was given for a graph through the origin with approximately the correct shape of their velocity.
	c		As $t \rightarrow \infty$, $v \rightarrow \frac{9.8}{k} = 7$ $k = 1.4$ Alternative solution Limiting value when $\frac{dv}{dt} = 0$ $k = \frac{9.8}{7} = 1.4$	M1 (AO 3.4) A1 (AO 1.1b) [2] M1 A1	Recognises the limiting value Any form from correct v Could be implied Examiner's Comments Part (c) can be answered directly from the differential equation, with the understanding that the velocity tends to zero when the acceleration tends to zero also. Candidates who were successful in part (a) were able to access these marks.
	d		$v = 3.5 \Rightarrow 1 - e^{-1.4t} = \frac{3.5}{7}$ Hence $-1.4t = \ln 0.5 \Rightarrow t = 0.495$	M1 A1 B1 (AO 3.4) [2]	May substitute in other versions of the equation relating v and t Allow for 0.50 or awrt 0.495 No FT from incorrect expression for v
			Total	12	

Question	Answer/Indicative content	Marks	Guidance
27	DR $\int_4^9 \left(2x + x^{\frac{1}{2}} \right) dx = \left[x^2 + \frac{x^{\frac{3}{2}}}{\frac{3}{2}} \right]_4^9$ $\left(9^2 + \frac{2}{3} \times 9^{\frac{3}{2}} \right) - \left(4^2 + \frac{2}{3} \times 4^{\frac{3}{2}} \right)$ $\left[81 + 18 - 16 - \frac{16}{3} \right] = \frac{233}{3}$	M1* (AO 2.1) M1 (dep) (AO 2.1) A1 (AO 2.1) [3]	Attempt to integrate using fractional power Use of limits must be seen AG Any interim working must be correct Examiner's Comments This was generally well answered with candidates showing at least the indefinite integral, the substitution of the limits and the final answer; it is however good practice to show some intermediate steps as well. Marks were sometimes lost where intermediate steps were given in decimals and a small number of candidates struggled to use fractional indices correctly. Exemplar 1  This exemplar shows sufficient detail for a 'Show that' question. The candidate has thought about the surd; changing this into a fractional index, then finding the indefinite integral. They have then clearly shown the substitution and evaluation to get the given answer.
	Total	3	

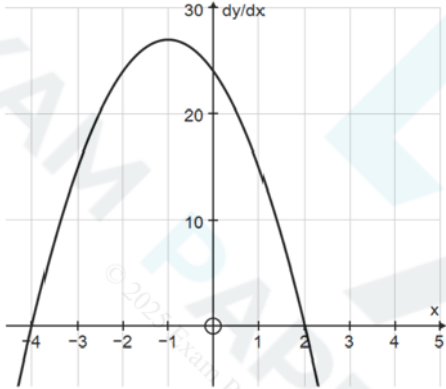
Question			Answer/Indicative content	Marks	Guidance
28			$F[8] - F[0] = 3 \times 8^5 + 3 \times 2^{11} [-0]$ eg $(3 \times 2^4 + 3) \times 2^{11}$ 51×2^{11}	M1 *(AO2.1) *M1 (AO2.4) A1 (AO1.1) [3]	or $3 \times 2^{15} + 3 \times 2^{11}$ $[-0]$ ft (a) any correct intermediate step OR $104448 \div 2^{11}$ (M2) <u>Examiner's Comments</u> Most candidates gained a mark for attempting to evaluate their answer to part (a) at the limits, and many gained a second mark for a correct attempt to then find the value of a . A few candidates wrongly applied the limits to the original expression, before it had been integrated.
			Total	3	

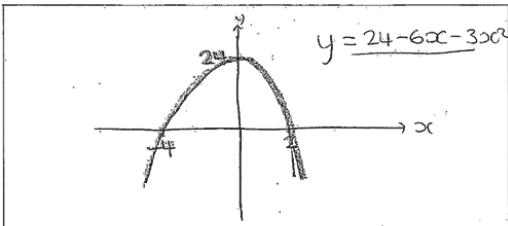
Question			Answer/Indicative content	Marks	Guidance
29			$x = 2, y = 3$ $\frac{dy}{dx} = \frac{2x}{4} - \frac{2}{x^2}$ oe Evaluaton of their derivative at $x = 2$ $y - 3 = \frac{1}{2}(x - 2)$ $y - 3 = -2(x - 2)$ intercepts are $(-4, 0)$ and $(3.5, 0)$ $\text{Area} = \frac{1}{2} \times 7\frac{1}{2} \times 3$ $\frac{45}{4}$ cao	B1 (AO1.1) M1 (AO2.1) A1 (AO1.1) M1 (AO1.1) M1 (AO2.4) M1 (AO1.1) A1 (AO1.1) A1 (AO1.1) M1 (AO2.1) A1 (AO1.1) [10]	soi Differentiation with 1 term correct NB $\frac{1}{2}$ equation of tangent $y = \frac{1}{2}x + 2$ $y = -2 + 7$ and equation of normal; FT their tangent gradient OR $\frac{1}{2} \times \sqrt{(6^2 + 3^2)} \times \sqrt{(3^2 + 1.5^2)}$ - lengths of tangent and normal or 11.25 <u>Examiner's Comments</u> This question was found accessible by almost all candidates, with a considerable number giving completely correct answers. However a small minority of candidates who integrated the equation of the curve – presumably triggered by the word ‘area’. Candidates started by differentiating the equation of the curve. Candidates making errors in this step were still able to gain marks for attempting to find the slope of the tangent when $x = 2$, attempting to find the equations of the tangent and normal, and for attempting to find the area between the tangent, normal and the x-axis. They were also able to gain a mark for finding the value of y when $x = 2$. It was noted that candidates drawing a sketch of the tangent, normal and x-axis were more successful in identifying the area required and to calculate it efficiently and correctly.
			Total	10	

Question			Answer/Indicative content	Marks	Guidance
30			$\int (15x^4 + 11x^{\frac{8}{3}}) \, dx$ $\frac{15x^5}{5} + \frac{11x^{\frac{11}{3}}}{\frac{11}{3}} [+c]$ $3x^5 + 3x^{\frac{11}{3}} + c^{\text{oe}}$	M1 (AO3.1a) A1 (AO1.1) M1 (AO1.1) A1 (AO1.1) A1 (AO3.2a)	Expands bracket with one term correct All correct integration with one term correct FT both terms correct coefficients must be simplified and must see + c either here or in previous step Examiner's Comments [5] Some candidates attempted to integrate each term before expanding the bracket while others were unable to expand the bracket to obtain two correct terms to integrate. The problem was in multiplying x^3 by $\frac{11}{3}x^{\frac{8}{3}}$ and the result was that a considerable number of candidates gained a mark for showing $15x^4$ and another mark for integrating this to $\frac{15}{5}x^5$ but no other marks. Those candidates who gained the first 2 marks for expanding the bracket generally ended up with full marks for this part, except for a few who omitted the arbitrary constant.
			Total	5	

Question		Answer/Indicative content	Marks	Guidance
31	a	DR Decreasing function when $\frac{dy}{dx} < 0$ $\{x:x < -4\} \cup \{x:x > 4\}$	M1 (AO1.1a) A1 (AO2.5) [2]	Attempt to give the values of x for which $\frac{dy}{dx} < 0$ from their graph FT their graph if quadratic Condone use of for M mark $\leq$ Allow for "$x < -4$" or 2" Must be correct use of language or set notation here. Do not allow A1 for $x < -4, x > 2$ $x < -4$ and $x > 2$ $-4 > x > 2$ Examiner's Comments Many candidates wrongly assumed that further differentiation was required. The phrase "set of values" in the question had been used to encourage candidates to use proper set notation. As an alternative, the use of the word "or" between two correct inequalities was accepted. "And" was not accepted as this implies the intersection rather than the union of the two sets. Some scientific calculators will solve a quadratic inequality, so it this question has been used to test clear understanding and good use of notation as well as an understanding of decreasing functions.



Question	Answer/Indicative content	Marks	Guidance
b	DR $\frac{dy}{dx} = 24 - 6x - 3x^2$ When, $x = 0$, $\frac{dy}{dx} = 24$ When $\frac{dy}{dx} = 0, -3(x^2 - 8) = 0 \Rightarrow (x + 4)(x - 2) = 0$ $x = -4, 2$ 	M1 (AO1.1a) A1 (AO1.1) M1 (AO1.1a) A1 (AO3.1a) B1 (AO1.1) [5]	Expression for derivative seen May be shown on graph or in the working Method for solving their quadratic equation (allow any algebraic method) Must be seen on graph Correct shape. Maximum point should be to the left of the y-axis but need not be exact. Examiner's Comments A minority of candidates sketched the function and not the gradient function. The majority attempted to sketch the derived function but as in Question 7(b) did not show enough detailed reasoning. Often even when a full method was shown, the final mark was lost for careless positioning of the maximum point – the symmetry of the quadratic graph is an important part of its characteristic shape.

Question			Answer/Indicative content	Marks	Guidance
					Exemplar 3  $y = 24x - 6x^2 - 3x^3$ $\frac{dy}{dx} = 24 - 6x - 3x^2$ y intercept = $(0, 24)$ x-axis intercepts: $(-x^2 - 2x + 8)$ $x = 2, x = -4$ This exemplar indicates that where the question requires detailed reasoning, marks can easily be lost. Graphical calculators can be used rather than reasoning to establish the shape of the graph, so the solution must show sufficient detail to be awarded full marks. The derivative and the product of linear factors should have been equated to zero to fully establish the values $x = -4$ and $x = 2$. The final mark was also lost as the maximum point should have been clearly to the left of the y-axis – its exact position was not needed for full credit.
			Total	7	

Question	Answer/Indicative content	Marks	Guidance
32	DR $y = x^2 - 4x^{\frac{1}{2}} \Rightarrow \frac{dy}{dx} = 2x - 4 \times \frac{1}{2} x^{-\frac{1}{2}}$ $\frac{dy}{dx} = 0 \text{ gives } x^{\frac{3}{2}} = 1 \Rightarrow x = 1$ There is only one solution to this so there is only one stationary point on the graph When $x = 1$, $y = 1 - 4 = -3$ $\frac{d^2y}{dx^2} = 2 - 2\left(-\frac{1}{2}\right)x^{-\frac{3}{2}}$ When $x = 1$, $\frac{d^2y}{dx^2} = 3 > 0$ So the stationary point is a minimum point Alternative for final three marks Attempt to evaluate $\frac{dy}{dx}$ at a point $x \neq 1$ Attempt to evaluate $\frac{dy}{dx}$ at a point the other side of $x = 1$ Correct conclusion from correct values	M1 (AO2.1) M1 (AO3.1a) A1 (AO2.2a) B1 (AO2.1) M1 (AO1.1a) M1 (AO1.1a) A1 (AO2.2a) [7] M1 M1 A1	Uses fractional power in an attempt to differentiate Attempt to solve their $\frac{dy}{dx} = 0$ Must obtain $x = 1$ from correct working and indicate that this is the only stationary point From correct working seen (AG) Attempt to find second derivative Substituting into their expression Conclusion from correct working (AG) As there is only one stationary point, allow for similarly evaluating y and comparing with -3 Examiner's Comments Most candidates looked for the points where $\frac{dy}{dx} = 0$ and many found it difficult to solve the resulting equation. Some introduced a spurious root at $x = 0$ and many others did not comment that the root they had found was the only root to be found. Verifying that the given point gives

Question			Answer/Indicative content	Marks	Guidance
					$\frac{dy}{dx} = 0$ was awarded one mark as a special case, as the method did not allow candidates to establish it as the only root. Some also argued (incorrectly) that this is a quadratic graph or a parabola so has only one stationary point. Many also lost marks by not establishing that the stationary point was a minimum point. In this case where there is only one stationary point, full credit was given to candidates who checked the value of y either side of the stationary point even though the more usual method is to check the sign of the derivative.  AfL This mark was worth 7 marks and candidates could have used this to realise how much evidence would be needed to obtain full credit.
			Total	7	

Question			Answer/Indicative content	Marks	Guidance
33		a	DR $\frac{dy}{dx} = 4x^2$ At point of contact, gradient = 1 so $4x^2 = 1$ DR $x = \frac{1}{2}$ $y = \frac{1}{2}$ $\frac{1}{2} = \frac{4}{3}\left(\frac{1}{2}\right)^3 + a$ $a = \frac{1}{3}$	M1 (AO 1.1a) E1(AO 2.4) [2] B1 (AO 3.2a) B1(AO 2.2a) M1(AO 3.1a) A1(AO 1.1) [4]	AG Convincing completion needed Taking the positive root soi
		b	DR When curve crosses line $\frac{4}{3}x^3 + \frac{1}{3} = x$ $4x^3 - 3x + 1 = 0$ $(2x - 1)^2(x + 1) = 0$ $x = -1$ $\int_{-1}^{\frac{1}{2}} \left(\frac{4}{3}x^3 + \frac{1}{3} - x \right) dx$ $\left[\frac{1}{3}x^4 + \frac{1}{3}x - \frac{1}{2}x^2 \right]_{-1}^{\frac{1}{2}}$ $\frac{9}{16}$	M1 (AO 3.1a) M1(AO 1.1) M1(AO 2.2a) A1(AO 1.1) M1(AO 1.1) M1(AO 1.1) A1(AO 1.1) [7]	oe; equation with zero on one side Need not include limits Need not include limits FT <i>their</i> (a), or working in terms of <i>a</i>, for all M marks throughout this part
			Total	13	

Question			Answer/Indicative content	Marks	Guidance
34		a	$-\frac{1}{x^2} \cos\left(\frac{1}{x}\right)$	B2 (AO 1.1a, 1.1) [1]	B1 for $-\frac{1}{x^2}$ or $\cos\left(\frac{1}{x}\right)$
		b	$\frac{1}{x} > 0$ $\frac{1}{x} < \frac{\pi}{2}$	B1 (AO 2.2a) B1 (AO 2.2a) [2]	Correct lower limit 0 stated Correct upper limit $\frac{\pi}{2}$ stated
		c	$0 < \frac{1}{x} < \frac{\pi}{2} \Rightarrow \cos\left(\frac{1}{x}\right) > 0$ $-\frac{1}{x^2} < 0$ Hence $h'(x) < 0$ so $h(x)$ is a decreasing function	M1 (AO 2.2a) M1 (AO 1.1) A1 (AO 2.4) [3]	AG Convincing completion needed
			Total	7	

Question			Answer/Indicative content	Marks	Guidance
35		a	DR $\frac{dy}{dx} = 1 + \ln x$ At stationary point $\ln x = -1$ $x = \frac{1}{e}$ $y = -\frac{1}{e}$	B1(AO 1.1a) B1(AO 1.1) M1(AO 1.1) A1(AO 2.2a) A1(AO 2.2a) [5]	B1 for each term correct $\left(\frac{1}{e}, -\frac{1}{e}\right)$
		b	DR $\frac{d^2y}{dx^2} = \frac{1}{x}$ At stationary point $\frac{d^2y}{dx^2} = e$ positive so minimum	M1(AO 1.1a) E1(AO 2.4) [2]	AG Convincing conclusion needed
			Total	7	

Question			Answer/Indicative content	Marks	Guidance	
36			$u = 2x - 1 \Rightarrow \frac{du}{dx} = 2$ $\int_5^{15} \frac{u+1}{2u} du$ $\frac{1}{2} \int_5^{15} \left(1 + \frac{1}{u}\right) du$ $\frac{1}{2} \left[u + \ln u \right]_5^{15}$ $\frac{1}{2} \{15 + \ln 15 - (5 + \ln 5)\} = \frac{1}{2} \{10 + \ln(\frac{15}{5})\}$ $= 5 + \frac{1}{2} \ln 3$ Alternative solution $\int_3^8 \frac{2x}{2x-1} dx = \int_3^8 \frac{2x-1+1}{2x-1} dx$ $\int_3^8 \left(1 + \frac{1}{2x-1}\right) dx$ $\left[x + \frac{1}{2} \ln 2x-1 \right]_3^8$ $8 + \frac{1}{2} \ln 15 - \left(3 + \frac{1}{2} \ln 5\right)$ $5 + \frac{1}{2} \ln(\frac{15}{5}) = 5 + \frac{1}{2} \ln 3$	M1(AO 1.1a) A1(AO 1.1) B1(AO 1.1) M1(AO 1.1) A1(AO 1.1) E1(AO 2.1) M1 A1 B1 B1 M1 E1 [6]	oe Integrand correct Limits correct (may be seen later or may later convert back to x instead) Dividing through by u Integration correct, condone missing modulus sign and missing factor half AG Convincing correct completion oe attempt at algebraic division x and $\ln(2x-1)$ Factor $\frac{1}{2}$ Correct substitution of limits AG Convincing correct completion	Limits may be wrong or missing for these two marks Limits may be wrong or missing for these two marks Modulus not needed
			Total	6		

Question			Answer/Indicative content	Marks	Guidance
37		a	$\ln a = b$ Area B: $\int_0^b e^y dy = e^b - e^0$ $= a - 1$ Area A: $\int_1^a \ln x dx = [x \ln x - x]_1^a$ $= a \ln a - a + 1$ Their $a - 1 =$ their $a \ln a - a + 1$ $a \ln a - 2a + 2 = 0$	B1 (AO 1.1) M1(AO 3.1a) A1(AO 1.1) M1(AO 2.1) A1(AO 1.1) M1(AO 3.1a) A1(AO 1.1) [7]	or $a \times b$ – their $a \ln a - a + 1$
		b	Evaluation of their $f(4.921554 - \delta)$ and their $f(4.921554 + \delta)$ eg -0.000000079882 and 0.000000513742 seen to 2 or more sf plus correct conclusion: sign change, so Heidi is correct	M1 (AO 2.1) A1(AO 2.2a) [2]	$\delta \leq 0.000\ 000\ 5$
			Total	9	
38			DR $\frac{dy}{dx} = \frac{\cos x \times \frac{1}{x+2} - \ln(x+2)(-\sin x)}{(\cos x)^2}$ $x = 0 \ y = \ln 2$ $x = 0, \frac{dy}{dx} = \frac{1}{2}$ So equation of tangent is $y = \frac{1}{2}x + \ln 2$	M1(AO 3.1a) A1(AO 1.1) B1(AO 1.1) M1(AO 1.1) A1(AO 1.1) [5]	Use of quotient rule
			Total	5	

Question			Answer/Indicative content	Marks	Guidance	
40			$\int \frac{5}{y^2 - y - 6} dy = \int \frac{1}{x} dx$ $\frac{5}{y^2 - y - 6} = \frac{A}{y - 3} + \frac{B}{y + 2}$ $A = 1, B = -1$ $\int \left(\frac{1}{y - 3} - \frac{1}{y + 2} \right) dy = \int \frac{1}{x} dx$ $\Rightarrow \ln(y - 3) - \ln(y + 2) = \ln x + c$ When $x = 1, y = 8$ so $\ln 5 - \ln 10 = \ln 1 + c$ $c = \ln \frac{1}{2}$ $\ln(y - 3) - \ln(y + 2) = \ln x + \ln \frac{1}{2}$ So $\ln \left(\frac{y - 3}{y + 2} \right) = \ln \left(\frac{x}{2} \right) \Rightarrow \frac{y - 3}{y + 2} = \frac{x}{2}$ $2y - 3 = xy + 2x \quad y(2 - x) = 6 + 2x$ $\Rightarrow y = \frac{6 + 2x}{2 - x}$	M1 (AO 3.1a) M1 (AO 3.1a) A1 (AO 1.1b) M1 (AO 1.1a) A1 (AO 1.1b) M1 (AO 1.1b) A1 (AO 1.1b) M1 (AO 3.1a) M1 (AO 1.1a) A1 (AO 1.1b) [10]	Attempt to separate variables Attempt to use partial fractions All correct Integrating, obtaining natural logs FT their partial fractions Use of initial conditions to find c oe Attempt to remove logs from their eqn Attempt to make y the subject oe, but must be $y = f(x)$ form	With their factors Of their equation
			Total	10		

Question			Answer/Indicative content	Marks	Guidance
41			$\frac{dy}{dx} = 5x(-2e^{-2x}) + 5e^{-2x}$ $e^{-2x}(-10x + 5) = 0$ $x = \frac{1}{2}$ $\text{Area} = \int_0^{\frac{1}{2}} 5xe^{-2x} dx$ $= \left[5x \left(-\frac{1}{2} e^{-2x} \right) \right]_0^{\frac{1}{2}} - \int_0^{\frac{1}{2}} \left(-\frac{5}{2} e^{-2x} \right) dx$ $= -\frac{5}{4} e^{-1} + \left[-\frac{5}{4} e^{-2x} \right]_0^{\frac{1}{2}}$ $= -\frac{5}{2} e^{-1} + \frac{5}{4}$	M1 (AO 3.1a) A1 (AO 1.1) M1 (AO 1.1a) A1 (AO 1.1) M1 (AO 3.1a) M1 (AO 1.1a) A1 (AO 1.1) A1 (AO 1.1) A1 (AO 1.1) A1 (AO 1.1) [9]	For use of product rule For correct (unsimplified) derivative For equating derivative to zero and attempting to solve for x oe Limits 0 and (their) $\frac{1}{2}$ must be seen at some stage for this mark to be awarded For integration by parts with $u = 5x$ and $v = e^{-2x}$ First stage all correct; ignore limits for this mark Integration completed correctly oe, but must be exact simplified form Not e.g. 0.3303...
Total				9	

Question			Answer/Indicative content	Marks	Guidance	
42		a	Using $\sin \theta \approx \theta$ with $\theta = 2x^2$ and $\cos \theta \approx 1 - \frac{1}{2}\theta^2$ with $\theta = 5x$ gives $2x^2 - \left(1 - \frac{1}{2}(5x)^2\right) = 0$ $\frac{29}{2}x^2 = 1 \Rightarrow x = \sqrt{\frac{2}{29}} = 0.2626K$ So estimate is 0.26 to 2 decimal places	M1 (AO 2.1) M1 (AO 2.1) A1 (AO 2.1) [3]	Allow slip in $(5x)^2$ Attempt to solve for x AG Must be rounded to 2 dp following either exact answer or answer to more than 2 dp seen	
		b	Denoting $\sin 2x^2 - \cos 5x$ by $f(x)$: $f(0.255) = -0.1618...$ and $f(0.265) = -0.1033...$ These are both negative so the root does not lie between 0.255 and 0.265, so the estimate is not correct to 2 decimal places	M1 (AO 2.1) A1 (AO 2.2a) [2]	Search for sign change using their 2dp value ± 0.005 Complete argument from correct figs	oe, e.g. comparing values of $\sin 2x^2$ and $\cos 5x$ at end-points
		c	$f'(x) = 4x \cos 2x^2 + 5 \sin 5x$ $x_1 = 0.2 - \frac{\sin(2 \times 0.2^2) - \cos(5 \times 0.2)}{4 \times 0.2 \cos(2 \times 0.2^2) + 5 \sin(5 \times 0.2)}$ $= 0.2 - \frac{-0.4603876119}{5.004796289} = 0.29198928K$ so $x_1 = 0.2919893$ correct to 7 dp	M1 (AO 1.1a) M1 (AO 1.1a) A1 (AO 1.1) M1 (AO 2.1) A1 (AO 2.1) [5]	Attempt to find $f'(x)$ Use of chain rule for $\sin 2x^2$ Derivative fully correct Use of iterative formula to find x_1 AG Answer must follow from clear working	

Question			Answer/Indicative content	Marks	Guidance
		d	$x_2 = 0.282\ 338\ 3$ $x_3 = 0.282\ 285\ 3$ Root is 0.282 correct to 3sf	M1 (AO 1.1a) A1 (AO 1.1) B1 (AO 2.2b) [3]	Finds at least one more iteration For correct x_3
			Total	13	

Question			Answer/Indicative content	Marks	Guidance
43		a	$v = 0.1$ when $t = 1$ gives $a + b = 0.1$ $\frac{dv}{dt} = 4at^3 + 3bt^2$ Maximum v when $t = 1$ gives $4a + 3b = 0$ Solving simultaneous equation for a and b $a = -0.3$ and $b = 0.4$	M1 (AO 3.3) M1 (AO 3.1b) M1 (AO 3.3) M1 (AO 1.1a) A1 (AO 1.1) [5]	Using given information to find an equation linking a and b Equating the derivative to zero to find an equation linking a and b Method may be implied, e.g. if BC cao
		b	Changes direction when $v = 0$, so $-0.3t^4 + 0.4t^3 = 0 \Rightarrow 0.3t^4 \Rightarrow 0.4t^3 \Rightarrow t = \frac{4}{3}$ (as $t > 0$) Particle changes direction when $t = \frac{4}{3}$	M1 (AO 3.4) A1 (AO 1.1) [2]	For equating v to 0 and solving for t (may be BC); ignore any inclusion of $t = 0$ at this point cao
		c	Model is not suitable for large values of t as the object's velocity would increase without limit	B1 (AO 3.5b) [1]	oe, e.g. 'velocity gets very large'
			Total	8	

Question			Answer/Indicative content	Marks	Guidance
44		a	$3x^2 + 3y^2 \frac{dy}{dx} = y + x \frac{dy}{dx}$ $\frac{dy}{dx}(3y^2 - x) = y - 3x^2$ $\frac{dy}{dx} = \frac{y - 3x^2}{3y^2 - x}$	M1 (AO 1.1a) B1 (AO 1.1a) M1 (AO 1.1a) A1 (AO 1.1) [4]	For attempted implicit differentiation Correct use of product rule on RHS Collecting terms and factorising Correct expression for $\frac{dy}{dx}$
		b	$x = 0$ and $y = 0$ gives $\frac{0}{0}$ which cannot be evaluated	B1 (AO 2.4) [1]	Or for stating that the curve goes through the origin twice, with different gradients
		c	Maximum occurs where $\frac{y - 3x^2}{3y^2 - x} = 0 \Rightarrow y - 3x^2 = 0$ So P lies on $y = 3x^2$ giving $k = 3$	M1 (AO 3.1a) A1 (AO 2.2a) [2]	For equating derivative to zero and attempting to rearrange FT provided it is the equation of a suitable parabola
			Total	7	

Question			Answer/Indicative content	Marks	Guidance	
45		a	The formula gives the gradient of the chord PQ	B1 (AO 2.4) [1]		
		b	Column F suggests that the limit as $h \rightarrow 0$ is 28, and this is the gradient of (the tangent to) the curve at P	B1 (AO 2.4) [1]	Comment must include reference to gradient and the idea of a limit	Not essential to mention that the limit is $f'(2)$
			Total	2		

Question			Answer/Indicative content	Marks	Guidance
46		a	$\int dV = \int \left(\frac{k}{\sqrt{t+1}} \right) dt$ $V = 2k(t+1)^{\frac{1}{2}} + c$ Substitution of both conditions $k = 200 \text{ or } c = -400$ $V = 400\sqrt{t+1} - 400 \text{ oe}$	M1 (AO3.1a) M1 (AO1.1) M1 (AO2.1) M1 (AO3.3) M1 (AO1.1) [5]	Separation of variables. Allow omission of integral signs and or dt or dV Allow one error eg omission of + c $0 = 2k + c$ $400 = 4k + c$ For either of these All correct
		b	$y = 5, V = 162.5$ $T = 0.9775(390625 \dots) \text{ BC}$	B1 (AO3.4) B1 (AO1.1) [2]	Rounded to 2, 3 or 4 sf, awrt 0.9775
		c	$\frac{dV}{dt} = \frac{dV}{dy} \times \frac{dy}{dt} \text{ used}$ $\frac{200}{\sqrt{1+0.9775\dots}} = (20 + 6 \times 5 - 0.3 \times 5^2) \frac{dy}{dt}$ $3.346(40522876 \dots)$	M1 (AO3.1a) M1 (AO2.1) A1 (AO1.1) [3]	Allow one slip in derivative Rounded to 2, 3 or 4 sf, awrt 3.35

Question			Answer/Indicative content	Marks	Guidance
		d	Either at $t = 3$, $\frac{dV}{dt} = 100\text{cm}^3\text{s}^{-1}$ or $\frac{dy}{dt} = 2\text{cms}^{-1}$ Therefore volume/height still increasing so the student is correct	B1 (AO3.4) E1 (AO3.5a) [2]	
			Total	12	
47		a	$\frac{du}{dx} = \frac{1}{2}x^{-\frac{1}{2}}$ $x = (u - 1)^2$ $\int \frac{(u-1)^2}{u} \times 2(u-1) du$ $2 \int \left(u^2 - 3u + 3 - \frac{1}{u} \right) du$ oe $2 \left[\frac{u^3}{3} - \frac{3u^2}{2} + 3u - \ln u \right]$ $\frac{2(1+\sqrt{x})^3}{3} - 3(1+\sqrt{x})^2 + 6(1+\sqrt{x}) - 2\ln(1+\sqrt{x})$ [+c]	B1 (AO1.1) M1 (AO2.1) M1 (AO3.1a) A1 (AO1.1) A1 (AO1.1) A1 (AO1.1) A1 (AO3.2a) [7]	allow sign error FT their x and their derivative must be in a form ready to integrate three terms correct all four terms correct allow full marks if + c omitted
		b	Evaluation of $F[1] - F[0]$ $\frac{5}{3} - \ln 4$	M1 (AO2.1) A1 (AO1.1) [2]	
			Total	9	

Question			Answer/Indicative content	Marks	Guidance
48		a	$f(2) = 16 - 2 = 14$. Since $f(x)$ passes through $(2, 14)$, $f^{-1}(x)$ must pass through $(14, 2)$	E1 (AO2.4) [1]	
		b	$f'(x) = 4x^3 - 1$ $f'(2) = 31$ $\frac{1}{31}$	B1 (AO2.1) M1 (AO1.1) A1 (AO1.1) [3]	Accept BC
			Total	4	
49		a	$\int \frac{1}{P} dP = \frac{1}{k} \int \frac{1}{Q} dQ$ $\ln P = \frac{1}{k} \ln Q + c$ $Q = AP^k$ [where A is a constant]	M1 (AO 1.1a) M1 (AO 1.1) A1 (AO 2.5) [3]	Separation of variables Integration oe; must include constant here
		b	$P_1 = 1.05P_0$ $Q_0 = AP_0^k$ and $Q_1 = A \times 1.05^k P_0^k$ Percentage change in Q is $100 \left(\frac{A \times 1.05^k P_0^k - AP_0^k}{AP_0^k} \right)$ $= 100(1.05^k - 1)$ and so is independent of original price	M1 (AO 3.1b) M1 (AO 3.4) E1 (AO 2.1) [3]	notation may vary, e.g. just $1.05P$ oe Convincing completion
			Total	6	

Question			Answer/Indicative content	Marks	Guidance	
50		a	Revenue R is $P \times Q$, so $R = mQ^2 + cQ$	B1 (AO 3.3) [1]		
		b	$\frac{dR}{dQ} = 2mQ + c$ $\frac{dR}{dQ} = 0 \text{ when } Q = -\frac{c}{2m}$ Maximum since $\frac{d^2R}{dQ^2} = 2m < 0$ as m is negative $\text{When } Q = -\frac{c}{2m}, P = -\frac{c}{2} + c = \frac{c}{2}$ $\text{So PED} = \frac{P}{mQ} = \frac{\frac{1}{2}c}{-\frac{1}{2}c} = -1$	M1 (AO 3.1b) M1 (AO 1.1) A1 (AO 2.4) M1 (AO 1.1) E1 (AO 2.1) [5]	Or attempt to complete the square oe identification of $Q = -\frac{c}{2m}$ oe justification for maximum AG Correct working needed	$m \left[\left(Q + \frac{c}{2m} \right)^2 - \left(\frac{c}{2m} \right)^2 \right]$
			Total	6		

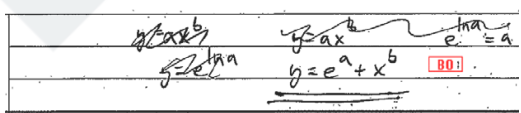
Question			Answer/Indicative content	Marks	Guidance	
51		a	x is not defined for this value	E1 (AO 2.4) [1]		
		b	$\cos 2\theta$ has max value 1 when $\theta = 0$ so max y is 2	E1 (AO 2.4) [1]	Or using $\frac{dy}{d\theta} = 0$ correctly	
		c	$y = 2 \cos^2 \theta$ $1 + x^2 = \sec^2 \theta$ So $1 + x^2 = \frac{2}{y} \Rightarrow y = \frac{2}{1 + x^2}$	M1 (AO 2.2a) M1 (AO 1.1a) E1 (AO 1.1) [3]	Relevant use of $\cos 2\theta = 2 \cos^2 \theta - 1$ Relevant use of $\sec 2\theta = 1 + \tan^2 \theta$ AG Successful completion	At any stage At any stage With use of $\sec \theta = \frac{1}{\cos \theta}$

Question			Answer/Indicative content	Marks	Guidance	
		d	DR $\frac{dy}{dx} = \frac{-4x}{(1+x^2)^2}$ Gradient OP = $\frac{y}{x}$ $\frac{-4x}{(1+x^2)^2} \times \frac{y}{x} = -1$ $\frac{4x}{(1+x^2)^2} \times \frac{2}{x(1+x^2)} = 1 \Rightarrow (1+x^2)^3 = 8 \Rightarrow 1+x^2 = 2$ $x = 1$ Point P is (1, 1)	M1 (AO 1.1) A1 (AO 1.1) B1 (AO 3.1a) M1 (AO 2.2a) M1 (AO 2.1) A1 (AO 2.2a) A1 (AO 3.2a) [7]	Use of chain rule or quotient rule Correct derivative May be implied Form and solve equation for x Allow ± 1 at this stage	Or, using parameters: $\frac{dy}{dx} = \frac{dy}{d\theta} \bigg/ \frac{dx}{d\theta} = \frac{-2 \sin 2\theta}{\sec^2 \theta}$ $\frac{-2 \sin 2\theta}{\sec^2 \theta} \frac{(1 + \cos 2\theta)}{\tan \theta} = -1$ $\frac{4 \sin \theta \cos^3 \theta}{1} \times \frac{2 \cos^3 \theta}{\sin \theta} = 1$ $\cos \theta = (\pm) \frac{1}{\sqrt{2}}$
			Total	12		

Question			Answer/Indicative content	Marks	Guidance
52			$\frac{du}{dx} = 1$ $\int (5u - 3)u^{\frac{1}{2}} du$ $\int \left(5u^{\frac{3}{2}} - 3u^{\frac{1}{2}} \right) du$ $2u^{\frac{5}{2}} - 2u^{\frac{3}{2}} [+c]$ $2(x+1)^{\frac{5}{2}} - 2(x+1)^{\frac{3}{2}} + c$ $\{2(x+1) - 2\}(x+1)^{\frac{3}{2}} + c$ $2x(x+1)^{\frac{3}{2}} + c$	B1 (AO 1.1a) M1 (AO 1.1) M1 (AO 1.1) M1 (AO 1.1) M1 (AO 1.1) M1 (AO 1.1) A1 (AO 2.1) [7]	Or $du = dx$ Complete substitution for x and dx Taking out factor $(x+1)^{\frac{3}{2}}$ Correct answer in correct form
			Total	7	

Question			Answer/Indicative content	Marks	Guidance	
53		a	$1 - 0.5^2$	B1 (AO 2.2a) [1]	oe	
		b	$\frac{0.5}{2} \{ (2.718282 + 0.049787) + 2(2.117 + 1 + 0.286505) \}$ $= 2.39$ (correct to 3 significant figures)	M1 (AO 1.1a) A1 (AO 1.1) [2]		
		c	$\frac{dy}{dx} = -2xe^{1-x^2}$ $\frac{dy}{dx} = 0$ only when $x = 0$, giving $y = e^{1-0} = e$	M1 (AO 1.1a) E1 (AO 2.1) [2]	AG Convincing completion	
			Total	5		

Question			Answer/Indicative content	Marks	Guidance	
54		a	DR $108 - 108 - 3 + 3 = 0$	B1 (AO 2.4) [1]	Sub. $x = 3$ and correct completion	
		b	DR $(x - 3)(4x^2 - 1)$ $(x - 3)(2x - 1)(2x + 1)$ $(3, 0), \left(\frac{1}{2}, 0\right), \left(-\frac{1}{2}, 0\right)$	M1 (AO 1.1a) M1 (AO 1.1) A1 (AO 1.1) A1 (AO 2.2a) [5]	For at least 2 correct x -values For all three correct points	
		c	DR $\int_{-\frac{1}{2}}^{\frac{1}{2}} (4x^3 - 12x^2 - x + 3) dx - \int_{\frac{1}{2}}^3 (4x^3 - 12x^2 - x + 3) dx$ $\left[x^4 - 4x^3 - \frac{1}{2}x^2 + 3x\right]_{-\frac{1}{2}}^{\frac{1}{2}} - \left[x^4 - 4x^3 - \frac{1}{2}x^2 + 3x\right]_{\frac{1}{2}}^3$ $\left(\frac{15}{16} - \frac{(-17)}{16}\right) - \left(-22\frac{1}{2} - \frac{15}{16}\right)$ $25\frac{7}{16}$	M1 (AO 3.1a) M1 (AO 1.1) A1 (AO 1.1) M1 (AO 1.1) A1 (AO 2.1) [5]	May be awarded at any stage for adding <i>their</i> two absolute areas Integration: at least one term correct All terms correct Use of <i>their</i> two pairs of limits awrt 25.4	Ignore limits (or absence of limits) for these marks
			Total	10		

Question			Answer/Indicative content	Marks	Guidance
55		a	$y = kx^n$	B1 (AO3.3) [1]	Allow any letters used for constant of proportionality and power. Examiner's Comments Many candidates seemed unprepared for this question which is covered by the specification point Ma14. Some tried to work backwards from the given answer in part (b) but many made fundamental errors in the laws of logarithms and gave an expression which was the sum of two terms. There was some evidence that candidates who had had $y = ax^b$ returned to their answer and changed it when they obtained $y = \ln a + b \ln x$ in part (b) which they did not recognise as being in the correct form. These two exemplars illustrate a successful and an unsuccessful attempt to use the given answer. Exemplar 4  
		b	$\ln y = \ln (kx^n)$ $\ln y = \ln k + \ln (x^n) = \ln k + n \ln x$ SCHEME FOR CANDIDATES USING LOG BASE 10	M1 (AO2.1) E1 (AO2.1) [2] SC1	Taking natural logs of both sides and one correct use of laws of logs used. Convincing argument. AG Correct use of log instead of $\ln$ and no other error



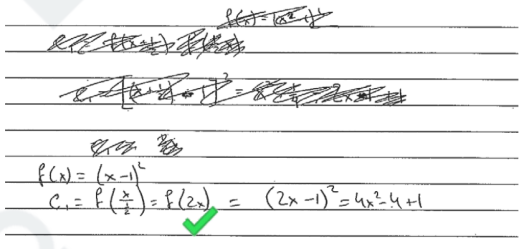
Question	Answer/Indicative content	Marks	Guidance
			Examiner's Comments The method mark here was credited to candidates who took logarithms on both sides and demonstrated at least one correct use of the laws of logarithms, usually for correctly dealing with $\ln x^b$ or similar. Candidates who had the sum of terms in part (a) were not credited the mark for adding the logarithms of separate terms. Candidates who did not know how to answer part (a) could access many of the subsequent marks by working from the equation given in here in part (b). There was some evidence of candidates giving up on all of question 11 without realising that there were many marks accessible. Prepare candidates to work through questions working from a given result even if they are unable to show where that result came from. Exemplar 5 $\ln y = \ln e^a + \ln x^b$ $\ln y = a \ln e + b \ln x$ $\ln y = a + b \ln x$ $\ln y = \ln(e^a x^b)$ $\ln y = \ln(e^a) + \ln x^b$ $\ln y =$ $\ln e^a = a \text{ so } \ln y = a + b \ln x$

Question			Answer/Indicative content	Marks	Guidance																		
		c	<table><tr><td></td><td>lnx</td><td>lny</td></tr><tr><td>Mercury</td><td>−1.179</td><td>9.575</td></tr><tr><td>Jupiter</td><td>1.599</td><td>4.022</td></tr></table> SCHEME FOR CANDIDATES USING LOG BASE 10 <table><tr><td></td><td>lnx</td><td>lny</td></tr><tr><td>Mercury</td><td>−0.5122</td><td>4.158</td></tr><tr><td>Jupiter</td><td>0.6946</td><td>1.747</td></tr></table>		lnx	lny	Mercury	−1.179	9.575	Jupiter	1.599	4.022		lnx	lny	Mercury	−0.5122	4.158	Jupiter	0.6946	1.747	 B1 (AO1.1b) B1 (AO1.1b) [2] B0 B1	 At least 2 correct values All correct and to 4sf <
	lnx	lny																					
Mercury	−1.179	9.575																					
Jupiter	1.599	4.022																					
	lnx	lny																					
Mercury	−0.5122	4.158																					
Jupiter	0.6946	1.747																					

Question			Answer/Indicative content	Marks	Guidance	
		d	EITHER $b = \frac{9.575 - 4.022}{-1.179 - 1.599} = -1.999 \text{ (-2.00 to 3sf)}$ $a = 7.218$ OR $9.575 = a - 1.179b$ $4.022 = a + 1.599b$ Giving $a = 7.218$ $b = -1.999 \text{ (-2.00 to 3sf)}$ SCHEME FOR CANDIDATES USING LOG BASE 10 As in main scheme $a = 3.1336$ $b = -1.999$	M1 (AO1.1a) A1 (AO3.1a) A1 (AO1.1b) [3] M1 A1 A1 [3] M1 A1 A1	using gradient formula Allow -2 a correct to at least 2 sf Setting up pair of equations by substitution of their values. Allow one slip. a correct to at least 2 sf b correct to at least 2 sf Examiner's Comments Candidates using the gradient method were often successful here. Many candidates using the alternative method were daunted by the complexity of the arithmetic here and made errors in the solution of their simultaneous equations.	These values could be found using the calculator STATS mode, so allow without working Simultaneous equations can be solved using calculator

Question			Answer/Indicative content	Marks	Guidance
		e	$y = 1363x^{-2.00}$ SCHEME FOR CANDIDATES USING LOG BASE 10 $y = 1360x^{-2}$	B1 (AO2.2a) B1 (AO3.3) [2] B1 B1	FT their equation in (a) awrt 1300 or 1400, or $e^{7.2}$ or better. FT their a Allow for x^{-2} or better. FT their b FT their equation in (a) awrt 1300 or 1400, or $e^{3.1}$ (= 22.2), $10^{3.1}$ or better. FT their a allow for x^{-2} or better. FT their b <u>Examiner's Comments</u> Many candidates were hampered by incorrect work in parts (a) and (b) and so did not realise that e^a was required. It was possible to obtain both marks here using their values in the given equation in part (b) but this was rarely seen. Follow-through marks were given for using e^a and b in whatever equation had been seen in part (a).

Question			Answer/Indicative content	Marks	Guidance
		f		B1 (AO1.2) B1 (AO1.1b) [2]	Appropriate curve with at least one horizontal asymptote or vertical asymptote shown Both asymptotes correct Ignore $x > 0$ if shown FT their equation in (e) provided their function is a decreasing function Examiner's Comments When attempted, many candidates correctly drew a graph with the positive x-axis as an asymptote but many thought their curve would cross the vertical axis at their a. As two points on the curve were given in the question, no marks were given for an increasing function.
		g	Earth $x = 1$, $y = 1363 \times 1^{-2} = 1360 \text{ W m}^{-2}$ (3sf) SCHEME FOR CANDIDATES USING LOG BASE 10 Earth $x = 1$, $y = 1363 \times 1^{-2} = 1360 \text{ W m}^{-2}$ (3sf)	B1 (AO3.4) [1] B1	FT their (e) FT their (e) Examiner's Comments This was an easy mark to obtain for candidates who substituted $x = 1$ into their equation even when the answer made little sense in the context.
			Total	13	

Question			Answer/Indicative content	Marks	Guidance
56		a	Using $y = f\left(\frac{x}{a}\right) \quad y = \left(\frac{x}{\frac{1}{2}} - 1\right)^2 = (2x - 1)^2$ $= 4x^2 - 4x + 1$	M1 (AO1.1a) A1 (AO2.1) [2]	Allow for 2 instead of $\frac{1}{2}$ used for method mark or attempt to write equation of quadratic that touches axis at (0.5, 0) AG Must be a convincing argument that references either stretch or $f(2x)$ or similar ($2x - 1$)² seen is sufficient for M1 <u>Examiner's Comments</u> Many candidates were credited one of two marks for this question as ($2x - 1$)² was seen. The best answers used function notation to explain the effect of the stretch in the x-direction. The following exemplar shows a candidate who is not sure what algebra is needed to achieve the given transformation. The given answer is used to identify the correct method and the notation makes it very clear that the full argument is given. Exemplar 3 
		b	EITHER C_2 is $y = 4.25x - x^2 - 3$ Normal to $y = 4x^2 - 4x + 1$ $\frac{dy}{dx} = 8x - 4$	B1 (AO3.1a) M1 (AO1.1a)	Finding the equation of C_2. Any form Finding the derivative

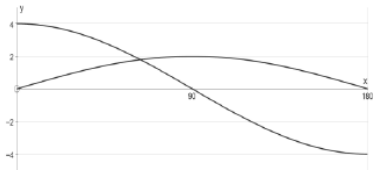
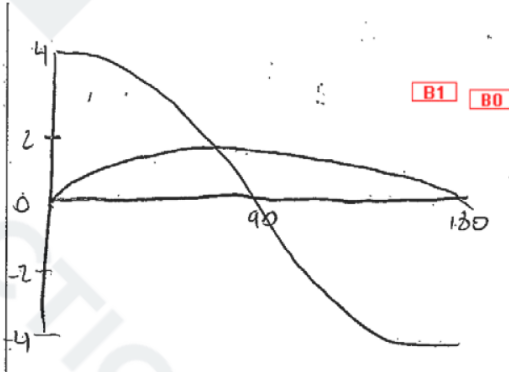
Question			Answer/Indicative content	Marks	Guidance	
			At (0.1) $\frac{dy}{dx} = -4$ Gradient of normal is $\frac{1}{4}$ (0, 1) on line so equation of normal is $y = \frac{1}{4}x + 1$ Intersection of normal and C_2 $\frac{1}{4}x + 1 = 4.25x - x^2 - 3$ $4x^2 - 16x + 16 = 0$ EITHER $(x - 2)^2 = 0$ OR discriminant $16^2 - 4 \times 4 \times 16 = 0$ Repeated root so the normal is a tangent to C_2 OR C_2 is $y = 4.25x - x^2 - 3$ Normal to $y = 4x^2 - 4x + 1$ $\frac{dy}{dx} = 8x - 4$ At (0.1) $\frac{dy}{dx} = -4$	M1 (AO1.1b) A1 (AO1.1a) M1 (AO3.1a) A1 (AO1.1b) E1 (AO3.2a) [7] B1 M1	Finding negative reciprocal of their gradient FT their value for derivative Attempt to solve simultaneous equations Repeated factor or root, or zero discriminant seen. Must interpret their solution in the context. Finding the equation of C_2. Any form Finding the derivative	

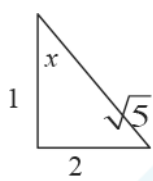
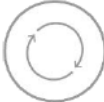
Question			Answer/Indicative content	Marks	Guidance	
			Gradient of normal is $\frac{1}{4}$ Equation of normal is $y = \frac{1}{4}x + 1$ Point on C_1 where gradient is $\frac{1}{4}$ $\frac{dy}{dx} = 4.25 - 2x = \frac{1}{4}$ giving $x = 2$ $y = 1.5$ EITHER So the equation of the tangent is $y - \frac{3}{2} = \frac{1}{4}(x - 2)$ Which is the same equation as the normal to C_1 OR show that point (2, 1.5) lies on normal So the normal to C_1 is a tangent to C_2 SPECIAL CASE when the candidate tries to show that the normal to C_2 is a tangent to C_1 C_2 is $y = 4.25x - x^2 - 3$ Normal to $y = 4.25x - x^2 - 3$ $\frac{dy}{dx} = 4.25 - 2x$	M1 A1 M1 A1 E1 (E1) [7] B1 M1	Finding negative reciprocal of their gradient FT their value for derivative Attempting to find the point on C_1 where tangent parallel to the normal found. Both coordinates required Correct equation for the tangent in form that makes it clear it is the same line as the normal. Finding the equation of C_2. Any form Finding the derivative	

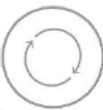
Question			Answer/Indicative content	Marks	Guidance	
			At (0, 1) $\frac{dy}{dx} = 4.25$ Gradient of normal is $-\frac{4}{17}$ Equation of normal is $y = -\frac{4}{17}x + 1$ EITHER point of intersection with C_1 $4x^2 - 4x + 1 = -\frac{4}{17}x + 1$ OR Attempt to find both coordinates of the point on C_1 with gradient $-\frac{4}{17}$ $\frac{dy}{dx} = 8x - 4 = -\frac{4}{17}$	A1 A0 M1 (M1)	Finding negative reciprocal of their gradient Attempt to solve simultaneous equations Attempting to find the point on C_1 where tangent parallel to the normal found. No further marks are available 4/7 maximum <u>Examiner's Comments</u> There were many good solutions to this question. Some candidates correctly obtained the equation of the normal and the equation of C_2 but then did not know how to proceed. Some incorrectly assumed that the point (0, 1) was a point of intersection of the two curves. Some candidates found the point on C_2 which had the correct gradient but then did not go on to show that the tangent here was the same line as the normal to C_1 and not simply parallel to it.	(0, 1) does not lie on C_2
			Total	9		

Question	Answer/Indicative content	Marks	Guidance
57	EITHER Equation of the form $y = k(x + 1)(x - 2)$ (0, -4) on curve so $k = 2$ OR Equation of the form $y = ax^2 + bx + c$ (0, -4) on curve $c = 4$ (-1, 0) on the curve $0 = a - b - 4$ (2, 0) on the curve $0 = 4a - 2b - 4$ Solving simultaneous equations $a = 2, b = -2$ BOTH Area = $\int_{-1}^2 (2x^2 - 2x - 4) dx$ $\left[\frac{2x^3}{3} - x^2 - 4x \right]_{-1}^2$ $\left(\frac{2 \times 2^3}{3} - 2^2 - 4 \times 2 \right) - \left(\frac{2 \times (-1)^3}{3} - (-1)^2 - 4 \times (-1) \right)$ $= -\frac{20}{3} - \frac{7}{3} = -9$	M1 (AO1.1a) M1 (AO3.1a) A1 (AO1.1b) (M1) (M1) (A1) M1 (AO1.1a) A1 (AO1.1b) M1 (AO1.1a) A1 (AO2.1)	DR Allow with $k = 1$ and without $y =$ Attempt to find $k \neq 1$ All correct Uses one point to form an equation Uses both other points and attempts to solve simultaneous equations All correct Integration – allow without limits – condone one error FT their quadratic Substitution of limits clearly seen Complete argument leading to exact answer. Allow for 9 if there is an argument to explain the change of sign even if -9 not seen. Ignore = 0 if seen Allow for $c = -4$ seen

Question			Answer/Indicative content	Marks	Guidance
			Area is 9 below the x-axis.	E1 (AO2.4) [8]	Must give modulus and explain the change of sign. FT if their definite integral is negative. “Area must be positive” is not sufficient explanation. Examiner’s Comments There were many very good answers to this question but many lost the final mark as they did not explain why their area is given as positive when the definitive integral gives a negative value. Only a few candidates used their calculators to evaluate their definite integral but lost marks as this was a detailed reasoning question that required all the lines of working to be clear. Examiners needed to see the indefinite integral and the substitution of limits. Some candidates made their answers unnecessarily complicated by splitting the required area into two or more regions. Many candidates struggled to obtain the correct equation of the curve, either using $y = (x + 1)(x + 2)$ or $y = x^2 - 4$ but most of the rest of the marks in this question were obtained following through their equation if it was quadratic.  Make sure you do not write $-9 = 9$ without explaining the change of sign. Candidates needed to comment that the area is below the x-axis.  Do not abandon a long question if there is a problem with the first part. Use any vaguely sensible equation to demonstrate your ability to integrate and use limits – it is not enough to describe this process in words.
			Total	8	

Question	Answer/Indicative content	Marks	Guidance
58	a $y = 4\cos x$ $y = 2\sin x$ 	B1 (AO1.2) B1 (AO1.2) [2]	Correct shapes and relative sizes Graphs labelled with correct amplitudes clear Ignore any graph beyond the given interval Examiner's Comments The graphs were generally well done although many candidates lost a mark as they did not indicate which graph was which. Some just drew the basic graphs of $y = \sin x$ and $y = \cos x$ with an amplitude of 1. Only rarely did candidates draw the graph of $y = \cos 4x$. Some candidates who used their calculators to produce points to plot $y = 4\cos x$ gave too few points and drew a line from (0, 4) to (180, -4)  Make sure all graphs are fully labelled. Exemplar 1  B1 B0

Question			Answer/Indicative content	Marks	Guidance
		b	$4\cos x = 2\sin x$ $\tan x = 2$ $x = \arctan 2$ $4\cos x = 4 \times \frac{1}{\sqrt{5}} = \frac{4}{\sqrt{5}}$  Point of intersection is $\left(\arctan 2, \frac{4}{\sqrt{5}} \right)$	M1 (AO3.1a) A1 (AO1.1b) M1 (AO1.1a) A1 (AO1.1b) [4]	Forming equation leading to $\tan x = k$ oe Award for 63.4° for this first A mark Attempting to find exact value for $\cos x$ or $\sin x$ Must be a coordinate in this form <u>Examiner's Comments</u> Many candidates divided by $\cos x$ to obtain an equation for $\tan x$, but there were many who got mixed up and obtained $\tan x = \frac{1}{2}$ or even $\tan x = 0$. Most did not think to use Pythagoras' theorem to find the y coordinate of the point of intersection.  Where a value for $\tan x$ is known, draw a simple right angled triangle and using Pythagoras' theorem to give the values for $\sin x$ or $\cos x$. Do not allow for using 63.4°

Question			Answer/Indicative content	Marks	Guidance
		c	The student's argument is invalid as the period of $\tan x$ is 180° .	E1 (AO2.3) [1]	Must state or imply that the argument is invalid and give a correct reason Allow for "roots occur every 180°" oe Examiner's Comments Few candidates used the periodicity of $\tan x$ to answer this question but many correct answers explained that there would be another point of intersection for $180^\circ \leq x \leq 360^\circ$.  Make sure that you clearly state whether the argument is valid or not and make sure this does not contradict your evidence.
			Total	7	

Question			Answer/Indicative content	Marks	Guidance
59		a	$\frac{dx}{dt} = \frac{(1+t^3) - t(3t^2)}{(1+t^3)^2}$ $\frac{dx}{dt} = \frac{(1-2t^3)}{(1+t^3)^2} \text{ oe}$ $\frac{dy}{dt} = \frac{2t(1+t^3) - t^2(3t^2)}{(1+t^3)^2}$ $\frac{dy}{dx} = \frac{(2t - t^4)}{(1+t^3)^2} \text{ oe}$ $\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt} = \frac{2t - t^4}{1 - 2t^3}$ $t=1 \Rightarrow \frac{dy}{dx} = -1$	M1 (AO 1.1a) A1 (AO 1.1) A1 (AO 1.1) M1(AO 1.1a) A1(AO 2.1) [5]	DR Use of quotient rule – may be gained for $\frac{dy}{dt}$ if $\frac{dx}{dt}$ not seen(allow $\pm$) Substitution into <i>their</i> $\frac{dy}{dx}$ dep on earlier M1 Examiner's Comments This part saw many attempts at the correct process, but there were some mistakes seen in applying the quotient rule generally with confusion over signs. More difficulty was experienced by those who chose to use the product rule but did not appreciate the need to apply the chain rule to the differentiation of $(1+t^3)^{-1}$.  Afl Possibly the use of the product rule to prove the quotient rule leads some candidates to assume that that is the method to use to differentiate quotients. Candidates should know the difference and be able to apply both the product rule and

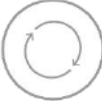
Question			Answer/Indicative content	Marks	Guidance
					the quotient rule accurately as appropriate.
		b	$\text{LHS} = \frac{t^3 + t^6}{(1+t^3)^3} \quad \text{oe}$ $= \frac{t^3(1+t^3)}{(1+t^3)^3}$ $\text{RHS} = \frac{t^3}{(1+t^3)^2} = \text{LHS}$	M1 (AO 1.1a) M1(AO 1.1) A1 (AO 2.1) [3]	AG Expression for LHS Factorising Completion of argument Examiner's Comments This part was generally started well, but many candidates struggled to complete the argument to score fully. The majority tackled this question by starting with finding an expression for $x^3 + y^3$ in terms of t, but a few chose to use the expression for x or y and make t the subject then continue to eliminate t from the expressions and generally complete the argument satisfactorily. Exemplar 4 $x = \frac{t}{1+t^3} \quad y = \frac{t^2}{1+t^3}$ $\frac{1}{y} = \frac{1+t^3}{t^2}$ $\frac{x}{y} = \frac{t}{1+t^3} \times \frac{1+t^3}{t^2}$ $= \frac{t}{t^2} = t^{-1} = \frac{1}{t}$ $t = \frac{y}{x}$ $x = \frac{y}{1 + \left(\frac{y}{x}\right)^3} = x \left(1 + \left(\frac{y}{x}\right)^3\right) = \frac{y}{x}$ $x + \frac{xy^3}{x^3} = \frac{y}{x}$ $x^4 + xy^3 = x^2y$ divide by x $x^3 + y^3 = xy$ $x^3 + y^3 = xy$

Question			Answer/Indicative content	Marks	Guidance
					This candidate does full proof of the result gaining all 3 marks. Exemplar 5 $\left(\frac{t}{1+t^3}\right)^3 + \left(\frac{t^2}{1+t^3}\right)^3 = \frac{t^3}{(1+t^3)^3} + \frac{t^6}{(1+t^3)^3}$ $= \frac{t^3 + t^6}{(1+t^3)^3} = \frac{t^3(t^3+1)}{(1+t^3)^3} = \frac{t^3}{(1+t^3)^2} = xy$ Because $\frac{t}{1+t^3} \cdot \frac{t^2}{1+t^3} = \frac{t^3}{(1+t^3)^2}$ $\frac{dx}{dt} = (1+t^3)^{-1} - 3t^3(1+t^3)^{-2}$ $\frac{dy}{dt} = 2t(1+t^3)^{-1} - 3t^4(1+t^3)^{-2}$ At $t=1$, $\frac{dx}{dt} = -\frac{1}{4}$ $\frac{dy}{dt} = \frac{1}{4}$ $\frac{dy}{dx} = \frac{1}{4} \div -\frac{1}{4} = -1$ This candidate shows appropriate initial working to verify the equation, but misses out on the final mark, as they do not state what they have shown.
			Total	8	

Question			Answer/Indicative content	Marks	Guidance
60		a	$\frac{dy}{dx} = 1 - \frac{1}{(x-2)^2}$ $1 - \frac{1}{(x-2)^2} = 0$ at stationary points $x - 2 = \pm 1$ so $x = 1, 3$ $(1, -5) (3, -1)$	M1(AO 1.1a) A1(AO 1.1) M1(AO 1.1a) A1(AO 2.2a) A1(AO 1.1) [5]	Attempt to differentiate with one term correct Correct derivative Both values of x Both values of y – ft <i>their</i> x <u>Examiner's Comments</u> Most candidates were able to score full marks here following correct differentiation and solution of what ended up as a quadratic equation. Many different ways were used to solve the equation usually without any wrong working. Examiners were pleased to see correct notation used in this question.

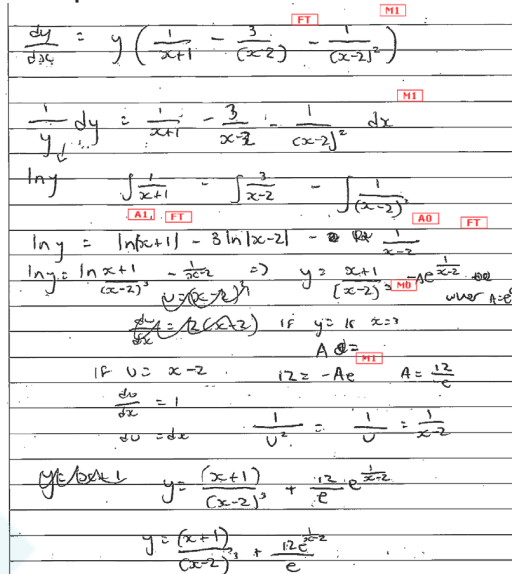
Question			Answer/Indicative content	Marks	Guidance																																																
		b	$\frac{d^2y}{dx^2} = \frac{2}{(x-2)^3}$ $x = 3 \quad \frac{d^2y}{dx^2} > 0 \quad (2) \text{ so minimum}$ $x = 1 \quad \frac{d^2y}{dx^2} < 0 \quad (-2) \text{ so maximum}$	M1(AO 1.1a) A1(AO 2.4) A1(AO 2.4) [3]	OR Allow consideration of gradient either side of stationary point for M1 Correct gradients above and below each tp A1 Correct convincing conclusions (possibly with sketches) A1 <table><tr><td>x</td><td>f(x)</td><td>x</td><td>f(x)</td></tr><tr><td>0.5</td><td>0.56</td><td>2.5</td><td>-3.00</td></tr><tr><td>0.6</td><td>0.49</td><td>2.6</td><td>-1.78</td></tr><tr><td>0.7</td><td>0.41</td><td>2.7</td><td>-1.04</td></tr><tr><td>0.8</td><td>0.31</td><td>2.8</td><td>-0.56</td></tr><tr><td>0.9</td><td>0.17</td><td>2.9</td><td>-0.23</td></tr><tr><td>1</td><td>0.00</td><td>3</td><td>0.00</td></tr><tr><td>1.1</td><td>-0.23</td><td>3.1</td><td>0.17</td></tr><tr><td>1.2</td><td>-0.56</td><td>3.2</td><td>0.31</td></tr><tr><td>1.3</td><td>-1.04</td><td>3.3</td><td>0.41</td></tr><tr><td>1.4</td><td>-1.78</td><td>3.4</td><td>0.49</td></tr><tr><td>1.5</td><td>-3.00</td><td>3.5</td><td>0.56</td></tr></table> Examiner's Comments Again most candidates were successful in classifying the stationary points with use of the second derivative being the most common method. A few considered the gradient either side of each turning point and then reasoned their way to a correct conclusion.	x	f(x)	x	f(x)	0.5	0.56	2.5	-3.00	0.6	0.49	2.6	-1.78	0.7	0.41	2.7	-1.04	0.8	0.31	2.8	-0.56	0.9	0.17	2.9	-0.23	1	0.00	3	0.00	1.1	-0.23	3.1	0.17	1.2	-0.56	3.2	0.31	1.3	-1.04	3.3	0.41	1.4	-1.78	3.4	0.49	1.5	-3.00	3.5	0.56
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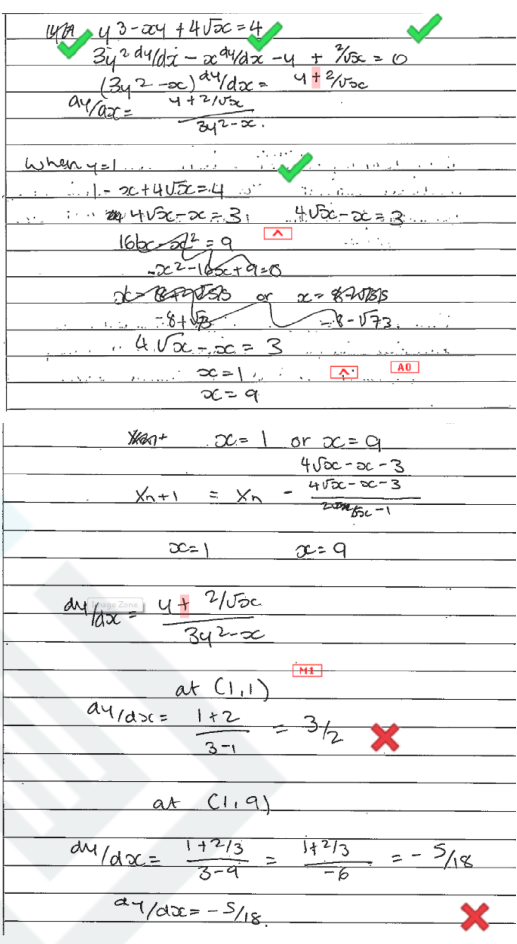
Question			Answer/Indicative content	Marks	Guidance
		c	$x = 2$	B1(AO 1.2) [1]	Examiner's Comments There appeared to be confusion as to the meaning of vertical asymptote which led to a low success rate for this part.
		d	$x > 2$	A1(AO 2.2a) [1]	FT <i>their</i> (c) if region is to right of <i>their</i> x value Examiner's Comments As in part (c), a large proportion of the candidates struggled with this part.  AfL The OCR B (MEI) H640 specification defines the terms “concave upwards” and “concave downwards” as those that will be used in examination questions.
			Total	10	

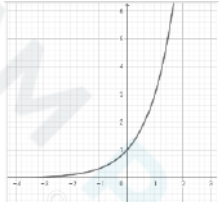
Question			Answer/Indicative content	Marks	Guidance
61			$y = (x - 1)^3 - 2(x - 1) - 4$ oe	B1(AO 1.1) B1(AO 1.1) [2]	Both factors $(x - 1)$ –4 <u>Examiner's Comments</u> Those candidates who knew to substitute $x - 1$ generally scored fully here.  AFL Note that the question did not require the answer to be expanded and some candidates wasted time doing this. Whilst this is an important skill, if required the question will specify a specific format for the answer.
			Total	2	

Question			Answer/Indicative content	Marks	Guidance	
62		a	$\frac{A}{(x+1)} + \frac{B}{(x-2)} + \frac{C}{(x-2)^2}$ $x^2 - 8x + 9 = A(x-2)^2 + B(x+1)(x-2) + C(x+1)$ $A = 2$ $B = -1$ $C = -1$	B1 (AO 3.1a) M1 (AO 2.1) A1 (AO 1.1) A1 (AO 1.1) A1 (AO 1.1) [5]	may be seen later $\frac{2}{(x+1)} - \frac{1}{(x-2)} - \frac{1}{(x-2)^2}$ <u>Examiner's Comments</u> Candidates who did well recognised the correct form of partial fractions and were able to work successfully to find the coefficients. Candidates who did less well made algebraic slips in clearing the fractions or made slips in arithmetic when finding the coefficients.	
		b	$\int \frac{dy}{y} = \int \frac{x^2 - 8x + 9}{(x+1)(x-2)^2} dx \text{ soi}$ use of their partial fractions in integration $\ln y = 2\ln x+1 - \ln x-2 + \frac{1}{x-2} + c$	M1* (AO 3.1a) M1* (AO 2.1) A1 (AO 1.1) A1 (AO 1.1)	allow omission of integral signs and/or omission of dy and/or dx allow one sign error and/or one coefficient error A1 for any correct natural log integral on RHS FT <i>their</i> $\frac{2}{x+1}$ or <i>their</i> $\frac{-1}{x-2}$ A1 for $\frac{1}{x-2}$ FT <i>their</i> $\frac{k}{(x-2)^2}$	condone use of brackets instead of modulus signs; these two A marks are only available following the award of both M marks

Question			Answer/Indicative content	Marks	Guidance	
			substitution of $y = 16$ and $x = 3$ correctly exponentiate both sides of their equation $y = \frac{(x+1)^2}{x-2} e^{\frac{3-x}{x-2}}$ oe	M1dep* (AO 1.1) M1 (AO 1.1) A1 (AO 2.1)	expression must include $+c$ and must include at least one natural log term; may be awarded after exponentiating eg $\frac{(x+1)^2}{x-2} e^{\frac{1}{x-2}} e^{-1}$	may be awarded following collection of like terms, which may contain errors NB $c = -1$
				[7]	Examiner's Comments Candidates who did well recognised the need to use their result from part (a). They separated the variables successfully and were then able to integrate and substitute the values of x and y to find the constant of integration. Candidates who did very well were able to go on and find a correct expression for y . Candidates who did less well rearranged incorrectly when they attempted to separate the variables, or were unable to integrate the quadratic term correctly. They made slips in exponentiating both sides of their equation, usually assuming that the operation is distributive.	

Question			Answer/Indicative content	Marks	Guidance
					Exemplar 4  In this response FT marks have been credited for the use of their partial fractions and separation of variables. One A mark has been credited FT, but the integration of the quadratic term went astray. The exponentiation of both sides was incorrect, but in spite of this, the method mark for substitution was subsequently earned.
			Total	12	

Question	Answer/Indicative content	Marks	Guidance
			Exemplar 3  This candidate differentiated correctly, but made a sign error when rearranging, which cost accuracy marks at the end. Note that detailed reasoning was required in this question. It was not clear how the (correct) values of x were obtained so A marks were withheld.
	Total	9	

Question			Answer/Indicative content	Marks	Guidance
64			$kx^{\frac{3}{2}}$ kx^{-2} $\frac{8}{3}x^{\frac{3}{2}}$ or $+3x^{-2}$ seen $\frac{8}{3}x^{\frac{3}{2}} + 3x^{-2} + c$ isw	M1 (AO 1.1) M1 (AO 1.1) A1 (AO 1.1) A1 (AO 1.1) [4]	Examiner's Comments Generally answered well, but the + c was often missed.
			Total	4	
65	a	(i)		B1 (AO 1.2) [1]	correct shape in both quadrants condone touching the x-axis, but not cutting it Examiner's Comments Sketches should make clear that the function tends towards $y = 0$ but does not cut the x-axis
		(ii)	(0, 1)	B1 (AO 1.1) [1]	do not allow just $y = 1$ Examiner's Comments A clear indication of the coordinates (0,1) was required and not just $y = 1$.
	b		$(f(x) =) \log_3 x$	B1 (AO 1.1) [1]	allow eg $\frac{\log x}{\log 3}$ Examiner's Comments There was an expectation that candidates would use base 3 logarithms, but a variety of correct functions were given and gained credit.
			Total	3	

Question			Answer/Indicative content	Marks	Guidance
66		a	$f'(x) = \frac{1}{3}(27 - 8x^3)^{-\frac{2}{3}} \times (-24x^2)$ $\left[= \frac{-8x^2}{(27 - 8x^3)^{\frac{2}{3}}} \right]$ $f'(1.5) = -\frac{8 \times 1.5^2}{0} \text{ and dividing by zero gives the error.}$	M1 (AO1.1a) A1(AO 1.1) E1 (AO2.4) [3]	Using the chain rule Allow unsimplified Sufficient to say "can't divide by zero" oe Examiner's Comments This was well answered as most candidates used the chain rule successfully and realised that substituting $x = 1.5$ gives a zero in the denominator
		b	$(27 - 8x^3)^{\frac{1}{3}} = 27^{\frac{1}{3}} \left(1 - \frac{8}{27}x^3 \right)^{\frac{1}{3}}$ $= 3 \left(1 + \left(\frac{1}{3} \right) \left(-\frac{8x^3}{27} \right) + \frac{\left(\frac{1}{3} \right) \left(-\frac{2}{3} \right)}{2!} \left(-\frac{8x^3}{27} \right)^2 + \dots \right)$ $= 3 - \frac{8x^3}{27} - \frac{64x^6}{2187} + \dots$	B1 (AO 3.1a) M1 (AO 1.1a) A1 (AO 1.1b) [1]	Dealing with the 27 correctly Using the Binomial expansion substantially correctly Cao Examiner's Comments Many candidates dealt successfully with the 27, but when that was done without clear working shown, could cost 2 marks here. Some candidates simplified their coefficients early and incorrectly, so it was not always clear that the binomial expansion had been used, costing the method mark.  AfL Make your method clear by writing down the factors of each term before simplifying.

Question			Answer/Indicative content	Marks	Guidance	
		c	The binomial expansion is valid for $\left -8 \frac{x^3}{27} \right < 1$ $x < 1.5$ and the limits of the integral are completely in this interval.	B1 (AO 2.4) E1 (AO 2.3) [2]	Allow unsimplified but must use correct modulus notation or equivalent Must indicate that the limits of the integral lie in their interval for which the expansion is valid. Examiner's Comments One of the assessment objectives in the specification is to test the ability of a candidate to assess the validity of an argument as in this question. Not many candidates realised that the key to this explanation was to find the range of values for which the binomial expansion is valid. The limits lie well within the valid range so the method is valid.	
		d	$\frac{0.25}{2} (3 + 2.6684 + 2(2.9954 + 2.9625 + 2.8694))$ $= \frac{0.25}{2} \times 23.3224 = 2.9153$	B1 (AO 1.1a) M1 (AO 1.1b) A1 (AO 1.1b) [3]	$h = 0.25$ used For sum in the bracket – condone one slip. Allow for 2.92 or better Examiner's Comments This was generally done very well.	Values from candidates own calculators may differ in the last decimal place.

Question			Answer/Indicative content	Marks	Guidance
		e	There is area between the curve and the top of the trapezia, so some area is missing from the estimate.	E1 (AO 2.4) [1]	Allow for any sensible explanation eg the trapezia are under the curve “The curve is concave downwards” on its own is not quite enough Examiner's Comments Most candidates were able to explain this clearly. Some had learned that a curve being concave downwards would give an underestimate but gave no indication as to why that would be, so lost the mark.  AfL Clear annotated sketches can support a mathematical explanation better than an extended written response.
			Total	12	

Question	Answer/Indicative content	Marks	Guidance
67	Curve crosses the x-axis when $y = 0$ $y = (k - x)\ln x = 0$ Either $k - x = 0$ or $\ln x = 0$ $x = k$ or 1 EITHER Area = $\int_1^k (k - x)\ln x \, dx$ Let $u = \ln x$, $\frac{dv}{dx} = k - x$, $\frac{du}{dx} = \frac{1}{x}$, $v = kx - \frac{1}{2}x^2$ Area = $\left[\left(kx - \frac{1}{2}x^2 \right) \ln x \right]_1^k - \int_1^k \frac{1}{x} \left(kx - \frac{1}{2}x^2 \right) dx$ $\left[\left(kx - \frac{1}{2}x^2 \right) \ln x \right]_1^k - \int_1^k \left(k - \frac{1}{2}x \right) dx$ $\left[\left(kx - \frac{1}{2}x^2 \right) \ln x - \left(kx - \frac{1}{4}x^2 \right) \right]_1^k$ $\left(\left(k^2 - \frac{1}{2}k^2 \right) \ln k - \left(k^2 - \frac{1}{4}k^2 \right) \right) - \left(\left(k - \frac{1}{2} \right) \ln 1 - \left(k - \frac{1}{4} \right) \right)$ $= \frac{1}{2}k^2 \ln k - \frac{3}{4}k^2 + k - \frac{1}{4}$ OR Integral split into two separate integrals $\int_1^k k \ln x \, dx$	M1 (AO 3.1a) A1 (AO 1.1b) M1 (AO 2.1) A1 (AO 1.1b) M1 (AO 3.1a) A1 (AO 1.1b) M1dep (AO 1.1a) A1 (AO 1.1b) [8]	Attempt to solve $y = 0$ Both roots required Using integration by parts with $u = \ln x$, $\frac{dv}{dx} = k - x$ clearly argued Allow without limits Simplifying the integrand Second part correct Using limits. Dependent on M mark for integration by parts Cao

Question			Answer/Indicative content	Marks	Guidance
			Let $u = \ln x$, $\frac{dv}{dx} = k$, $\frac{du}{dx} = \frac{1}{x}$, $v = kx$ $= [kx \ln x]_1^k - \int_1^k \frac{1}{x} kx \, dx$ $[kx \ln x]_1^k - \int_1^k k \, dx = [kx \ln x - kx]_1^k$ $(k^2 \ln k - k^2) - (k \ln 1 - k) = k^2 \ln k - k^2 + k$ And Area $= \int_1^k x \ln x \, dx$ Let $u = \ln x$, $\frac{dv}{dx} = x$, $\frac{du}{dx} = \frac{1}{x}$, $v = \frac{1}{2}x^2$ $= [\frac{1}{2}x^2 \ln x]_1^k - \int_1^k \frac{1}{x} \times \frac{1}{2}x^2 \, dx$ $[\frac{1}{2}x^2 \ln x]_1^k - \int_1^k \frac{1}{2}x \, dx$ $[\frac{1}{2}x^2 \ln x - \frac{1}{4}x^2]_1^k$ $(\frac{1}{2}k^2 \ln k - \frac{1}{4}k^2) - (\frac{1}{2} \ln 1 - \frac{1}{4}) = \frac{1}{2}k^2 \ln k - \frac{1}{4}k^2 + \frac{1}{4}$ Area $= (k^2 \ln k - k^2 + k) - (\frac{1}{2}k^2 \ln k - \frac{1}{4}k^2 + \frac{1}{4})$ $= \frac{1}{2}k^2 \ln k - \frac{3}{4}k^2 + k - \frac{1}{4}$	M1 M1 M1dep A1 A1 A1	Using integration by parts with $u = \ln x$, $\frac{dv}{dx} = k$ or $u = \ln x$, $\frac{dv}{dx} = \pm x$ clearly argued Simplifying the integrand Substitution of limits seen in at least one integral. Dependent on M mark for integration by parts Both integrals correct at this stage Allow without limits Both integrals fully correct Allow without limits Cao Examiner's Comments This is an exemplar of a question requiring an extended answer – there are 4 method marks in the scheme. Candidates had to structure their answer. Had a value for k been given, the definite integral would

Question			Answer/Indicative content	Marks	Guidance
					have been possible on many calculators and the question may have become a “detailed reasoning” question. Successful candidates generally used integration by parts with $\frac{dv}{dx} = k - x$. It needed much more work to expand the bracket and split the integral in two. Some candidates lost the final mark, as they did not tidy up their answer so had too many terms.  AfL In an unstructured question like this one, do not give up or leave it blank because you do not know how to calculate the limits. Find the indefinite integral to make sure of 4 out of 8 marks. Using incorrect limits could also have been credited a method mark.
			Total	8	

Question			Answer/Indicative content	Marks	Guidance	
68		a	EITHER $8\sin^2 x \cos^2 x = 2(1 - \cos 2x)(1 + \cos 2x)$ $= 2(1 - \cos^2 2x) = 2 - (1 + 2 \cos 4x)$ $= 1 - \cos 4x$ OR $8\sin^2 x \cos^2 x = 2(2\sin x \cos x)^2$ $= 2\sin^2 2x$ $[= 1 - \cos 2(2x)] = 1 - \cos 4x$ OR $1 - \cos 4x = 1 - (1 - 2\sin^2 2x)$ $= 2\sin^2 2x$ $= 2(2\sin x \cos x)^2$ $= 8\sin^2 x \cos^2 x$	M1 (AO 3.1a) M1 (AO 3.1a) E1 (AO 2.1) [3] M1 M1 E1 [3] M1 M1 E1 [3]	AG Using a double angle formula Second use of a double angle formula Clearly shown Using a double angle formula Another use of a double angle formula Clearly shown Using a double angle formula Another use of a double angle formula Clearly shown	Allow any other valid sequence of identities used. Examiner's Comments This question posed considerable difficulty to many candidates. Some had success by writing both $\sin^2 x$ and $\cos^2 x$ in terms of $\cos 2x$ and multiplying out. Many then did not go on to complete the proof. The first 2 marks were credited for any two applications of the double angle formulae and the final mark only when there was a convincing proof so many candidates were credited 2 out of 3 marks.

Question			Answer/Indicative content	Marks	Guidance
		b	$\int \sin^2 x \cos^2 x \, dx = \frac{1}{8} \int 1 - \cos 4x \, dx$ $= \frac{1}{8}x - \frac{1}{32}\sin 4x + c$	M1 (AO 1.1a) A1 (AO 1.1b) A1 (AO 1.1b) [3]	Attempt to integrate both terms $\frac{1}{4}\sin 4x$ seen or implied All correct. Must include +c <u>Examiner's Comments</u> Most candidates realised they needed to use the identity from part (a) although many omitted the factor of $\frac{1}{8}$. Some candidates lost the final mark for omitting the +c.
			Total	6	

Question			Answer/Indicative content	Marks	Guidance	
69			$F[x] = 3x + \frac{4x^{\frac{3}{2}}}{\frac{3}{2}}$ oe $F[9] - F[0]$ $27 + \frac{4 \times 2 \times 27}{3} = 99$	M1(AO2.1) A1(AO1.1) B1(AO1.1) B1(AO2.4) [4]	Attempt at integration; sight of (first term) kx or (second term) $kx^{\frac{3}{2}}$ Dep M1; ft their $F(x)$; Accept $27 + 72 = 99$ AG	Allow +c Must see convincing arithmetic for award of final mark
			Total	4	Examiner's Comments Most candidates were able to integrate the expression, though the square root did cause some problems. Quite a number of candidates seemed uncertain as to the point at which the integral sign with limits before the expression should be replaced with square brackets to show that the expression should now be evaluated. In order to gain full marks candidates were required to show that they had worked out $F(9)$ and $F(0)$ and had calculated $F(9) - F(0)$. They were also required to show that the value of 99 for $F(9)$ came from $27 + 72$. The question asked for the given result to be shown; writing $\left[3x + \frac{8}{3}x^{\frac{3}{2}}\right]_0^9 = 99 - 0 = 99$ does not show this sufficiently well to be given full marks.	

Question			Answer/Indicative content	Marks	Guidance
70		a	$\frac{dy}{dx} = 3x^2 + 2bx + c$ At stationary points $3x^2 + 2bx + c = 0$ No real roots if $(2b)^2 < 4 \times 3c$ hence if $3c < b^2$	B1(AO1.1a) B1(AO3.1a) B1(AO2.1) [3]	AG
		b	The gradient is never zero, hence is always positive or always negative so the graph must cross the x-axis in one place	E1(AO2.4) [1]	Convincing explanation
			Total	4	

Question			Answer/Indicative content	Marks	Guidance
71		a	DR At intersection $\cos x = \sin x$ $\tan x = 1$ $x = \frac{1}{4}\pi, -\frac{3}{4}\pi$ Between intersections $\cos x - \sin x > 0$, so area = $\int_{-\frac{3}{4}\pi}^{\frac{1}{4}\pi} (\cos x - \sin x) dx$	M1(AO3.1a) M1(AO1.1) A1(AO1.1) E1(AO2.4) [4]	Both values Explanation and completion
		b	DR $[\sin x + \cos x]_{-\frac{3}{4}\pi}^{\frac{1}{4}\pi}$ $\sin\left(\frac{1}{4}\pi\right) - \sin\left(-\frac{3}{4}\pi\right) + \cos\left(\frac{1}{4}\pi\right) - \cos\left(-\frac{3}{4}\pi\right) = \frac{4}{\sqrt{2}}$ $2\sqrt{2}$	M1(AO1.1a) M1(AO2.1) A1(AO1.1) [3]	Limits not needed for method mark
			Total	7	

Question			Answer/Indicative content	Marks	Guidance
72		a	$\frac{dx}{dt}$ is the rate at which x is increasing Mass of B is x , so mass of A is $(1 - x)$ $\frac{dx}{dt} \propto x(1 - x)$, so $\frac{dx}{dt} = kx(1 - x)$	B1(AO2.1) [1]	Must indicate where terms come from AG
		b	$\int \frac{1}{x(1-x)} dx = \int k dt$ $\frac{1}{x(1-x)} = \frac{A}{x} + \frac{B}{1-x}$ $1 \equiv A(1-x) + Bx \Rightarrow A = 1, B = 1$ $\int \left(\frac{1}{x} + \frac{1}{1-x} \right) dx = \int k dt \Rightarrow \ln x - \ln(1-x) = kt + c$ $t = 0, x = 0.2 \Rightarrow c = \ln \frac{1}{4}$ (oe) $\frac{4x}{1-x} = e^{kt}$ (oe) $x = \frac{e^{kt}}{4 + e^{kt}}$	M1(AO3.1a) M1(AO3.1a) A1(AO2.2a) M1*(AO1.1a) M1dep*(AOs3.1a) M1dep*(AO1.1b) A1(AOs2.5) [7]	Separation of variables Find partial fractions (may be implied) Condone sign error, but must have two $\ln$ terms and $+c$ Use of initial conditions; may be done after equation is rearranged Rearrange equation to remove logs; may be done before finding c oe, but must be of the form $x = f(t)$
		c	$t = 15, x = 0.9 \Rightarrow 36 = e^{15k}$ (oe) $k = 0.239$ to 3sf	M1(AO3.1) A1(AO1.1b) [2]	Substitute values in their solution

Question			Answer/Indicative content	Marks	Guidance
		d	$t = 30 \Rightarrow$ mass of B is $\frac{e^{0.239 \times 30}}{4 + e^{0.239 \times 30}} = 0.997 \text{ kg}$	B1(AO3.4) [1]	
		e	As $t \rightarrow \infty$, $x \rightarrow 1$ and so $1 - x \rightarrow 0$, so the model predicts there is a very small amount of A remaining when t is large	B1(AO3.5a) [1]	May evaluate x for large t (eg $t = 100$)
			Total	12	

Question			Answer/Indicative content	Marks	Guidance
73		a	At stationary point: $4x^3 - 9x^2 + 6x = 0$ $x(4x^2 - 9x + 6) = 0 \Rightarrow x = 0$ or $4x^2 - 9x + 6 = 0$ Discriminant of quadratic is $(-9)^2 - 4 \times 24 = -15 < 0$ So $x = 0$ gives the only stationary point $f(0) = 0^4 - 3 \times 0^3 + 3 \times 0^2 = 0$, so the point is the origin $f''(x) = 12x^2 - 18x + 6 \Rightarrow f''(0) = 6 > 0$ So the origin is a minimum point	M1(AO1.1a) M1(AO1.1a) M1(AO1.1a) A1(AO2.2a) B1(AO2.1) M1(AO1.1a) A1(AO2.2a) [7]	Attempt to differentiate & equate to 0 Factorising the cubic oe (quadratic formula, completing the square, complex roots from calculator) $f(0)$ or y must be shown to be zero Or other valid method, e.g. sign of $f'(x)$ for $x < 0$ and $x > 0$ AG Must establish minimum clearly
		b	f is not a one-to-one function	B1(AO2.4) [1]	

Question			Answer/Indicative content	Marks	Guidance
		c	$g(0) = 0$ and $g(2) = 4$ Domain of $g^{-1}(x)$ is range of $g(x)$, so $0 \leq x \leq 4$ Range of $g^{-1}(x)$ is domain of $g(x)$, so $0 \leq g^{-1}(x) \leq 2$	B1(AO2. 2a) B1(AO1. 2) [2]	Must not be in terms of y Must not be an interval for x
			Total	10	
74			Use $x = e^u$ and $\frac{dx}{du} = e^u$ $\int (\ln x)^2 dx = \int (\ln e^u)^2 e^u du = \int u^2 e^u du$ Use integration by parts $= u^2 e^u - \int 2ue^u du$ $= u^2 e^u - 2(ue^u - \int u^2 e^u du)$ $= u^2 e^u - 2ue^u + 2e^u + c$ $= x(\ln x)^2 - 2x \ln x + 2x + c$	M1(AO1. 1b) A1(AO1. 1b) M1(AO3. 1a) A1(AO1. 1b) M1(AO1. 1a) A1(AO1. 1b) [6]	Using substitution including dx Simplifying correctly First use of integration by parts First stage all correct Second use of integration by parts Must be in terms of x for final mark
			Total	6	
75			Product rule with $u = 3x^2$ and $v = \sin 2x$ $\frac{dy}{dx} = 6x \sin 2x + 6x^2 \cos 2x$	M1(AO1. 1a) A1(AO1. 1b) A1(AO1. 1b) [3]	Need not be written explicitly For either of the two terms correct For completely correct answer
			Total	3	

Question			Answer/Indicative content	Marks	Guidance	
76			$\frac{\ln x \times 2 - 2x \times \frac{1}{x}}{(\ln x)^2}$ indicates $\frac{dy}{dx} = 0$ $x = e$ and $y = 2e$ shown www <i>either</i> finds second derivative <i>or</i> considers <i>their</i> gradient function either side of $x = e$ 2nd derivative is $\frac{4 - 2 \ln x}{x(\ln x)^3}$ $\frac{2}{e}$ (0.7357588...) > 0 so minimum or signs of gradient function either side of $x = e$ identified and explained	M1(AO3.1a) A1(AO1.1b) A1(AO1.1b) M1(AO2.1) A1(AO2.4) A1(AO3.1a) M1(AO2.1) A1(AO1.1) E1(AO2.4) [9]	Quotient Rule allow sign errors only numerator correct denominator correct AG Allow FT if previous M1 awarded, but must be correctly evaluated	or Product & Chain Rules $2 \times \frac{1}{\ln x} +$ $2x \times (-1) \times (\ln x)^{-2} \times \frac{1}{x}$ Allow decimal equivalent to 2 sf or better
			Total	9		

Question			Answer/Indicative content	Marks	Guidance
77			$-4x^2 \times \frac{\cos 2x}{2} - \int (-8x) \times \frac{\cos 2x}{2} dx$	M1(AO3.1a)	Integration by parts, allow sign errors only for M1
				A1(AO1.1b)	all correct
			$4x \times \frac{\sin 2x}{2} - \int 2 \sin 2x dx$	M1(AO2.1)	Integration by parts, allow sign errors only for M1
			$-2x^2 \cos 2x + 2x \sin 2x - \int 2 \sin 2x dx$	A1(AO1.1b)	
			$-2x^2 \cos 2x + 2x \sin 2x - \left(\frac{-2 \cos 2x}{2} \right) + c$	M1(AO2.1)	convincing attempt, allow sign error and/or omission of + c at this stage
			$-2x^2 \cos 2x + 2x \sin 2x + \cos 2x + c$	A1(AO1.1b)	all correct AG
				[6]	
Total				6	

Question			Answer/Indicative content	Marks	Guidance
78		a	$A = 22 \text{ ms}^{-1}$ $25.3 = 22 + B(1 - e^{-0.2 \times 2})$ $B = 10$	B1(AO1.1b) M1(AO3.3) A1(AO1.1b) [3]	from $t = 0$ NB from 10.0097...
		b	$22 + 10(1 - e^{-0.2 \times 5})$ 28.3(212...) so good fit	M1(AO3.4) A1(AO3.5a) [2]	
		c	$-10 \times (-0.2) \times e^{-0.2t}$ oe soi substitution of $t = 5$ in <i>their</i> derivative 0.74 ms^{-2}	M1*(AO3.1a) M1dep(AO1.1b) A1(AO1.1b) [3]	0.73575888...to 2 sf or better

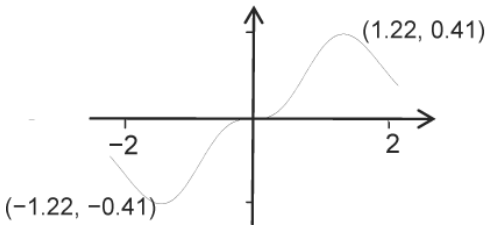
Question			Answer/Indicative content	Marks	Guidance
		d	as $t \rightarrow \infty$ oe $V \rightarrow 32$ $32 \times 3.6 = 115.2 \text{ kmh}^{-1}$ $115.2 < 121$ so not 10% above the speed limit	M1(AO3.5a) A1(AO3.4) B1(AO1.1b) E1(AO3.2a) [4]	FT their long term V or 5.2 is 4.7% of 110
			Total	3	

Question			Answer/Indicative content	Marks	Guidance
79		i	$\frac{dP}{dt} = k\sqrt{P}$	B1 [1]	oe – condone $dP = k\sqrt{P}dt$ Examiner's Comments In part(i) most candidates correctly wrote down the differential equation relating P, the time t, and the constant k. The most common errors in this part were those candidates who wrote $\frac{dt}{dP} = k\sqrt{P}$ or $\frac{dP}{dt} = kP^2.$

Question			Answer/Indicative content	Marks	Guidance
		ii	$P = (A + Bt)^2 \Rightarrow \frac{dP}{dt} = 2(A + Bt)B$ $\frac{dP}{dt} = 2B\sqrt{P}$ $k = 2B$ OR $\int \frac{dP}{\sqrt{P}} = k \int dt \Rightarrow 2\sqrt{P} = kt(+c)$ $P = \left(\frac{1}{2}kt + A\right)^2$ $k = 2B$	M1 A1 A1 [3] M1 A1 A1	Attempt at chain rule – allow this mark if B absent or incorrect but must include $2(A + Bt)$ – if multiplied out condone one error in differentiation oe – correctly showing that the rate of increase of P is proportional to the square root of P Not for $B = \frac{k}{2}$ Separates their variables correctly and attempt to integrate for their differential equation given in (i) – for an attempt powers must increase by 1 (oe) but not for k – condone lack of $+c$ Correct integration – accept any (correct) constant for A – coefficient of t maybe implicit stated e.g. $P = (Ct + A)^2$ with $k = 2C$ seen Not for $B = \dots$ Examiner's Comments Even though in part (ii) the question asked for candidates to verify that $P = (A + Bt)^2$ was a solution to the differential equation many decided instead to solve the differential equation by separating the variables and integrating. It was disappointing how many candidates gave the final answer in this part as $B = \frac{k}{2}$ even though the question specifically asked for k in terms of B.
			Total	4	

Question			Answer/Indicative content	Marks	Guidance	
80		i	$f(-x) = (-x)^3 e^{-(-x)^2}$ $= -x^3 e^{-x^2} = -f(x)$ Rotational symmetry of order two about the origin.	M1 A1 B1 [3]	substituting $-x$ for x in $f(x)$ must have $f(-x) = (-x)^3 e^{-(-x)^2}$ for A1 or point or half-turn (180°) symmetry about O	at least once allow description of symmetry, e.g. 'fits its outline if rotated etc...'
Examiner's Comments Most candidates scored 2 or 3 here. We required to see $f(-x) = (-x)^3 \exp(-x)^2$ in the proof that $f(x)$ was an odd function, with the brackets correctly placed. For the 'B' mark describing the property of the graph, we needed to see reference to 'symmetry', 'half-turn, 180° or order 2', and 'about the origin'.						

Question			Answer/Indicative content	Marks	Guidance	
		ii	$f'(x) = 3x^2 e^{-x^2} + x^3 (-2x) e^{-x^2}$ $f'(x) = 0$ when $3x^2 e^{-x^2} - 2x^4 e^{-x^2} = 0$ $\Rightarrow 3x^2 = 2x^4$ $\Rightarrow x = 0, \sqrt{1.5}, -\sqrt{1.5}$ $y = 0, 0.41, -0.41$ So (0, 0), (1.22, 0.41), (-1.22, -0.41)	M1 A1* M1 M1 A1dep A2dep [7]	product rule correct expression their deriv = 0 taking out or dividing by e^{-x^2} dep A1* or $x = \pm\sqrt{1.5}$ o.e. dep A1*	consistent with their derivatives - condone deriv of e^{-x^2} is e^{-x^2} for M1 must be 2 terms must be 2 terms Allow SC A1 if both x-coords correct or one point correct (dep A1*)
						Examiner's Comments The main problem with the product rule here was to get the correct derivative of $\exp(-x^2)$. A common mistake was to think this is $\exp(-x^2)$. Having found the correct derivative and equated it to zero, the next issue was dividing through by, or factorising, $\exp(-x^2)$. After this, not many candidates got all three turning points, either omitting the origin or (-1.22, -0.41) or both. Also, evaluating the y-coordinates was sometimes done incorrectly. Where these issues were overcome, half of the candidates scored 6 or over; of these, half scored full marks.

Question			Answer/Indicative content	Marks	Guidance
		iii		M1 A1dep [2]	correct shape for $-2 \leq x \leq 2$ with 2 TPs, through O, reasonable half turn symmetry need not show stationary inflexion at O. ignore shape outside $-2 \leq x \leq 2$ coords of TPs and stationary inflexion at origin shown dep (0,0) given as a stationary point in part (ii) condone plotting beyond $[-2, 2]$ provided shape is correct Examiner's Comments Very few candidates scored both marks here. Many omitted the inflection at the origin, and the graphs were often lacking the point symmetry stated in part (i).
		iv	(A) let $t = x^2$, $dt/dx = 2x \Rightarrow xdx = \frac{1}{2} dt$ o.e. $\int x^3 e^{-x^2} [dx] = \int x^2 e^{-x^2} x [dx] = \int \frac{1}{2} t e^{-t} [dt]$	M1 A1 [2]	$k = \frac{1}{2}$ Examiner's Comments Half the candidates scored these two marks. Using a substitution in this context was perhaps unexpected.

Question			Answer/Indicative content	Marks	Guidance
		iv	(B) $\int_0^2 x^3 e^{-x^2} dx = k \int_0^4 t e^{-t} dt$ let $u = t$, $v = e^{-t}$, $u' = 1$, $v = -e^{-t}$ $= [k] \left\{ \left[t(-e^{-t}) \right]_0^4 - \int_0^4 (-e^{-t}) dt \right\}$ $= [k] \left\{ \left[-e^{-t} - t e^{-t} \right]_0^4 \right\}$ $= -\frac{1}{2} e^{-4} - 2e^{-4} + \frac{1}{2} = \frac{1}{2} - \frac{5}{2e^4}$	M1 A1 A1 A1cao [4]	correct parts on $\int t e^{-t} [dt]$ or $\int k t e^{-t} [dt]$ ignore limits, ft their k limits must be correct here, ft their k oe but must evaluate $e^0 = 1$ and combine e^{-4} terms ft their k, condone $v = e^{-t}$ Examiner's Comments They could get three out of the four marks with a missing, or incorrect, value for k, but not many succeeded with this.
			Total	18	

Question			Answer/Indicative content	Marks	Guidance	
81		i	$(\frac{\pi}{2}, 0), (0, \frac{1}{2})$	B1B1 [2]	or $y = 0 \Rightarrow x = \pi/2$; $x = 0 \Rightarrow y = \frac{1}{2}$ (isw) Ignore incorrect labelling Examiner's Comments Some candidates lost marks here from working in degrees rather than radians. It is important that candidates state both the coordinates the right way round, so 'A = $\pi/2$' and 'B = $\frac{1}{2}$' scored zero.	
		ii	$\frac{dy}{dx} = \frac{(2 - \sin x)(-\sin x) - \cos x(-\cos x)}{(2 - \sin x)^2}$ when $\frac{dy}{dx} = 0$ $(2 - \sin x)(-\sin x) - \cos x(-\cos x) = 0$ $\Rightarrow 1 - 2\sin x = 0$ $\Rightarrow x = \frac{\pi}{6}, y = \frac{\sqrt{3}}{3}$ o.e.	M1 A1 M1 M1 A1 A1 A1 [7]	correct quotient or product rule correct expression (isw) setting (only) their numerator to zero use of $\sin^2 x + \cos^2 x = 1$ must be exact, isw Examiner's Comments Over half got full marks here. The quotient rule was well answered, and the subsequent simplification using $\sin^2 x + \cos^2 x = 1$ was good. The most common error was in the sign of the derivatives of $\sin x$ and $\cos x$, which could fortuitously lead to the correct turning point – the final 'A' marks here being withheld in this case.	denom must be correct at some stage missing brackets may be inferred from subsequent work not denominator withhold if denom is set to zero

Question			Answer/Indicative content	Marks	Guidance
		iii	(A) $\int_0^{\frac{\pi}{2}} \frac{\cos x}{2 - \sin x} [dx] = [-\ln(2 - \sin x)]_0^{\frac{\pi}{2}}$ or let $u = 2 - \sin x$, $du/dx = -\cos x$ $= \int_2^1 -\frac{1}{u} du$ $= [-\ln u]_2^1$ $= \ln 2$	B1ft M1 A1 M1 A1 A1 A1 [4]	correct integral and limits $c \ln(2 - \sin x)$ $c = -1$ $\int -\frac{1}{u} [du]$ (ignore limits) or $\int_1^2 \frac{1}{u} [du]$ $[-\ln u]$ (ignore limits) or $[\ln u]_1^2$ $-\ln(1/2)$ is A0, isw after $\ln 2$ Examiner's Comments Most candidates used a substitution $u = 2 - \sin x$. Errors thereafter were $du/dx = \cos x$, or getting the limits the wrong way round, perhaps under the misconception that the larger number must be the upper limit of the integral). ft their $\pi/2$, not 90°, limits may be implied from subsequent work or $u = \sin x$, $du/dx = \cos x$ $\int \frac{1}{2-u} [du]$ $[-\ln(2-u)]$ not $\ln 2 - \ln 1$

Question			Answer/Indicative content	Marks	Guidance	
		iii	(B) $\int_0^k \frac{\cos x}{2 - \sin x} [dx] = \frac{1}{2} \ln 2$ $\Rightarrow \ln 2 - \ln(2 - \sin k) = \frac{1}{2} \ln 2$ $\ln (2 - \sin k) = \frac{1}{2} \ln 2 = \ln \sqrt{2}$ $\Rightarrow 2 - \sin k = \sqrt{2}$ $\Rightarrow \sin k = 2 - \sqrt{2}$ $\Rightarrow k = \arcsin(2 - \sqrt{2})^*$ or $\int_0^{\arcsin(2-\sqrt{2})} \frac{\cos x}{2 - \sin x} [dx] = [-\ln(2 - \sin x)]_0^{\arcsin(2-\sqrt{2})}$ $= \ln 2 - \ln(2 - 2 + \sqrt{2}) = \ln 2 - \ln \sqrt{2}$ $= \ln 2 - \frac{1}{2} \ln 2 = \frac{1}{2} \ln 2^*$	M1 A1 M1 A1 A1cao M1 A1 A1 [5]	equating integral from 0 to k or from k to $\pi/2$ to $\frac{1}{2}$ their area o.e. e.g. $\ln(2 - \sin k) = \frac{1}{2} \ln 2$ eliminating logarithms correctly o.e. e.g $(2 - \sin k)^2 = 2$ NB AG SC: verifying (max 3 marks out of 5): attempt to find integral from 0 to $\arcsin(2-\sqrt{2})$ correct expression N.B AG Examiner's Comments This question was the most demanding in the paper, with nearly half the candidates scoring zero marks. Often the problem seemed to lie with getting expressions consistent with the limits of the integral, either in terms of x or u.	or equating integral from 0 to k to integral from k to $\pi/2$ $\ln 2 - \ln(2 - \sin k) = \ln(2 - \sin k)$ dep first M1 or from $\arcsin(2-\sqrt{2})$ to $\pi/2$
			Total	18		

Question			Answer/Indicative content	Marks	Guidance
82		i	$\frac{1}{2}x = \sin y$ $\Rightarrow x = 2\sin y$ $\frac{dx}{dy} = 2\cos y$	B1cao B1cao [2]	Examiner's Comments The majority of candidates scored both these marks. Occasionally they found dy/dx instead of dx/dy and lost a mark.

Question			Answer/Indicative content	Marks	Guidance
83		i	$2x^{\frac{1}{3}} + \frac{2}{3}y^{\frac{2}{3}} \frac{dy}{dx} = 0$ $\Rightarrow \frac{dy}{dx} = -3x^{\frac{1}{3}}y^{\frac{2}{3}} \text{ o.e.}$	M1 A1 A1 [3]	$\frac{d}{dx}(y^{\frac{1}{3}}) = \frac{1}{3}y^{-\frac{2}{3}} \frac{dy}{dx}$ seen correct equation must simplify 2 / (2/3) = 3 mark final answer Examiner's Comments The implicit derivative here was a straightforward example, and virtually all the candidates got the derivative equation correct. However, simplifying the fractional expression to get the final mark was often missing or incorrect: in particular, many learners made mistakes when dividing 2 by 2/3.
		ii	when $x = 1, y = 8, \Rightarrow \frac{dy}{dx} = -3 \times 4 = -12$	M1 A1cao [2]	substituting both $x = 1$ and $y = 8$ into their dy/dx NB check power of x is correct in part (i) Examiner's Comments There were two easy marks here, and virtually all candidates achieved the 'M' mark for substituting for x and y in their derivative.
			Total	5	

Question			Answer/Indicative content	Marks	Guidance
84			$\frac{1}{(5-2x^3)^2} = (5-2x^3)^{-2}$ $\frac{d}{dx}(5-2x^3)^{-2} = (-6x^2)(-2)(5-2x^3)^{-3}$ $= 12x^2(5-2x^3)^{-3} \text{ isw}$	M1 A1 A1cao [3]	Chain rule on $(5-2x^3)^{-2}$ correct expression, allow $(-6x^2)(-2)u^{-3}$ o.e. or $\frac{12x^2}{(5-2x^3)^3}$ isw or quotient (or product) e.g. $\frac{(5-2x^3)^2 \cdot 0 - 1 \cdot 2(5-2x^3)(-6x^2)}{(5-2x^3)^4}$ M1A1 [must have correct denom for M1] $u v' - v u'$ in QR is M0 Examiner's Comments This was a straightforward test of the chain rule, in which over three quarters of the candidates scored full marks. Occasionally we saw a quotient rule used, which required to be simplified to gain full marks. Another occasional error was to get the wrong sign, e.g. $-12x^2(5-2x^3)^{-3}$.
			Total	3	

Question			Answer/Indicative content	Marks	Guidance
85		i	correct rearrangement of $400 = \pi r^2 h$ seen, where h is not in the denominator substitution seen to obtain given answer $A = 2\pi r^2 + \frac{800}{r}$ not from wrong working	B1 B1 [2]	eg $h = \frac{400}{\pi r^2}, rh = \frac{400}{\pi r}, \pi rh = \frac{400}{r}$ or $2\pi rh = \frac{2 \times 400}{r}$ if B0B0 allow SC2 for eg $400 \pi r^2 h$ used $\frac{800}{r} = \frac{2 \times 400}{r}$ $\left(\text{or } \frac{2V}{r} \right) = \frac{2 \times \pi r^2 h}{r}$ used to obtain $A = 2\pi r^2 + 2\pi rh$ allow embedded versions of these must see all the steps if starting from $A = 2\pi r^2 + \frac{800}{r}$ Examiner's Comments Most candidates scored full marks here, but poor algebra let some candidates down. A wide variety of solutions were seen, some of which very elaborate.
		ii	$\left(\frac{dA}{dr} \right) = 4\pi r - \frac{800}{r^2}$ oe $\left(\frac{d^2 A}{dr^2} \right) = 4\pi + \frac{1600}{r^3}$ oe	B1 B1 B1 B1 [4]	for first term for second term FT to give non-zero first term FT negative power of r to give non-zero second term A maximum of B1B0B1B0 is available if 2nd term left in terms of h Examiner's Comments In spite of the correct expression being given in part (i), some candidates worked with an expression involving h, which inhibited much further progress. Some candidates worked with 800^{-r} and some disregarded π or treated it as a variable. The majority, however, differentiated successfully to obtain full marks.

Question			Answer/Indicative content	Marks	Guidance
		iii	their $\frac{dA}{dr} = 0$ seen $r = \sqrt[3]{\frac{200}{\pi}}$ or 3.99... isw $\frac{d^2 A}{dr^2} > 0$ justified so minimum oe or check gradient either side of <i>their</i> positive r $A = 300$ to 301	M1 A1 B1 A1 [4]	A0 for two or more values eg $r = 0$, 3.99 or ± 3.99 eg $4\pi > 0$ and $\frac{1600}{r^5} > 0$ NB 12π or 37.699... to 38 NB 300.530027931 Examiner's Comments A sizeable minority of candidates failed to score any marks in this part, beginning with an inequality in the second derivative. A good number of candidates started on the right track by setting the first derivative to zero, but then failed to make progress. Only rarely did candidates successfully find r and A and then use the second derivative correctly to establish that they had indeed found the minimum surface area.
			Total	10	

Question			Answer/Indicative content	Marks	Guidance	
86			kx^4	M1	$k > 0$	must not follow from use of $y = mx + c$
			$3x^4$	A1	may be seen later	
			$-7x + c$	B1	must follow from integration	
			$10 = (\text{their } 3) \times 2^4 - 7 \times 2 + c$ oe	M1	must be 3 terms on RHS including term in x^4 , term in x and “ c ”;	must not follow from use of $y = mx + c$
			$y = 3x^4 - 7x - 24$	A1	or $y = 3x^4 - 7x + c$ and $c = -24$ stated isw	must see “ $y =$ ” or “ $f(x) =$ ” at some point for A1
				[5]		
					Examiner’s Comments The vast majority of candidates tackled this question successfully. A few slipped up with the arithmetic in finding c , and a small minority worked with $y = mx + c$ with $m = 12 \times 3 - 7$ and failed to score.	
Total				5		

Question			Answer/Indicative content	Marks	Guidance	
87			$\left[\frac{dy}{dx} = \right] kx^2$ soi When $x = 2$, $\left[\frac{dy}{dx} = \right] 24$ $-\frac{1}{\text{their } 24}$ $x = 2, y = 16$ $x + 24y = 386$ oe	M1 A1 M1 B1 A1 [5]	$k > 0$ their 24 must come from evaluating their derivative NB $y - 16 = -\frac{1}{24}(x - 2)$ coefficients in any exact form eg $\frac{1}{24}x + y = \frac{193}{12}$ but not rounded or truncated decimals	NB $6x^2$ M0 if their 24 from elsewhere eg integration
			Total	5		

Question			Answer/Indicative content	Marks	Guidance
88		i	$\frac{\log_{10} 0.2 - \log_{10} 0.1}{0.2 - 0.1}$ or eg $\frac{-0.7 - -1}{0.2 - 0.1}$ seen 3.01 to 3.0103 isw or $10 \log_{10} 2$ isw oe	M1 A1 [2]	NB $\frac{\log_{10} 2}{0.1}$ or $\frac{0.3}{0.1}$ allow -0.69 to -0.7 for $\log_{10} 2$ in gradient formula for M1 Examiner's Comments Most knew what to do, but many slipped up by making a sign error in the numerator or by working with a rounded or truncated value of $\log_{10} 0.2$, thus losing the accuracy mark. condone omission of base 10; B2 for 3.01...unsupported
		ii	one point C marked on curve between A and B or before A	B1 [1]	condone omission of label of C Examiner's Comments Nearly all candidates correctly identified a suitable point on the curve. A few guessed wrongly and placed C to the right of B, and a very small number placed C off the curve altogether.
			Total	3	

Question			Answer/Indicative content	Marks	Guidance
89		i	$2x^2$ oe $F[5] - F[1]$ 48 cao	B1 M1 A1 [3]	ignore + c for the first two marks where $F[x] = kx^2$ no marks for 48 unsupported A0 for $48 + c$ Examiner's Comments Most candidates successfully integrated and went on to obtain the correct answer. A few spoiled this by leaving "+ c" in the final answer, and a small number either differentiated or simply evaluated the integrand.
		ii	$kx^{\frac{1}{2}+1}$ seen $4x^{\frac{3}{2}} + c$ or $4\sqrt{x^3} + c$ or $4(\sqrt{x})^3 + c$ isw	M1 A1 [2]	Examiner's Comments Nearly all candidates achieved the method mark by integrating, but a surprising number omitted the constant of integration thereby losing an easy mark.
			Total	5	